Ram Calculus and the Exponential Time Hypothesis

Revision 7: Four-State Boolean Reduction and Closed-Form Higher-Mode Summation

Exact Matched Tests, Runtime, Allocation Peaks, and Limits

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Abstract

We implement two exact packet simplifications previously proposed in v6: the four-state Boolean observation quotient and closed-form summation of all higher geometric modes. A factorial ablation compares each change and their combination against frozen v6 under both conservative source precision and degree-plus-symmetry settings. All 324 fresh worker outputs on four original packet fixtures match independent exact counts and, where applicable, full local polynomial oracles. Separate controls include eight no-op fork comparisons, 48 additional weighted-case validations, and modular-ring identities and negative tests. Five untraced timing repeats and three separate allocation-profile repeats are used per variant. Combined source-schedule median speed ratios range from 5.24× to 6.46×; within the v6 DG schedule they range from 1.52× to 2.07×. The report distinguishes tracked Python allocation from process high-water memory, documents timing variation, and charges the folding cost. Precision and interpolation schedules are unchanged within each ablation. Direct enumeration remains faster on every tiny fixture. These are one-active-level two-edge-packet tests, not new deep recursive tests or an improved general SAT exponent. The scientific body of v6 is retained as historical context. The new contribution is matched implementation evidence for exact local reductions, not a claim that classical Boolean algebra or geometric summation is exclusive to Ram Calculus.

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1 Revision 7: implemented Boolean quotient and closed higher-mode sum

1.1 Scope and comparison contract

This revision implements two opportunities proved algebraically but not used by the timed v6 packet routine: the four-state Boolean quotient and closed-form higher-mode summation. It tests each separately and jointly against the *frozen v6 implementation*, with the same four packet fixtures, exact root counts, local interpolation points, target maps, creation-time lifts, and bounded child cache. The comparison is against a finite rescaled source-form routine, not an executable supplied by the source authors and not the complete algorithm for SAT.

There are two schedules. S retains the conservative source-form working precision and does not use symmetry. DG retains v6's degree-certified precision and exact packet-interchange certificate. Within each schedule, 6L is the unmodified v6 six-state/geometric-loop routine; 4L changes only the observation basis; 6C changes only the kernel summation; and 4C uses both changes. Direct enumeration is a separate small-instance reference. The characters 4 and 6 count observation states per packet, not edges, recursion levels, or output bits. C means closed summation, not omission of higher orders.

All measured roots have four Boolean edge variables in two packets. Each packet has two real positions, 0 and 2, and internally coupled two-port tensors; the root decomposes across packets. The four generated two-edge children are leaves. The three-active-level singleton family of v6 is **not changed or rebenchmarked**: its kernel and packet type differ, so assigning the new speedups to that experiment would be unjustified. No test here combines general multi-edge packets with deep recursion or hard large SAT inputs.

1.2 Exact four-state reduction over a ring

Let R be a commutative coefficient ring, including a ring with zero divisors. The six-state row polynomial used by v6 is

$$P(Z_0, Z_+) = c_{00} + c_{10}Z_0 + c_{01}Z_+ + c_{20}Z_0^2 + c_{11}Z_0Z_+ + c_{02}Z_+^2.$$

For the two Boolean column bits x, y, the observed variables are $Z_0 = x + y$ and $Z_+ = y$.

Proposition 1.1 (Four-state quotient)**.** *On $\{0,1\}^2$, the same function is*

$$\begin{aligned} P(x+y, y) &= d_{00} + d_{10}x + d_{01}y + d_{11}xy, \\ d_{00} &= c_{00}, \\ d_{10} &= c_{10} + c_{20}, \\ d_{01} &= c_{10} + c_{01} + c_{20} + c_{11} + c_{02}, \\ d_{11} &= 2c_{20} + c_{11}. \end{aligned} \tag{1}$$

The identity uses no division and is valid over R.

Proof. In $R[x,y]/(x^2-x, y^2-y)$, $(x+y)^2 = x+y+2xy$, $(x+y)y = xy+y$, and $y^2 = y$. Substitution and collection of coefficients give (1). Evaluation of $1, x, y, xy$ at the four Boolean points is an invertible triangular transform after ordering the points by inclusion; its determinant is a unit, 1 or -1. Thus this is an exact coordinate representation of every function on the Boolean square, not an approximation or a rank test over an incorrectly chosen field. □

This quotient preserves each fixed row-bit/column-bit kernel product, not merely the final scalar count. It therefore preserves contraction against arbitrary old row and column tensors at each interpolation point. In the implementation, the six coefficients are first constructed by the unchanged v6 routine, and their folding cost is then paid explicitly. The column table changes from $Z_0^a Z_+^b$ to $x^a y^b$ with $(a,b) \in \{0,1\}^2$. Reusing the old column table with the new row coefficients would be incorrect.

Worked example and a failure control. For $P = Z_0^2$, the correct four-state expression is $x + y + 2xy$. At $(x,y) = (1,1)$ modulo eight, its value is four. Incorrectly dropping the factor two gives two. Although the residue field has characteristic two, the coefficient ring modulo 2^B generally does not. The quotient also cannot be applied to arbitrary non-Boolean observations: at $(x,y) = (2,3)$ modulo 16, $(x+y)^2 = 9$ but $x + y + 2xy = 1$.

1.3 Closed summation of all higher modes

Keep v6's $L = 4$ kernel, centers $t = -3, -1, 1, 3$, scale eight, and five-coordinate extension

$$R_B = (\mathbb{Z}/2^B\mathbb{Z})[U]/(U^5 + U^2 + 1), \qquad T = t + 8\zeta.$$

Define

$$C(T) = \prod_{s=-4}^{3} (T - s), \quad C'(0) = -144, \quad N = C(T)/(-144), \quad A_x = N(T+x)^{-1}, \quad r_x = 2(T+x)^{-1}.$$

Both $T + x$ and $T + x - 2$ are units for $x = 0, 2$. The two column coordinates are $2j$ with $j = 0, 1$, so every positive-order moment $\sum_j j^b X_{2j}$ equals y. V6 explicitly sums its coefficient $B_x = \sum_{b=1}^{B-1} A_x r_x^b$ until the modular tail vanishes.

Proposition 1.2 (Exact finite closed sum)**.** *In R_B,*

$$B_x = N\big((T + x - 2)^{-1} - (T + x)^{-1}\big). \tag{2}$$

Replacing the geometric loop by (2) preserves every packet kernel coefficient modulo 2^B.

Proof. The element r_x is divisible by two, hence $r_x^B = 0$ in R_B. Consequently $\sum_{b=0}^{B-1} r_x^b = (1 - r_x)^{-1}$. Since $1 - r_x = (T + x - 2)(T + x)^{-1}$, multiplying by A_x and subtracting the order-zero term gives (2). No convergence limit, floating-point approximation, or division by a nonunit is involved. □

The original loop already computes both unit inverses and the second closed kernel value even in a non-diagnostic timed worker. Thus the new closed variant removes the geometric multiplications and additions; it does not gain an unfair advantage by removing a separate diagnostic cost charged only to the comparator. In diagnostic workers an independent call to the old loop checks the new closed result, outside timing samples. The closed routine has zero geometric-loop iterations, but this does *not* mean that its higher-mode contribution is zero.

For fixed extension degree and the specified two-position support, the old row calculation may require a number of extra modular products linear in B; the closed row calculation uses a constant number of products and differences in addition to its two unit inversions. The implemented Newton inversion takes a number of lifting steps logarithmic in B, and those inversions were already paid by the old routine. This is a statement about the local arithmetic operation sequence. Each product still has a bit cost depending on B, and the contraction, interpolation, and full source recursion remain outside this local count.

Protected division is retained. Since $-144 = 16(-9)$, the code computes $C(T)$ modulo 2^{B+4}, divides its canonical coordinates by 16, and multiplies by the unit inverse of -9 modulo 2^B. All numerator coordinates are divisible by 16. At these centers they are in fact divisible by 2^7. The four guard bits cannot be dropped merely because the geometric sum is closed. For $T = 5$ and $B = 5$, premature reduction of the numerator modulo 32 yields quotient zero, whereas the correct $C(5)/(-144)$ is eight modulo 32. This is an executed negative control.

A complete two-bit kernel example. At $T = 5$ and row bits $(0, 0)$, take a modulus larger than the integers below. The two rows have

$$(A_0, B_0) = (-504, -336), \qquad (A_2, B_2) = (-360, -144).$$

Thus

$$P(Z_0, Z_+) = (1 + 504Z_0 + 336Z_+)(1 + 360Z_0 + 144Z_+).$$

Its four-state form is

$$1 + 182304x + 424704y + 556416xy.$$

It evaluates on $(0, 0), (1, 0), (0, 1), (1, 1)$ to $1, 182305, 424705, 1163425$, respectively, exactly as the unfactored kernel expression. The value at $(0, 1)$ is the v6 higher-mode counterexample: discarding the higher contribution would incorrectly give 182305. Closed summation and Boolean folding instead retain 424705. This example distinguishes changing coordinates from throwing away information.

1.4 Orbit counts and recursive interface preservation

With two packets, the observation assignments fall from $6^2 = 36$ to $4^2 = 16$. When whole-packet interchange is certified, unordered representatives fall from 21 to ten. Orbit sums use integer multiplicity, not division by group order. The same certificate preserves ordered real positions, tensor values and axes, side, creation precision, and grouping up to a bijection. Four-state labels are not silently permuted inside a packet. P01 admits interchange; P02–P04 do not.

The generated row tensor has four entries indexed by its two Boolean ports, as does its corresponding column tensor. Child-network construction, canonical creation-time lifts, precision requests, and the exact-key 16-entry cache are unchanged. A changed basis can nevertheless change the sequence of cache keys and the number of distinct child values. It is therefore not valid to infer a runtime factor of exactly 36/16 or a memory factor of exactly 6/4 from observation counts alone.

1.5 Validation and benchmark design

The timed cases are the original v6 P01–P04 definitions. P01 has OR tables at all four vertices; P02 replaces the first row by XOR; P03 zeros that row; P04 assigns entry j of vertex i the integer $2^{15} + 7i + 3j + 1$. Their exact answers are respectively

$$9, \quad 6, \quad 0, \quad 18482797547933526180.$$

P04 is a weighted contraction, not a Boolean SAT count. Output precisions five and 65 bits exceed proven nonnegative root-count upper bounds. All non-direct outputs are checked against the independently constructed rational polynomial oracle at all four centers and all 29 coefficients, including every extension coordinate. The checks are not limited to agreement of a single final residue.

There are nine variants, including direct enumeration. Each case/variant has five untraced timing workers, three separately traced memory workers, and one diagnostic worker: 180 timing, 108 memory, and 36 diagnostic outputs, 324 in total. Workers are sequential fresh Python processes; method order is rotated across repeats and fixtures. Startup and imports are outside the pipeline interval. Network construction, precision plans, symmetry discovery, guarded kernel arithmetic, folding, child construction, cache operations, interpolation, and target recovery are included. Independent oracle computation and diagnostic extra checks are excluded from the performance samples. CPU time is recorded alongside elapsed time.

The no-op new-engine fork is separately checked against the frozen v6 source and DG implementations on all four cases, including full local polynomials, counter dictionaries, plans, and node records. The timed 6L baselines execute frozen v6 itself, rather than that fork. This detects accidental changes outside the two intended ablations.

A separate record audit also checks the independently expected working precision and plan, canonical coefficient coordinates, basis-specific logical terms and orbit counts, and folding or closed-kernel counters. It deliberately mutates otherwise valid records to ensure that a wrong precision or a relabeled ablation is rejected. A correct scalar answer, especially on the zero fixture, does not certify that the intended optimization was actually exercised. The accepted experimental domain is explicitly the four-edge, two-edge-packet family with leaf threshold two, not arbitrary lower-level calls with a different leaf threshold. The completed mutation suite rejects 398 deliberately malformed records or domain requests. Cache counters are checked through conservation and arithmetic accounting, not by replaying every transient cache entry.

Untimed controls compare 192 closed/loop row kernels at six precisions and nonbase extension points; 1,536 row-bit/column-bit products after folding; and 192 evaluations with arbitrary extension-valued polynomial coefficients, including characteristic two and higher powers of two. Twelve additional deterministically generated weighted cases use a fixed random seed and are tested under the four DG variants, giving 48 further full-polynomial validations. These are additional small correctness checks, not a random workload performance study.

Memory metric. The primary new memory measure is peak `tracemalloc` allocation since starting tracing after imports and fixture selection and before the pipeline call. Garbage collection occurs before each call. The peak, baseline, and final traced bytes are saved. This includes tracked Python allocations during the call, not all native allocation, the interpreter's full resident set, or the full input representation already constructed before tracing. The returned result and any surviving allocation are included. Profiling times are not used as timing samples. Three memory repetitions allow a descriptive median and range, not an asymptotic space bound. Untraced workers also retain Windows process working-set high-water measurements for context, with the same baseline limitation as v6. These two memory metrics must not be interchanged. An independent instrumentation self-test allocates a temporary 2 MiB buffer and frees it before returning; tracing detects its peak while the final live increment is below 64 KiB. This checks that freed transients are observable, not that all native or operating-system allocations are captured.

1.6 Matched timing results

Every recorded worker passed its exact-count check; all non-direct workers also passed the complete local-polynomial check. Elapsed times below are seconds. Ratios use unrounded medians of five fresh untraced workers; a ratio below one is a slowdown. The S and DG schedules are shown separately so a bit-budget change is not confused with either new ablation.

Case	Exact integer target	Count matches	Polynomial checks
P01	9	81	72
P02	6	81	72
P03	0	81	72
P04	18482797547933526180	81	72

The 72 polynomial checks per fixture exclude the nine direct workers, which do not interpolate. Repeats validate repeated executions of the same four fixtures, not 324 distinct mathematical instances.

S schedule.

Case	S6L	S4L	S6C	S4C
P01	3.0953	2.6546	0.8731	0.5908
P02	3.1716	2.6410	0.7737	0.5204
P03	2.6004	2.4440	0.6097	0.4544
P04	3.9597	4.1678	0.8958	0.6126

Case	6L/4L	6L/6C	6L/4C
P01	1.17	3.55	5.24
P02	1.20	4.10	6.10
P03	1.06	4.26	5.72
P04	0.95	4.42	6.46

DG schedule.

Case	DG6L	DG4L	DG6C	DG4C
P01	0.5333	0.4415	0.4158	0.3508
P02	0.6242	0.5234	0.5642	0.4088
P03	0.5882	0.4414	0.4824	0.3465
P04	0.8534	0.6548	0.6055	0.4120

Case	6L/4L	6L/6C	6L/4C
P01	1.21	1.28	1.52
P02	1.19	1.11	1.53
P03	1.33	1.22	1.70
P04	1.30	1.41	2.07

For the combined change, source-schedule ratios range from 5.24× to 6.46×; within DG they range from 1.52× to 2.07×. These are local host-specific observations on the four fixtures, not a general complexity bound.

CPU times and variability.

Case	S6L CPU	S4C CPU	DG6L CPU	DG4C CPU
P01	3.0781	0.5781	0.5312	0.3438
P02	3.1562	0.5312	0.6250	0.4062
P03	2.6094	0.4531	0.5938	0.3438
P04	3.9219	0.6094	0.8438	0.4062

The next table gives elapsed minima and maxima across the five samples. It is a range of observations, not a confidence interval. Processor scheduling, frequency changes, and other activity can affect the ratios; this experiment does not establish a hardware-independent constant speedup.

Case	S6L range	S4C range	DG6L range	DG4C range
P01	2.709–6.459	0.517–0.706	0.493–0.670	0.311–0.746
P02	2.688–3.731	0.496–0.588	0.601–0.779	0.385–0.502
P03	2.535–2.784	0.441–1.024	0.561–0.690	0.321–0.366
P04	3.840–6.128	0.537–0.623	0.833–0.885	0.402–0.457

For a second view of variability, the following table pairs 6L and 4C by fixture and repeat index, and summarizes the five elapsed-time ratios. These are sequential matched-repeat comparisons, not simultaneous runs. A median of ratios need not equal the ratio of medians reported above; neither removes host variation.

Case/schedule	Paired minimum	Paired median	Paired maximum
P01/S	4.45	5.38	9.14
P02/S	5.42	5.95	7.17
P03/S	2.48	5.77	5.90
P04/S	6.16	6.35	11.42
P01/DG	0.90	1.52	1.83
P02/DG	1.52	1.56	1.65
P03/DG	1.53	1.68	2.15
P04/DG	1.88	2.07	2.15

Variants slower than their same-schedule v6 comparator by these medians are: P04/S4L (0.95). These results are retained; a smaller state space does not guarantee a faster measured implementation.

Direct enumeration remains faster in every case: its median ranges from 0.000201 to 0.000245 seconds. All roots have only 16 assignments, so the interpolation/recursive machinery remains substantially more expensive than the independent direct reference.

1.7 Tracked allocation peaks and process-memory context

The following tables give median peak incremental tracked Python allocation in KiB, measured in three separate traced workers per case/variant. They are not Windows process RSS and do not use

the much slower profiling elapsed times as timing samples.

S schedule: tracked KiB.

Case	S6L	S4L	S6C	S4C	4C change (%)
P01	818.82	819.26	817.56	819.20	+0.05
P02	809.08	807.85	807.82	807.78	-0.16
P03	783.57	786.61	783.51	786.54	+0.38
P04	835.19	828.46	833.85	828.40	-0.81

DG schedule: tracked KiB.

Case	DG6L	DG4L	DG6C	DG4C	4C change (%)
P01	537.80	537.38	537.16	537.35	-0.08
P02	534.86	532.72	534.23	532.69	-0.41
P03	526.78	525.97	526.75	525.94	-0.16
P04	568.49	563.64	567.77	563.61	-0.86

Across the eight combined-change comparisons the signed allocation-peak change ranges from -0.86% to $+0.38\%$; a negative sign means a smaller measured traced peak. State-count reductions need not translate into memory reductions because the complete live interpolation basis, folded coefficient construction, child cache, and retained result also contribute.

All eight combined-change allocation effects are below one percent in magnitude. Accordingly these measurements do not establish a substantial practical peak-memory advantage, despite the large decrease in observation-state and loop counts. Small byte differences should not be promoted to a general storage theorem.

Case	S6L range	S4C range	DG6L range	DG4C range
P01	818.8–818.8	819.2–819.2	537.8–537.8	537.4–537.4
P02	809.1–809.1	807.8–807.8	534.9–534.9	532.7–532.7
P03	783.6–783.6	786.5–786.5	526.8–526.8	525.9–525.9
P04	835.2–835.2	828.4–828.4	568.5–568.5	563.6–563.6

Untraced process high-water peaks span 25.65–26.34 MiB; 161 of 180 timing workers have zero increase over the pre-call process high-water mark. Such process peaks include imports and interpreter baseline. They do not replace allocation-specific measurements or prove the source algorithm's asymptotic space bound.

1.8 Executed structural work

The next tables use diagnostic counters. The basis construction and 116 interpolation evaluations are retained. Closed-kernel diagnostics additionally compare the loop using a separate counter object; they are not included in timing medians or in the optimized loop count.

P01.

Variant	Reps	Child requests	Leaf calls	Loop orders	Ring mult.
S6L	36	16704	570	107976	162358
S4L	16	7424	345	107976	151278
S6C	36	16704	570	0	54382
S4C	16	7424	345	0	43302
DG6L	21	9744	570	11432	55896
DG4L	10	4640	345	11432	48992
DG6C	21	9744	570	0	44464
DG4C	10	4640	345	0	37560

P04.

Variant	Reps	Child requests	Leaf calls	Loop orders	Ring mult.
S6L	36	16704	1819	121416	186776
S4L	16	7424	913	121416	170248
S6C	36	16704	1819	0	65360
S4C	16	7424	913	0	48832
DG6L	36	16704	1819	24872	87274
DG4L	16	7424	913	24872	70746
DG6C	36	16704	1819	0	62402
DG4C	16	7424	913	0	45874

The ring-multiplication counters include guarded numerator operations and interpolation. Changes in leaf calls or cache hits can reflect the changed observation basis as well as the number of representatives; not every saved child request is a saved direct contraction. Closed-mode counters measure a different implementation from the old higher-order counters, and absence of loop iterations must not be reported as absence of higher-mode mathematics.

1.9 Precision is unchanged by these two ablations

S uses 485 coefficient bits for P01–P03 and 545 for P04; DG uses 54 and 114. The numerator uses four additional guard bits in each case. The new four-state basis and closed sum request exactly these same precisions and the same 116 local evaluations. Thus v7's within-schedule effects are not additional bit-budget reductions. V6's 480-to-49 increment improvement and 581-to-45 deep-path maximum remain prior results; the latter is not an executed v7 path.

The closed identity replaces up to hundreds of nonzero modular orders by two unit-inverse kernel evaluations per row, operations the old loop also computed. This is a finite nilpotent-series simplification. The four-state quotient removes redundant Boolean observations, not bits of information needed to represent arbitrary functions on the Boolean square.

1.10 Interpretation, advantages, and limitations

The new contribution relative to v6 is an implemented and matched-tested version of the two previously unbenchmarked simplifications. The exact identities were already in v6; v7 does not reclassify them as newly discovered mathematics. Its new evidence concerns executable correctness, runtime, allocation peaks, counters, and interactions with the existing precision and symmetry schedules.

These operations fit the Ram Calculus viewpoint of choosing an explicit observation quotient and evaluating only the target information. They are also conventional Boolean quotient-ring algebra

and finite geometric summation. An otherwise identical conventional implementation obtains the same benefits. The experiments do not establish a gain exclusive to Ram Calculus terminology or historical priority for these algebraic identities.

Advantages include an exact four-dimensional representation, removal of precision-proportional geometric loops, fully auditable identities, and compatibility with the existing modular target certificate. Costs include the paid folding step, changed cache behavior, inverse arithmetic, interpolation, and symmetry certificates. Process memory and tracked Python allocation need not move in the same direction as state count. The unchanged interpolation stage also imposes a practical floor on the benefit from optimizing just packet summation.

The two-coordinate identity $j^b = j$ is special to $j = 0, 1$. Larger position sets may have independent higher moments. The Boolean quotient theorem is valid for arbitrary two-bit packet functions, but a small uniform state bound for arbitrary packet size does not follow; a generic Boolean packet of k bits has 2^k independent functions. Nor does the exact closed expression solve the full cost of producing, coupling, and recursively contracting these functions.

1.11 Requirements for improving the full source algorithm

A complete improvement needs uniform certificates on the source's actual packet families, all local/crossing interactions, and every generated child. It needs a bit-complexity recurrence charging inverse construction, extension arithmetic, degree maps, orbit discovery, and all child creation, plus a space proof that includes caches and interpolation data. A smaller finite state count does not by itself improve a SAT exponent. The literal source activation and leaf thresholds are unchanged; their enormous parameter scale is neither executed here nor refuted by a small test.

Next concrete steps are connected multi-packet families with higher degree, compatible deep multi-port recursion, nontrivial moment-rank certificates for larger supports, direct construction of the four coefficients without first constructing six, selective-channel recovery, and controlled allocation/streaming comparisons. Any extension must retain creation-time lifts and exact target checks. No product of v6's deep speedup with v7's packet speedup is a demonstrated combined result.

1.12 Reproducibility and preservation

All new implementation and evidence files reside in `eth-benchmark-v7/`. The runner is `run_v7.py`; the new engine is `reductions_v7.py`. The directory `results-20261027/` contains the raw worker outputs, full fixtures and exact polynomial oracles, controls, aggregate tables, and file-hash register. A fresh run uses a previously nonexistent destination:

```
python -B run_v7.py -out NEW_RESULTS_DIRECTORY
```

The runner refuses to overwrite an evidence directory. V6 and its dependencies are imported read-only with bytecode writing disabled. The report assembly retains the scientific v6 body below unchanged, while updating the title, abstract, and this new section. Statements below that an optimization is unbenchmarked describe the historical v6 boundary and are superseded only by the precisely scoped v7 packet results above. Previous PDFs, source reports, code, and raw results are preserved rather than relabeled as new measurements.

Retained v6 and earlier scientific material

The following body is retained from v6. Its experiment versions, dates, performance measurements, and limitations are historical; they are not additional v7 workers. Section 1 is the current extension.

2 Revision 6: two-edge packets, higher modes, and three active levels

2.1 New scope and the limit that remains

Revision 6 extends the executable tests in two complementary directions. The first family has *two real edges per packet*, nonzero off-diagonal kernel entries, and nonzero higher geometric orders. The second family extends the singleton construction to *three active recursive levels*, along the path $4 \to 3 \to 2 \to 1$. The root and the three- and two-edge children are active; one-edge children are leaves. These are implemented extensions, not extrapolations from a one-edge calculation.

The families must not be conflated: the timed two-edge-packet tests have one active level, while the timed three-active-level test has singleton packets. We have **not** tested three active levels of general multi-edge packets with independently varying high-order moments. Both families remain finite rescaled specializations, not executions of the source's literal active ETH schedule. The activation barrier and the absence of a full-algorithm complexity improvement from v5 remain.

The source comparator again means our finite source-form contraction and Hermite routine, with conservative precision. It is not an author-supplied full SAT executable. All variants share the same justified packet-mode coalescence described below, bounded exact cache, and interpolation schedule within a given test family. Consequently no observed ratio is a measured comparison with literal enumeration of every original high-dimensional mode tuple.

2.2 What changed from v5

1. The packet family uses $L = 4$, $a = 2$, $u = 1$, scale $\alpha = 8$, four centers $-3, -1, 1, 3$, and real packet positions $0, 2$. Its deformation has degree at most $7M$, rather than $3M$.
2. The code constructs full two-port packet tensors by coefficient extraction. It no longer replaces every packet tensor by a unary tensor.
3. Geometric orders $b = 1, 2$ and higher are explicitly evaluated; nonzero higher-mode products contribute to the contraction. An exact observation-space quotient makes this finite case tractable.
4. The Hermite planner supports both two- and four-center families, and correctly handles two-adically integral fractions with odd denominators in the independent polynomial oracle.
5. A separate deeper evaluator reuses the v5 algebra with smaller strict-shrinkage steps, reaching three active levels and inherited nonbase extension-valued tensors.
6. Memory is measured by the Windows process peak working set in untraced fresh workers. This is a different metric from v5's Python `tracemalloc` peaks; those numbers are not compared directly.

The v5 scientific body and bibliography are retained after this section. The author fields of the existing reports in this directory were updated separately at the user's request, with original

copies and hashes preserved. That authorized metadata change does not alter any benchmark implementation, raw result, mathematical argument, or historical hash register.

2.3 The full two-edge packet kernel

For $S = \{-4, -3, -2, -1, 0, 1, 2, 3\}$ put

$$\mathcal{C}(T) = \prod_{s \in S} (T - s), \qquad \mathcal{C}'(0) = -144, \qquad K_{xy}(T) = \frac{\mathcal{C}(T)}{-144(T - y + x)}.$$

The real row and column positions in each packet are $x, y \in \{0, 2\}$. Each entry is a degree-seven polynomial because $y - x \in S$. At zero, the diagonal entries equal one and the off-diagonal entries equal zero. Thus the restricted matrix is the identity at the target. The complete network deformation is

$$F(T) = \sum_{X,Y} A(Y)B(X) \prod_{\text{packets } P} \prod_{x \in \{0,2\}} \ell_{Y_{P,x}} \left(\sum_{y \in \{0,2\}} K_{xy}(T) X_{P,y} \right), \tag{3}$$

where $\ell_0(z) = 1 - z$ and $\ell_1(z) = z$. Consequently $\deg F \le 7M$ and $F(0) = \sum_X A(X)B(X)$ is the original count or weighted contraction.

The real positions are even. At an odd center t, every difference $y - x$ is even, so all four entries of the real submatrix vanish at t. Rows belong to one group and columns to a distinct group in the packet. All its real interactions are therefore crossing interactions in this specialization, and moving to $T = t + 8\zeta$ activates them. This tests nontrivial two-edge coupling without claiming to cover arbitrary mixtures of local and crossing packet entries.

For the four-edge root the interpolation degree is 28. Arithmetic uses

$$\mathcal{R}_B = (\mathbb{Z}/2^B\mathbb{Z})[U]/(U^5 + U^2 + 1),$$

whose modulo-two quotient is $\mathbb{F}_{32}$. The 29 interpolation points are distinct binary residue vectors. Their differences are units, giving exact interpolation at all four centers. The tests check inverses of all 31 nonzero residue elements. This fixed extension is sufficient for this degree, not a replacement for the source's size-dependent extension in a general theorem.

Proposition 2.1 (Local integrality and protected division)**.** *For odd $t \in \{-3, -1, 1, 3\}$ and any ζ in the unramified extension algebra, $\mathcal{C}(t + 8\zeta)$ is divisible by 2^7. Consequently $N = \mathcal{C}(t + 8\zeta)/(-144)$ and every real kernel entry at this point are divisible by 2^3. Computing the numerator modulo 2^{B+4} is sufficient to compute N modulo 2^B.*

Proof. The four factors with even s are units. Among the four factors with odd s, the factor $s = t$ is divisible by 8, while the other three constant differences have valuations $1, 1, 2$ in some order. Adding 8ζ preserves these three valuations. Their product is therefore divisible by $2^{3+1+1+2} = 2^7$. Since $-144 = 2^4(-9)$ and -9 is a unit, division leaves at least three powers of two. The real denominators $T - y + x$ are units because x, y are even and t is odd. Changing a numerator representative modulo 2^{B+4} changes its quotient by a multiple of 2^B; multiplication by the inverse of -9 preserves that congruence. This proves both integrality and the guard-bit claim without treating division by two as a modular inverse. □

2.4 Nonzero higher modes and exact coalescence

At a fixed center and evaluation point, write

$$T = t + 8\zeta, \quad N = \mathcal{C}(T)/\mathcal{C}'(0), \quad A_x = \frac{N}{T+x}, \quad r_x = \frac{2}{T+x}.$$

All $T + x$ are units because t is odd and x is even. The two column positions lie in the tile $y = 2j$ with $j \in \{0, 1\}$, so

$$K_{x,2j}(T) = A_x \sum_{b \geq 0} r_x^b j^b \pmod{2^B}.$$

The modes are

$$L_0(X) = X_0 + X_2, \qquad L_b(X) = X_2 \quad (b \geq 1).$$

The latter equality is an equality of functions on the real column bits, not an approximation and not the assertion that higher orders vanish. It permits the exact coalescence

$$\sum_y K_{xy}(T) X_y = A_x L_0(X) + B_x L_+(X) \pmod{2^B}, \qquad L_+(X) = X_2, \qquad B_x = \sum_{b=1}^{B-1} A_x r_x^b. \quad (4)$$

The ratio r_x is divisible by two; A_x has the requisite positive valuation. Terms omitted beyond this finite range vanish modulo 2^B. The implementation explicitly multiplies and sums the geometric terms until they become zero, recording nonzero orders one and two and the largest nonzero order reached. It does not replace B_x by zero.

An independent closed expression is

$$B_x = K_{x,2}(T) - K_{x,0}(T) = N\left((T + x - 2)^{-1} - (T + x)^{-1}\right) \pmod{2^B}. \quad (5)$$

Multiplying the finite geometric sum by $1 - r_x$ proves this identity, since its remainder vanishes modulo 2^B and $1 - r_x$ is a unit. Diagnostic runs check it at every interpolation point. The timed variants all use the explicit geometric summation; the closed form is not used as a substitute. The implementation nevertheless computes the closed-kernel value in timed workers too; only the equality assertion is conditional on diagnostics. Its inversion and multiplication costs are therefore common timed overhead, not an uncharged audit operation.

Why coalescing is legitimate under the original weight rule. Before coalescence each order is a distinct formal mode whose coefficient has the source's valuation guarantee. Identifying modes that have the same observation function commutes with polynomial substitution and coefficient summation. Terms whose original total weight is at least B still vanish in the modular ring when the products are formed. We therefore preserve the evaluated polynomial modulo 2^B, without pretending that the original formal-mode enumeration has been executed. This is a small observation-space quotient specific to the two-column tile. For three or more distinct tile coordinates, the high moments need not coincide, and a different certificate would be required.

A concrete failure when higher modes are discarded. At $t = -3$, $\zeta = 1$, hence $T = 5$, choose column bits $(X_0, X_2) = (0, 1)$ and row bits $(Y_0, Y_2) = (0, 0)$. The full product of row factors is

$$(1 - K_{0,2}(5))(1 - K_{2,2}(5)) = 841 \cdot 505 = 424705.$$

Keeping only the zeroth mode instead gives

$$(1 - K_{0,0}(5))(1 - K_{2,0}(5)) = 505 \cdot 361 = 182305.$$

The test records this difference modulo 2^{40}. This is a local kernel/product counterexample, not a claim that an invalid decoder's final SAT count was meaningfully recovered. It establishes that the tested higher-mode content cannot simply be omitted.

2.5 Two-port child tensors and recursion

Replace L_0, L_+ by formal variables Z_0, Z_+. For each packet, form

$$\prod_{x \in \{0,2\}} \ell_{Y_x}(A_x Z_0 + B_x Z_+).$$

Its monomial indices are $(\beta_0, \beta_+) \in \mathbb{N}^2$ with $\beta_0 + \beta_+ \leq 2$, giving six states per packet. The row-side packet tensor is the coefficient of $Z_0^{\beta_0} Z_+^{\beta_+}$, indexed by the two row bits. The column-side packet tensor is

$$(X_0 + X_2)^{\beta_0} X_2^{\beta_+},$$

indexed by the two column bits, with zero powers interpreted as one. These are genuine two-port tensors. Their coefficients can couple the two old edges; they are not presumed to factor into unary entries.

For two packets there are $6^2 = 36$ coalesced mode assignments. Factoring the independent group sums gives

$$F(t + 8\zeta) = \sum_{\boldsymbol{\beta}} \prod_{g} Z_{g,\boldsymbol{\beta}} \pmod{2^B}.$$

The code constructs these child networks, retains old tensor entries with their original creation precisions, creates the packet tensors at precision B, and calls the evaluator on each child. In this family the four children have two edges each and are leaves at threshold two. The correctness proof is coefficient extraction followed by exact substitution, group factorization, and exact leaf contraction, followed by interpolation and certified recovery.

The finite decomposition deliberately keeps both real ports of each packet in the same group on a given side. Root tensors are independent across the two packets. This is a restricted disconnected network family, although the two edges *within* each packet are coupled. It is not a test of a general connected packet arrangement.

2.6 Three active levels with fixed creation-time lifts

The deeper family uses the frozen v5 singleton kernel and arithmetic, but groups four unary vertices into blocks of sizes $3 + 1$ on each side. An active three-edge child is then grouped into $2 + 1$ on each side; two-edge children split into singleton groups. The path is

$$M = 4 \longrightarrow 3 \longrightarrow 2 \longrightarrow 1.$$

Thus there are three active levels at depths zero, one, and two, with leaves at depth three. Each child has strictly fewer edges, and the total child port count remains twice the parent edge count.

The input consists of four independent row unary constraints $(1, 0)$ and column unaries $(1, 1)$, giving exact count one. This is still an intentionally easy instance used to stress the recursive

arithmetic, not a hard SAT distribution. It creates many child calls and nonbase extension-valued tensors. A vertex created with canonical entries at precision p_c carries those representatives unchanged to all later precisions; it is never re-evaluated symbolically from an ancestor. Cache keys and symmetry checks include these creation data.

The degree and symmetry methods are exercised at all active depths. Diagnostic runs compare every executed active child target with a direct ring-valued contraction. The deepest new test is therefore more than a root-only scalar check, but it does not combine this depth with the new two-real-edge packet family.

2.7 Precision for four centers and deeper paths

For $L \in \{2, 4\}$, use odd centers $-L < t < L$, scale $2L$, and degree $D = M(2L-1)$. For selected multiplicities q_t, define

$$E_t(T) = \prod_{s \neq t} \left(\frac{T-s}{t-s}\right)^{q_s}.$$

The canonical Hermite target weights are obtained from the truncated reciprocal of $E_t(t+2Lz)$ and evaluation at $z = -t/(2L)$. Exact monomial checks certify their action on the required degree class. The sufficient common extra-bit loss is the largest negative two-adic valuation of a nonzero weight, with strict positive increment retained.

Theorem 2.2 (Degree-class target and precision certificate)**.** *Let $Q = \sum_t q_t$, $\alpha = 2L$, and let $F \in \mathbb{Q}[T]$ have degree less than Q. Write $c_{tj} = [z^j]F(t+\alpha z)$ and $a_{tk} = [z^k]E_t(t+\alpha z)^{-1}$. Then*

$$F(0) = \sum_t \sum_{j=0}^{q_t-1} w_{tj} c_{tj}, \qquad w_{tj} = E_t(0) \sum_{k=0}^{q_t-1-j} a_{tk}(-t/\alpha)^{j+k}.$$

Suppose the c_{tj} are two-adically integral and available modulo 2^{p+e}, where

$$e \geq \max(0, \max_{w_{tj} \neq 0} -v_2(w_{tj})).$$

The formula determines $F(0)$ modulo 2^p. The same statement holds coefficientwise in the fixed unramified extension used by the implementation.

Proof. For each (t, j) use the Hermite basis polynomial

$$H_{tj}(T) = E_t(T) \left(\frac{T-t}{\alpha}\right)^j \sum_{k=0}^{q_t-1-j} a_{tk} \left(\frac{T-t}{\alpha}\right)^k.$$

Its degree is at most $Q-1$. At every other center it has the required zero jet; at t its jet through order $q_t - 1$ is exactly z^j, because the truncated inverse cancels E_t through those orders. Thus $F - \sum c_{tj} H_{tj}$ has all Q prescribed zeros counting multiplicity and degree less than Q, and is zero. Evaluating at zero gives the displayed weights. A change of $2^{p+e} d_{tj}$ in any integral input coefficient changes the target by $2^{p+e} w_{tj} d_{tj}$, which is divisible by 2^p by the valuation bound. Adding such errors preserves that divisibility. Rational constants act coefficientwise in the fixed extension, so the same argument applies there. Odd denominators are units; negative powers of two are accounted for by e. □

This is a certificate on the specified polynomial class and target functional. It does not assert that the required extra bits are unavoidable after using all additional structure of a particular counting instance. The exact program checks the identity on the monomial basis, and hence by linearity on the entire certified class.

The source-form schedule uses $q_t = 3M$ and increment

$$e_S = 24ML + 8M(a+1), \qquad a = \log_2 L.$$

The degree plan balances the multiplicities with total $D+1$ and uses the actual loss of the resulting target map. The relevant budgets are:

L	M	D	S increment used	Exact loss of S weights	Degree increment used
4	4	28	480	81	49
2	4	12	256	34	18
2	3	9	192	25	13
2	2	6	128	16	9

For the four-center degree plan, one center receives multiplicity eight and the others seven, totaling 29 channels. Both S and D still recover all 29 coefficients at each of four centers, requiring 116 local evaluations at an executed packet root. No gain from deleting those evaluations is claimed.

For the Boolean packet cases, $p = 5$, so the coefficient working precision is 485 bits for S/G and 54 bits for D/DG. The weighted case has $p = 65$, giving 545 and 114 bits. The numerator calculation $\mathcal{C}(T)/\mathcal{C}'(0)$ uses *four guard bits* because $\mathcal{C}'(0) = 16(-9)$: the corresponding numerator moduli have 489/58 or 549/118 bits. These guards are included in runtime and memory. The "working bits" counter denotes the coefficient modulus, not the largest temporary integer or rational numerator anywhere in the program.

For three active singleton levels the observed precision paths are

$$\begin{aligned} S/G: &\quad 5 \longrightarrow 261 \longrightarrow 453 \longrightarrow 581, \\ D/DG: &\quad 5 \longrightarrow 23 \longrightarrow 36 \longrightarrow 45. \end{aligned}$$

The large savings relative to S combine removal of a conservative schedule with a smaller degree-certified target map. There is no timed variant using only the exact loss of the unchanged S weights, so these effects are not separately identified by the runtime experiment.

Odd denominators are not precision failures. Four-center kernel coefficients and local oracle coefficients may be rational with odd denominators, because of the odd factor nine in $\mathcal{C}'(0)$. They are integral over $\mathbb{Z}_2$, not necessarily ordinary integers. The oracle therefore retains exact fractions, checks odd denominators, and maps n/d to $nd^{-1} \bmod 2^B$. Rejecting every noninteger fraction would incorrectly reject valid local polynomials. Division by powers of two, unlike odd factors, requires the proved guards and target-loss certificate.

2.8 Symmetry certification for packets and descendants

In the packet family only whole-packet permutations preserving ordered real positions are candidates. A candidate must also preserve tensor entries with the induced port axes, bipartite side, creation precisions, and grouping up to a group bijection. Arbitrary swaps of the two real positions are not

presumed to preserve the kernel. For identical packets, swapping them reduces 36 coalesced mode assignments to 21 orbits. Asymmetric packet data are correctly rejected as interchangeable.

The orbit identity is an integer-multiplicity sum

$$\sum_{\beta} Q(\beta) = \sum_{O} |O| Q(\beta_O),$$

not division by group order. The complete deformation, not just its value at zero, must be invariant. In the deeper singleton family, v5's edge-permutation certificate is recomputed for each active node using that node's fixed tensor representatives. No root symmetry is blindly propagated to descendants.

All variants share the same exact-key child cache with capacity 16 per active call. The packet family also uses the same full interpolation basis construction. The deeper family retains v5's within-evaluator degree/precision basis cache. These common implementation choices affect runtime and memory but do not prove a source-style space bound.

2.9 Cases, exact validation, and negative controls

The four packet cases have two two-edge packets, with a row and column tensor per packet. The default table is $(0, 1, 1, 1)$, the two-bit OR. The five measured cases are:

Case	Definition	Exact answer
P01	Two identical OR/OR packets	9
P02	First row changed to $(0, 1, 1, 0)$ (XOR)	6
P03	First row identically zero	0
P04	Entry j of vertex i is $2^{15} + 7i + 3j + 1$	See below
D01	Unary singleton input with path $4 \to 3 \to 2 \to 1$	1

P04 is a weighted contraction, not a SAT count. Its exact answer is

$$18482797547933526180.$$

All requested output moduli exceed an elementary nonnegative-answer bound. Exact modular agreement therefore identifies the integer answer, not merely an ambiguous residue class.

The independent oracle forms each full polynomial $K_{xy}(T)$ using exact rational products, expands the double assignment sum over X, Y, and computes every local polynomial coefficient. It does not reuse the geometric mode construction or the recursive evaluator. Worker validation requires every expected center and coefficient to be present and correct, including all nonbase coordinates (which vanish at the root). Direct workers check only the direct target and correctly have no local polynomial.

Diagnostic workers additionally check the closed kernel identity and every executed active target. Instrumentation requires nonzero first and second higher geometric orders for packet workers; nonzero higher-mode products are required except in the identically zero-output fixture. The deep workers must report active depth two, leaf depth three, and nonbase active returns. A coded sixth candidate, D02, is not part of the measured suite and is not reported as a validated benchmark.

Separate controls check 31 field inverses and 192 packet coefficient identities, compare full and coalesced forms, reject the false packet symmetry, and record the higher-mode omission counterexample above. The new precision planner passes 200 monomial-basis checks, including 148

through the engine degree bounds; 126 uncompiled source-form comparisons; 90 residue-recovery checks; 30 literal-source residue comparisons; and 84 two-node closed-form checks. Its 52 rejection tests and out-of-class/one-bit-short witnesses are fixed-map checks, not a universal lower bound for all realizable counting instances.

2.10 Measurement protocol and memory definition

Five methods are compared: Direct, S, D, G, and DG, as in v5. Each of five fixtures has three fresh untraced timing workers per method and one separate diagnostic worker per method. Thus there are 75 timing workers and 25 diagnostic workers, totaling 100 validated outputs. Three timing repeats are used for this costlier experiment, rather than v5's seven. Method order rotates across fixtures and repeats; workers are sequential. Interpreter startup and imports are outside the timed interval. Network construction, precision planning and checks, grouping/orbits, full geometric summation, child construction, caching, arithmetic, interpolation, and target recovery are included.

The original timing phase was interrupted during its first D01/S timing worker, which exceeded a configured 900-second subprocess timeout with a large elapsed-time overrun. Its exact cause was not established. All 47 completed original worker records were preserved: 25 diagnostics and 22 initial timings. The entire timing phase was then restarted, not selectively filtered, with a 3600-second worker timeout. The final tables use the 75 restarted timing workers and the 25 retained diagnostics; the 22 retired timing records remain available but are not mixed into the final medians. The failed worker returned no exact result and is not counted as a passed test. Recovery provenance records this event.

Each restarted timing worker also records process CPU time around the same pipeline call. This distinguishes scheduled computation from some forms of elapsed-time interruption, but it does not eliminate processor frequency variation or turn a small experiment into a controlled system benchmark. The zero or quantized CPU times of very short direct calls are not used to form CPU speed ratios.

Independent oracles are outside timing. Diagnostic times, which include extra direct checks, are never used in performance medians. The environment is recorded with the raw data (CPython 3.12.14, Windows 11 build 26300, Intel64 Family 6 Model 140 Stepping 1). No statistical population claim is inferred from three repeats on these hand-constructed instances.

Memory is the Windows *peak process working set*, recorded in each untraced timing worker, and reported in MiB. It includes interpreter, imports, allocator retention, and other process-resident memory. The separately reported increase over the process's pre-call high-water mark is a diagnostic difference, not an isolated allocation profile or a theoretical workspace bound. Medians of these differences and medians of absolute peaks need not subtract exactly. This metric cannot be compared numerically with v5's traced Python-allocation KiB figures. Avoiding tracing makes the deeper measurements feasible without including profiler slowdown in runtime.

2.11 Runtime results

Case	Method	Median (s)	Min (s)	Max (s)	MAD (s)
P01	Direct	0.000252	0.000163	0.000254	0.000002
P01	S	4.216	2.127	4.909	0.693
P01	D	0.780	0.438	1.094	0.314
P01	G	3.521	1.963	4.125	0.604
P01	DG	0.556	0.384	0.752	0.173
P02	Direct	0.000238	0.000168	0.000401	0.000070
P02	S	3.455	2.283	3.996	0.541
P02	D	0.715	0.513	0.872	0.156
P02	G	3.948	2.189	4.034	0.086
P02	DG	0.732	0.483	0.855	0.123
P03	Direct	0.000410	0.000165	0.000514	0.000104
P03	S	3.095	2.080	3.567	0.472
P03	D	0.643	0.449	0.810	0.166
P03	G	3.741	2.131	3.881	0.140
P03	DG	0.652	0.484	0.827	0.168
P04	Direct	0.000254	0.000190	0.000284	0.000030
P04	S	5.013	2.652	5.795	0.782
P04	D	0.931	0.616	1.056	0.126
P04	G	4.632	2.630	5.770	1.138
P04	DG	0.875	0.629	1.222	0.247
D01	Direct	0.000547	0.000380	0.000613	0.000066
D01	S	858.467	422.564	871.824	13.357
D01	D	749.932	329.560	1793.712	420.372
D01	G	32.927	28.972	72.069	3.955
D01	DG	22.094	21.437	22.709	0.615

Median source/variant speed ratios (a value below one is a slowdown):

Case	S/D	S/G	S/DG
P01	5.40	1.20	7.58
P02	4.83	0.88	4.72
P03	4.81	0.83	4.75
P04	5.39	1.08	5.73
D01	1.14	26.07	38.85

Process CPU time, measured separately in the same untraced workers:

Case	Method	CPU median (s)	CPU min (s)	CPU max (s)
P01	S	4.016	2.125	4.047
P01	D	0.766	0.438	0.938
P01	G	3.438	1.969	3.781
P01	DG	0.562	0.391	0.750
P02	S	3.453	2.281	3.891
P02	D	0.703	0.516	0.844
P02	G	3.859	2.188	3.938
P02	DG	0.719	0.484	0.812
P03	S	3.047	2.078	3.422
P03	D	0.641	0.453	0.750
P03	G	3.516	2.109	3.703
P03	DG	0.641	0.484	0.812
P04	S	4.906	2.656	5.719
P04	D	0.938	0.609	1.047
P04	G	4.516	2.625	5.719
P04	DG	0.859	0.625	1.203
D01	S	825.984	418.828	830.125
D01	D	649.062	324.516	715.516
D01	G	32.734	28.719	66.500
D01	DG	21.703	21.172	22.609

On D01 the source median is 858.467 s and DG is 22.094 s, giving a 38.85× source/DG ratio. G alone has median 32.927 s; its G/DG ratio is 1.49. Thus degree and symmetry effects are not assumed to combine multiplicatively. Every direct median is smaller than every recursive median on its fixture. The ranges reveal machine-level variation; three repeats do not justify ranking closely spaced methods or treating small differences as robust speedups. No timing workers from the completed suite were discarded as outliers. The first timing phase was retired as a whole after interruption, as explained in the protocol. Diagnostic times are excluded from these tables. The deeper source/DG median CPU-time ratio is 38.06×. CPU time excludes time when the process is not scheduled but remains sensitive to processor frequency and implementation details. Very short direct calls fall below the Windows CPU-accounting resolution, so their CPU ratios are not reported. The largest completed elapsed/CPU ratio occurs in D01/D, repeat 2: 1793.712 s elapsed versus 649.062 s CPU. The large gap shows that elapsed time contains substantial time outside scheduled computation. That observation does not identify whether the cause was suspension, scheduling, or another system condition. The completed sample is retained, not removed. Even CPU-time ranges show machine variation. These medians are descriptive measurements, not stable hardware-independent speedup estimates or controlled causal evidence.

The source comparator and the reduction variants all return the same integer answers and full root local polynomials on these tests. These are local implementation comparisons against the finite comparator, not evidence of a faster unrestricted SAT solver. Packet and deep-family ratios are not multiplied together because no measured run combines their hardest structural features.

2.12 Memory results

Case	Method	Peak MiB	Pre-peak MiB	Increase MiB
P01	Direct	25.82	25.82	0.00
P01	S	25.76	25.76	0.00
P01	D	25.66	25.66	0.00
P01	G	25.79	25.79	0.00
P01	DG	25.65	25.65	0.00
P02	Direct	25.95	25.95	0.00
P02	S	25.91	25.91	0.00
P02	D	25.75	25.75	0.00
P02	G	26.10	26.10	0.00
P02	DG	25.88	25.88	0.00
P03	Direct	25.93	25.93	0.00
P03	S	25.74	25.74	0.00
P03	D	25.71	25.71	0.00
P03	G	25.93	25.93	0.00
P03	DG	25.97	25.97	0.00
P04	Direct	25.86	25.86	0.00
P04	S	26.01	26.01	0.00
P04	D	25.84	25.84	0.00
P04	G	26.04	26.04	0.00
P04	DG	25.89	25.89	0.00
D01	Direct	25.90	25.90	0.00
D01	S	26.02	26.02	0.00
D01	D	25.94	25.94	0.00
D01	G	26.04	26.04	0.00
D01	DG	25.81	25.81	0.00

D01 has median process peaks 26.02 MiB for S and 25.81 MiB for DG. However, **all 75 timing workers have exactly zero increase over their own pre-call process high-water mark**. The displayed peak differences therefore reflect pre-call process baselines, not resolved algorithm-specific peak allocations. In particular, the numerical -0.8% S-to-DG difference is *not evidence of a memory saving*. These are whole-process resident peaks, not cumulative memory across workers. They show that execution did not exceed each worker's recorded baseline peak, but they cannot determine which method uses less algorithm-specific memory. A separate allocation profile or suitably controlled in-call measurement is needed for that comparison.

The values report the actual chosen process metric. A smaller coefficient modulus need not produce a proportional decrease in process peak memory; fixed interpreter costs and object/certificate storage can dominate. Similarly, orbit reduction can remove child-contraction requests and executed descendant calls while retaining additional group and cache state. This experiment does not establish the paper's asymptotic space claim or an optimal implementation memory bound.

2.13 Recorded packet modes and recursive activity

Case	Method	B	$b_{\max}$	Orders	Higher products	Reps
P01	S	485	481	107976	2352	36
P01	D	54	50	11432	2352	36
P01	G	485	481	107976	1344	21
P01	DG	54	50	11432	1344	21
P02	S	485	481	107976	2352	36
P02	D	54	50	11432	2352	36
P02	G	485	481	107976	2352	36
P02	DG	54	50	11432	2352	36
P03	S	485	481	107976	0	36
P03	D	54	50	11432	0	36
P03	G	485	481	107976	0	36
P03	DG	54	50	11432	0	36
P04	S	545	541	121416	560	36
P04	D	114	110	24872	560	36
P04	G	545	541	121416	560	36
P04	DG	114	110	24872	560	36

D01 diagnostic counters, each for one execution:

Method	Active calls	Leaf calls	Child requests	Cache hits	Nonbase returns
S	54913	981032	12648064	11612120	20892
D	54913	981032	12648064	11612120	20892
G	4303	75790	886168	806076	3142
DG	4303	75790	886168	806076	3142

Here $b_{\max}$ is the largest nonzero geometric order, not the coefficient precision or the number of independent moment functions. Every non-direct packet diagnostic has 116 local evaluations, 232 closed-kernel checks, and 224 nonzero coefficients of each of orders one and two. The zero-output fixture still computes nonzero kernel modes even though its contracted higher-mode products vanish. D01/DG executes 4,303 active calls, each checked against an independent direct target, and reaches leaf depth three. It records 3,142 nonbase active returns. The observed structural counts agree within S/D and within G/DG; the precision change primarily changes arithmetic size here. Symmetry substantially reduces executed children, but this is not a claim about uniform savings on arbitrary recursive inputs.

"Child requests" includes cache hits; "active calls" excludes them. Repeated mode products and geometric orders are implementation counters, not counts of independent mathematical instances. Coalescing all $b \geq 1$ moments into one observable does not mean only the $b = 1$ term was evaluated: the recorded nonzero orders and the exact-kernel checks explicitly distinguish those statements.

2.14 What is new, what is proved, and what remains untested

The new proved local results are the two-column moment coalescence under modular truncation, its coefficient-extracted two-port child identity, and the four-center target-loss certificates. Exact algebraic checks and matched executions demonstrate these constructions on the specified families. The deeper path also demonstrates valid precision propagation and creation-time lift inheritance over one additional active level.

The main limitations are substantive:

1. The source's literal global parameter schedule and full SAT reductions are still not implemented or independently verified.
2. The packet tests have only two real positions $0, 2$, so all positive tile powers coincide as observations. They do not test a large independent high-moment system, arbitrary positions, or general mixed local/crossing interactions.
3. The packet roots are disconnected across packets and have one active level; the deeper test is singleton and factorized. A general multi-edge deep recursion test remains to be constructed.
4. Full coalescence is common to the finite comparators. No measured speedup against uncoalesced enumeration of all original mode tuples is claimed. The source's asymptotic mode-count analysis is not replaced by the number 36 on arbitrary inputs.
5. Only four Boolean edge variables occur at a root. Direct counting is the natural small-instance reference; a local recursive-comparator speedup is not evidence of practical superiority at that size.
6. Symmetry discovery and storage are charged, but only very small candidate groups are considered. Generic inputs may have trivial symmetry.
7. Process peak working set is not a stand-in for a proof of workspace or for the earlier Python-traced memory metric.

For a full-algorithm result, one still needs uniform certificates for the actual packet families, invariance under every recursive construction, a complete recurrence including bit arithmetic and certificate costs, and a space-preserving implementation. The new identities alone do not change a general SAT exponent or independently settle ETH.

2.15 Further improvements and conventional interpretation

The exact closed coefficient formula (5) could replace the geometric loop in a further timed variant. That would measure resummation separately from precision and symmetry. Another useful ablation would retain the source target weights but use their exact denominator loss. Neither ablation is timed here.

A further exact Boolean quotient, not a measured optimization. Even the six coalesced packet monomials used in the measured code are not independent as functions of two Boolean inputs. Set $x = X_0$, $y = X_2$, and write the row polynomial as

$$c_{00} + c_{10}Z_0 + c_{01}Z_+ + c_{20}Z_0^2 + c_{11}Z_0Z_+ + c_{02}Z_+^2.$$

Substituting $Z_0 = x + y$, $Z_+ = y$, and $x^2 = x$, $y^2 = y$ gives exactly

$$\begin{aligned} &c_{00} + (c_{10} + c_{20})x \\ &\quad + (c_{10} + c_{01} + c_{20} + c_{11} + c_{02})y + (2c_{20} + c_{11})xy. \end{aligned}$$

No division is needed, so this identity holds over every tested modular ring. Evaluation of $1, x, y, xy$ on the four Boolean assignments has a unimodular matrix; these four observations form a basis of all two-bit functions, including over rings with zero divisors. Thus a further implementation could use four states per packet instead of six, reducing two-packet assignment counts from 36 to 16 (and, for exact whole-packet interchange, from 21 to 10 unordered representatives). These are proved

algebraic state counts, not measured time or memory improvements. Child tensors, symmetry checks, and cache behavior would need to be updated and benchmarked before attributing an end-to-end benefit to this additional quotient. The measured code deliberately retains the six-state form described earlier.

The next structural test should use at least three distinguishable tile coordinates, certify the rank of their moment observation matrix, and combine nontrivial multi-port children with deeper recursion. Such a test must establish new degree, interpolation-field, and precision bounds before reusing these decoders. It should also include connected networks, negative controls for false moment identification, and a direct target oracle as far as feasible. Selective-channel reconstruction and streaming evaluation remain proposals requiring their own cost proofs.

The Ram Calculus interpretation is an explicit choice of observation space, exact quotient, target functional, and precision certificate. The coalescence is also conventional equality of moment functions on a finite support, and the reconstruction is classical Hermite interpolation with valuation-aware arithmetic. A conventional implementation of the same identities would obtain the same benefits. The contribution is the specified integration and verified scope, not a claim that these underlying mechanisms are exclusive to Ram Calculus.

2.16 Evidence and preservation

New code and raw evidence are in `eth-benchmark-v6/`. The principal files are `recursive_v6.py`, `precision_v6.py`, and `run_v6.py`. The completed run is `results-20261027-resumed/`, containing `controls.json`, `instances-and-oracles.json`, `workers.jsonl`, `recovery-provenance.json`, and `results.json`. The interrupted original run is retained separately in `results-20261027/`. Reproduction from scratch uses

```
python -B run_v6.py --out results-new-run
```

with a new, nonexistent output directory. The raw metadata records all measured-source hashes. Frozen v4/v5 arithmetic and helper modules are imported read-only with bytecode writing disabled. The recovery driver `resume_v6.py` preserves completed diagnostics and restarts the timing phase with CPU-time measurement; it never changes the frozen mathematical workers.

The separate author-only update has its own original-file backups and provenance manifest. Old benchmark hashes remain historical evidence, not a demand to overwrite their records after a metadata-only edit to the associated reports. The new v6 assembly checks that the preceding v5 technical body and bibliography remain intact.

Revision 6 conclusion. Two-real-edge packets, nonzero higher geometric orders, exact moment coalescence, four-center recovery, and a separate three-active-level recursive path have now been implemented and checked. The measured benefits and costs are reported above with exact-answer validation. The experiment broadens the finite evidence, while leaving general multi-edge deep recursion and full-algorithm time/space improvement open.

Retained v5 analysis and earlier results

The following text preserves the preceding scientific report. Its experimental counts, precision budgets, page references, and limitations describe the historical v5 and earlier runs, not the new v6 suite.

3 Revision 5: recursive experiments and their precise scope

3.1 What the new experiment does and does not establish

Revision 5 implements the degree-aware precision and certified symmetry ideas inside an actual recursive contraction evaluator. Every nonleaf call builds child tensor networks, calls the evaluator on those networks, and reconstructs its target from local polynomial data. One fixture reaches two active levels before reaching leaves. This is a stronger test than revision 4's one-edge interpolation experiment, in which "two active nodes" meant two interpolation centers, not two recursive calls.

There is, however, a decisive qualification. **No executable instance of the paper's literal active parameter schedule was tested.** Its first positive recursion parameter already requires an input-size parameter with 39,457 decimal digits. The new experiments therefore use an explicitly rescaled, locally certified finite version of the source's contraction identity. They preserve the recursive algebra, fixed creation-time lifts, local interpolation, and exact recovery, but not the source's global parameters or asymptotic complexity guarantees. Throughout this section, the "source comparator" means this finite specialization with source-form Hermite reconstruction and conservative precision. It is *not* the paper's full SAT solver or an author-supplied executable implementation.

All earlier mathematical material, equation comparisons, and recorded experiments from revision 4 are retained after this new section. They remain historical results on their stated classes; they have not been relabelled as recursive benchmarks. The principal additions are:

1. An explicit feasibility calculation for the literal schedule, and a counterexample to blindly reusing v4's fixed decoders at larger nodes.
2. A finite recursive implementation with node-specific certificates, fixed tensor lifts, exact orbit tests, and independent polynomial oracles.
3. Seven timing repeats and two separate memory repeats for each of five variants on six fixtures, plus a diagnostic repeat: 300 fresh worker outputs, not 300 independent problem instances.
4. Successful exact checks on all fixtures and 128 additional method–instance checks on 32 deterministic random truth-table networks.
5. Measured speedups, memory penalties, and unprofitable symmetry cases, with no inference of a better general SAT exponent.

3.2 Why the literal active schedule cannot be benchmarked here

Use the notation of the pinned source's Appendix D [1]. The size parameter is s, $\lambda = \lceil \log_2 s \rceil$, and the depth-dependent parameters include

$$\varepsilon = \frac{1}{131072}, \qquad u_d = \left\lfloor \frac{\varepsilon \lambda}{2^d} \right\rfloor, \qquad a_d = 2 \sum_{j=0}^{d} u_j, \qquad L_d = 2^{a_d}.$$

The direct-summation leaf threshold is $H = \lceil s/\lfloor\sqrt{\lambda}\rfloor\rceil$. A nonleaf construction requires $M > H$ and the positive parameters on which its packet and interpolation formulas depend.

Proposition 3.1 (Necessary size for a positive source recursion parameter)**.** *For an integer $s \geq 2$, $u_d \geq 1$ holds exactly when*

$$\lambda \geq 131072\,2^d, \qquad s \geq 2^{131072\,2^{d-1}} + 1.$$

In particular, the smallest possible root size is $2^{131071} + 1$, a 39,457-digit integer. At that size $\lfloor\sqrt{\lambda}\rfloor = 362$, so a nonleaf also needs $M \geq \lceil(2^{131071} + 1)/362\rceil + 1$.

Proof. The floor defining u_d is positive precisely when $\lambda \geq 131072\,2^d$. For integral K, $\lceil\log_2 s\rceil \geq K$ is equivalent to $s > 2^{K-1}$. The decimal-digit statement follows from $\lfloor 131071 \log_{10} 2\rfloor + 1 = 39457$. The leaf inequality follows by substituting this s and λ. □

This is only a necessary barrier, not a sufficient activation certificate. For example, when $u_d \geq 1$, the earlier u_j obey $u_j \geq 2^{d-j}$, whence $a_d \geq 2^{d+2} - 2$. The smallest possible active root therefore has $a_0 \geq 2$, not $a = 1$. With the source's $r = 64$, the packet arrangement size $D_{\mathrm{arr}} = r^{2a}$ is at least 2^{24} at that root. Its packing conditions include

$$M \geq 2^{24a+1}\Delta, \qquad M \geq 2^{13a}, \qquad 2^{a+1} \leq \Delta.$$

Additional sufficient asymptotic inequalities may be still more demanding. For instance $u_d \geq \lambda^{4/5}$ implies the necessary condition $\lambda \geq 2^{85+5d}$. None of these calculations is an objection to an asymptotic theorem merely because its constants are large. They explain why rescaling must be disclosed in an implementation experiment.

Artificially assigning a huge s to a four-edge graph does not solve the problem: it also raises H, and the graph remains in the direct branch. The source further directly handles a fixed finite range of small sizes. The closest faithful source-schedule control on our tiny inputs is therefore the direct evaluator, which is reported alongside every recursive variant.

3.3 The tested finite model and all departures from the source

At each active call set $a = u = 1$, $L = 2$, $\alpha = 4$. Use one real edge per packet, at its zero slot; other slots are empty. Use disjoint row-side and column-side vertex groups. The number of groups is variable instead of 64, and each child is required to have strictly fewer edges than its parent. For the fixtures below the child edge counts are $(2, 2, 2, 2)$ at the root and, when a second level is active, $(1, 1, 1, 1)$ at its children. Leaf thresholds are specified per fixture rather than by s.

The coefficient algebra at precision b is

$$\mathcal{R}_b = (\mathbb{Z}/2^b\mathbb{Z})[U]/(U^4 + U + 1).$$

The polynomial is irreducible over $\mathbb{F}_2$: it has no root there and is not divisible by the only irreducible quadratic $U^2 + U + 1$. Thus its modulo-two quotient is $\mathbb{F}_{16}$. Thirteen distinct residue classes suffice for degree-12 interpolation, the largest degree in these tests. We use the binary coefficient vectors of the integers $0, \ldots, 3M$ as interpolation points. Pairwise differences are units because their reductions in $\mathbb{F}_{16}$ are nonzero. This fixed extension replaces the source's size-dependent extension; it is justified only for the finite degrees here.

Singleton packets eliminate the higher tile moments: their tile coordinate is $j = 0$, so $j^b = 0$ for $b \geq 1$. Every row factor is affine, hence a packet mode exponent at least two also has zero coefficient.

Consequently the only surviving mode assignments are $\beta \in \{0,1\}^M$. This is an exact zero-pruning of the identity, *not* a test of nonzero higher tile moments or general multi-edge packets. The working precision exceeds M, so all these binary assignments are inside the retained weight range.

All variants share an exact-key least-recently-used child cache of capacity 16 within each active call. The key contains the full child network, creation precisions, edge count, and requested output precision. No isomorphism-based cache key is used. Interpolation bases are also cached within the top-level evaluator by degree and precision. These are common implementation choices, not statements that the source's stated space algorithm uses those caches. Cache construction, lookups, exact arithmetic, and certificates are included in measured runtime and traced memory. The cached and uncached implementations agree on all four variants of the symmetric-cycle fixture.

3.4 The actual recursive identity

For the single real slot the restricted kernel is

$$W(T) = 1 + \frac{T}{2} - T^2 - \frac{T^3}{2}, \quad W(0) = 1, \quad W(-1) = W(1) = 0. \tag{6}$$

Its local forms are integral:

$$W(-1+4z) = 4z + 8z^2 - 32z^3, \tag{7}$$

$$W(1+4z) = -12z - 40z^2 - 32z^3. \tag{8}$$

For fixed tensors on the two sides, let $A(Y)$ and $B(X)$ denote their products for edge assignments $Y, X \in \{0,1\}^M$. Set $\ell_0(v) = 1 - v$, $\ell_1(v) = v$. The deformation is

$$F(T) = \sum_{X,Y} A(Y)B(X) \prod_{e=1}^{M} \ell_{Y_e}(W(T)X_e). \tag{9}$$

It has degree at most $3M$. At $T = 0$ the product enforces $X = Y$, so $F(0) = \sum_X A(X)B(X)$, the original contraction.

For a fixed local evaluation $w = W(t + 4\zeta)$, expand

$$\ell_y(wx) = \underbrace{\mathbf{1}_{y=0}}_{\beta=0} + \underbrace{w(2y-1)x}_{\beta=1}.$$

The exact child packet unary tensors are therefore

	$\beta_e = 0$	$\beta_e = 1$
row port, indexed by Y_e	$(1, 0)$	$(-w, w)$
column port, indexed by X_e	$(1, 1)$	$(0, 1)$.

For each group g, retain its old vertices with their original tensor entries, attach the indicated packet tensor to each incident port, and call the resulting child contraction $C_g(\beta, w)$. Group membership depends only on the fixed parent network. Then

$$F(t + 4\zeta) = \sum_{\beta \in \{0,1\}^M} \prod_g C_g(\beta, w). \tag{10}$$

This follows by distributing the binary expansion and factoring the remaining assignment sums by disjoint vertex groups. Both endpoints of each parent edge become child ports; hence $\sum_g M_g = 2M$.

Proposition 3.2 (Finite recursive correctness)**.** *Suppose every active node satisfies the singleton-packet construction, uses fixed creation-time lifts, has a valid degree and target-recovery certificate, and has child sizes $M_g < M$. Suppose all leaves are exactly summed. Then the recursively reconstructed target is the original contraction modulo its requested 2^p.*

Proof. Induct on M. The assertion is direct at leaves. At an active node, the induction hypothesis computes each child target to the working precision. Equation (10) gives the local evaluations modulo that precision. Unit-separated interpolation uniquely recovers each local polynomial of degree at most $3M$. The certified target functional then returns $F(0)$ modulo 2^p. Strict child shrinkage proves termination. No claim about the source's asymptotic time or space follows from this induction alone. □

3.5 Creation-time lifts are a mathematical contract

A vertex stores $(p_c, \mathbf{v})$, with each coordinate of $\mathbf{v}$ the canonical integer in $[0, 2^{p_c})$ chosen when that vertex was created. When a child later requests higher precision, these integers are embedded unchanged. Packet tensors newly formed at precision b acquire creation precision b. They must not later be regenerated from a symbolic ancestor at a different modulus.

For example, an entry represented by $-1 \pmod{2^5}$ is stored as 31. Its prescribed lift to precision seven is 31, not 127. Re-evaluating the expression -1 at precision seven changes the input tensor to the child. The implementation stores immutable vertices, checks canonical ranges and $p_c \le p$ at every call, and includes p_c in symmetry and cache tests. The nested fixture returns nonzero U, U^2, U^3 components in many internal calls, so its correctness is not merely a check that scalar inputs remain scalar throughout the recursion.

Different precision variants can create different canonical integer lifts for corresponding packet tensors. Their internal integer networks need not be numerically identical. What is matched is the same initial network, the same finite decomposition, and exact equality at the required quotient precision. Full root local polynomials are checked against an independent integer oracle at each variant's own working precision.

3.6 Node-specific degree-aware recovery

Let $c_{t,j} = [z^j]F(t + 4z)$ for $t \in \{-1, 1\}$, and choose positive multiplicities q_-, q_+. Define

$$E_t(T) = \left(\frac{T+t}{2t}\right)^{q_{-t}}, \qquad r_t(z) = \left[\frac{1}{E_t(t+4z)}\right]_{<q_t}.$$

The source-form Hermite identity, for $\deg F < q_- + q_+$, is

$$F(0) = \sum_{t=\pm 1} \sum_{j<q_t} w_{t,j} c_{t,j}, \tag{11}$$

$$w_{t,j} = E_t(0) \sum_{k=j}^{q_t-1} [z^{k-j}] r_t(z)(-t/4)^k. \tag{12}$$

Indeed the truncated reciprocal cancels E_t to the required local order; the resulting Hermite reconstruction matches the prescribed jets at both nodes and is the unique polynomial below the total multiplicity. This derives the weights rather than assuming that a low-degree decoder from v4 remains valid.

Proposition 3.3 (Precision of the selected target map)**.** *Set* $e = \max(0, \max_{t,j}[-v_2(w_{t,j})])$*. If all input channels and the target are integral over the two-adic coefficient algebra, knowing each channel modulo* 2^{p+e} *determines the target modulo* 2^p *through* (12)*. The statement holds coordinatewise in* $\mathcal{R}_b$*.*

Proof. Changing an input channel by $2^{p+e}a$ changes its weighted contribution by an element of $2^p\mathbb{Z}_2$ because $v_2(w_{t,j}) \geq -e$. Sum these errors. Rational denominators with odd factors are units; their inverses are computed modulo the output modulus. The powers of two are divided only after the accumulated numerator has the certified divisibility. The code checks that divisibility before recovery. □

For the two-node family the truncated reciprocal has coefficient $[z^k]r_t(z) = (-2t)^k\binom{q_{-t}+k-1}{k}$. The largest denominator loss is

$$e = \max_t(q_{-t} + 2q_t - 2).$$

For a fixed t, the highest-order channel has that exact loss; all earlier channels have no worse loss by the displayed formula. The finite source comparator sets $h = 3M$, $q_- = q_+ = h$ and keeps the conservative schedule increment $64M$ inherited from the source formula under this specialization. The degree-aware plan instead uses balanced prefixes totaling $3M + 1$ and increment $\lfloor 9M/2 \rfloor$. This gives:

M	Degree	Source channels	Source increment	Exact source-map loss	Degree increment
2	6	12	128	16	9
4	12	24	256	34	18
8	24	48	512	70	36

The $M = 8$ line is a decoder unit test only: the recursive engine is limited to $M \leq 4$. At $M = 2, 4$ the degree-aware channel counts are respectively 7 and 13. Both variants still compute *all* $3M + 1$ local coefficients at each of the two centers using $3M + 1$ evaluations per center. Fewer recovery channels have not been counted as fewer recursive evaluations. The measured precision improvement is therefore chiefly smaller arithmetic and smaller recovery computations, not a hidden deletion of interpolation work.

The increments are exact sufficient losses for specified linear maps. They are not lower bounds on every decoder exploiting extra relations among a network's coefficients. As a separate symbolic extension, the singleton construction implies $F(T) = G(W(T))$ with $\deg G \leq M$. Since $W(-1+4z) = 4R(z)$ and $R(z) = z + 2z^2 - 8z^3$ has an integral formal inverse, the prefix through degree M at -1 recovers $G(1)$ with loss $2M$. The witness $2^{p-1}W^M$ has all its local coefficients at both centers zero modulo 2^{p+2M-1} but nonzero target modulo 2^p. Thus $2M$ is sharp for residue-only observations of this unrestricted coefficient family. This additional plan is implemented and symbolically checked in `precision_plans.py`, but **not used in the timed recursive comparison**. It is not a lower bound for Boolean-count inputs, nor a result for general nonsingleton packets.

3.7 Why v4's fixed formulas cannot simply be transplanted

V4's degree-three map is

$$\frac{c_{-1,0} + c_{1,0}}{2} + \frac{c_{-1,1} - c_{1,1}}{16},$$

and its one-W restricted map is $c_{-1,0} + c_{-1,1}/4$. Take two independent real singleton packets with row and column unary tensors $(0, 1)$. Their deformation is $F(T) = W(T)^2$ and their count is $F(0) = 1$. Both constant and linear local coefficients vanish at each center because W has a zero there. Both old maps consequently give zero, even with unlimited working precision. This is a realizable two-edge counterexample, not an arbitrary polynomial outside the physical construction. It does not invalidate the v4 formulas on their original one-edge hypotheses. It invalidates extending those hypotheses silently.

The new planner checks its recovery identity exactly on every monomial of the certified degree range. Underprecision tests distinguish arbitrary lift-error witnesses for a fixed map from realizable-family lower bounds. The suite also tests out-of-degree and out-of-family inputs, rather than interpreting a successful rational fit as an unconditional decoder.

3.8 Certified symmetry inside recursive calls

For each current node, enumerate edge permutations and accept only those preserving: row/column side; tensor entries after the induced port-axis permutation; creation precision; and group membership up to a bijection of groups. All singleton packet kernels coincide, so these checks preserve the complete deformation, not just its target at $T = 0$. Permutations cannot interchange unequal weighted tensors merely because the unweighted graph looks symmetric.

Let Γ be the accepted group and $\mathcal{O}$ its orbits on binary mode assignments. Then

$$\sum_{\beta} \prod_{g} C_g(\beta, w) = \sum_{O \in \mathcal{O}} |O| \prod_{g} C_g(\beta_O, w). \tag{13}$$

The vertex and group bijections change only dummy summation variables in the child contractions; hence the product is constant on each orbit. Equation (13) uses integer multiplicities and does not divide by group order, which might be noninvertible modulo a power of two. The implementation additionally checks that the proposed orbits are disjoint and cover every mode. Testing all $M!$ permutations is acceptable only because $M \leq 4$; it is not a scalable general group-discovery method.

The symmetric cycle has four accepted permutations and seven mode orbits instead of sixteen assignments. The broken-symmetry fixture has only the identity. The zero-tensor fixture has a smaller useful group, while the weighted fixture has no nontrivial accepted symmetry. Certificates are recomputed at children using their actual stored tensors; root symmetry is not assumed to survive the recursion.

3.9 Fixtures, controls, and exact answers

There are four edge variables $x_0, \ldots, x_3$. In the cycle fixture, row tensors lie on $(0, 1)$ and $(2, 3)$, and column tensors on $(0, 2)$ and $(1, 3)$. All four tables are $(0, 1, 1, 1)$, using the first listed port as the least significant truth-table bit. This is the positive two-CNF cycle with count seven. The fixtures are:

ID	Input and purpose	Leaf H	Root p	Exact result
R01	Cycle, nonrecursive control	4	5	7
R02	Cycle, symmetric active root	2	5	7
R03	Cycle with first row $(0, 0, 1, 1)$	2	5	5
R04	Cycle with first row identically zero	2	5	0
R05	Four independent unary constraints, nested recursion	1	5	1
R06	Asymmetric positive 32-bit weighted cycle	2	129	See below

For R05 every row unary is $(1, 0)$ and every column unary is $(1, 1)$. Pairs of row vertices form two groups and pairs of column vertices form two more, producing four two-edge children. These children are active and split into one-edge leaves. Although nested, this is an intentionally easy, factorized test, not a hard SAT distribution.

For R06, table entry j of vertex i is $2^{31} + 17i + 3j + 1$ for $i, j \in \{0, 1, 2, 3\}$. Its exact result is

$$340282386569523140639734138112894692878.$$

This is a *weighted tensor contraction*, not a CNF assignment count. The root output modulus exceeds the elementary bound $2^M \prod_v \max |v|$, so equality modulo that modulus determines the nonnegative integer answer uniquely for all fixtures.

Independent validation computes the scalar count by exhaustive assignments and expands the complete deformation by a separate double sum over X, Y, using rational polynomial arithmetic. Every fresh worker's root count and all coefficients of both root local polynomials are checked against that oracle. Diagnostic workers also compare *every executed active child target* with exhaustive ring-valued contraction. Checks outside the timed region do not contribute to benchmark times or traced peaks.

3.10 Experimental protocol

The five variants are Direct (exhaustive target contraction), S (finite source comparator), D (node-specific degree-aware precision), G (certified mode-orbit symmetry at source precision), and DG (both changes). The interpreter is CPython 3.12.14 on Windows 11 build 26300, with processor identifier Intel64 Family 6 Model 140 Stepping 1. Each measurement uses a fresh process; tests are sequential, and variant order is rotated across repeats. Import and interpreter startup are outside the measured region. Network construction, plan compilation and exact basis checks, group certification, interpolation, child construction, caching, contraction, and target recovery are inside it.

Seven untraced timing runs, two separate `tracemalloc` runs, and one diagnostic run are performed for each variant and fixture. Memory tables report median peak traced Python allocations, in KiB; native allocator and interpreter overhead are not comprehensively measured by that metric. Raw records additionally contain the Windows process peak working set, including interpreter startup and imports. The latter is not treated as incremental algorithm memory. Tracing runs are never used as runtime measurements. Logical operation counters and bounded provenance records are common instrumentation included in the measurements.

This yields 210 timing workers, 60 memory workers, and 30 diagnostic workers. All 300 passed validation. Separately, all four recursive variants passed 128 checks on 32 seeded random Boolean truth-table networks, with full root polynomial comparisons and active-node target checks. Those networks test algebra, not a representative distribution of hard CNFs. The precision unit tests passed 167 monomial checks, 108 comparisons with the uncompiled source-form Hermite expression, 23 restricted-family basis checks, and 120 precision checks. Negative controls reject nonshrinking children, mixed-side grouping, invalid creation precision, and noncanonical stored values. The W^2 failure above and the broken-symmetry identity-only certificate are explicitly checked.

3.11 Measured runtime

	Median runtime, milliseconds					S/DG
ID	Direct	S	D	G	DG	speed ratio
R01	0.286	0.320	0.328	0.330	0.319	1.00
R02	0.272	51.920	45.371	38.368	28.331	1.83
R03	0.290	55.405	54.441	57.560	52.823	1.05
R04	0.285	54.865	50.988	43.091	44.003	1.25
R05	0.475	472.726	401.095	297.413	245.233	1.93
R06	0.286	62.502	56.971	76.165	60.481	1.03

Ratios divide unrounded medians. R01 does not activate either reduction, so its small differences are control noise and wrapper overhead. For R02, precision alone gives 1.14× and symmetry alone 1.35×; the combined observed ratio is 1.83×. R05, with two active levels, gives 1.18× for precision alone, 1.59× for symmetry alone, and 1.93× combined. These are descriptive timings of this implementation, not asymptotic speedup factors or statistical population estimates.

The negative evidence matters. R03 has no useful permutation, and G is about 3.9% slower than S. R06 likewise has no useful symmetry and G is about 21.9% slower than S; DG is about 6.2% slower than D. R04's combined median is slightly slower than its symmetry-only median despite using fewer bits. Source-comparator R02 timing ranges from 42.446 to 136.355 ms; R05 S ranges from 436.557 to 520.125 ms and DG from 226.328 to 265.102 ms. All minima, maxima, medians, and median absolute deviations are saved. Small 1.03–1.05 ratios should not be interpreted as established robust gains in the presence of this variability.

On every active fixture, even DG is roughly 104–517 times slower than the direct control. These instances are far below any credible asymptotic cross-over. The experiment demonstrates local implementation savings relative to an expensive recursive comparator, not a practical advantage over direct counting at this size.

3.12 Measured memory and recursive activity

	Median peak traced Python allocations, KiB				
ID	Direct	S	D	G	DG
R01	8.20	8.51	8.51	8.51	8.51
R02	8.20	228.09	204.15	230.85	206.92
R03	8.20	226.19	203.10	228.02	204.93
R04	8.20	220.75	199.84	223.33	202.28
R05	14.12	286.84	246.86	356.92	317.31
R06	8.39	236.38	211.78	238.15	213.56

Precision alone reduces the traced peak on all active fixtures. Symmetry adds group/orbit and provenance overhead; the effect is most visible in the nested case. R05 G uses about 24.4% more memory than S, while DG uses about 10.6% more than S and 28.5% more than D. Thus fewer recursive calls do not imply smaller peak memory. At these small sizes, Python objects, temporary tensor tuples, and certificate data dominate much of the allocation profile; this is not a bound for a low-level implementation.

ID	Active calls S	Active calls DG	Child requests S	Child requests DG	Represented mode products S	Represented mode products DG
R02	1	1	1664	728	416	182
R03	1	1	1664	1664	416	416
R04	1	1	1664	1040	416	260
R05	84	57	20256	11544	5064	2886
R06	1	1	1664	1664	416	416

"Active calls" counts executed nonleaf calls, excluding cache hits. "Child requests" includes hits. Mode counts sum over all evaluated active calls, so their totals can also decrease because fewer children are executed. The nested fixture has one active root and 83 executed active children in S/D, versus one root and 56 executed active children in G/DG. Its maximum active depth is one and leaf depth two when the root is depth zero. The source-comparator diagnostic records 66 nonbase active returns. Nonzero crossing-mode products also occur, excluding a vacuous experiment where every term except the zero mode disappears.

3.13 Precision propagation actually observed

For R02–R04 the output precision is five bits. S/G use 261 bits at the active root; D/DG use 23. For R06 these values are 385 and 147 bits, starting from the required 129-bit output. R05 follows the path

$$\begin{aligned} \text{S/G:} \quad & 5 \longrightarrow 261 \longrightarrow 389, \\ \text{D/DG:} \quad & 5 \longrightarrow 23 \longrightarrow 32. \end{aligned}$$

The reductions $256 \to 18$ and $128 \to 9$ are proved decoder-budget changes for the finite construction; the maxima above are also runtime observations. No underprecision was hidden by wrapping an incorrect answer modulo the final modulus: full root local polynomials and diagnostic child targets were independently compared. Nonetheless the bit-width ratios are not runtime ratios, because control flow, interpolation-point count, object creation, and group discovery remain.

3.14 Failure cases, limitations, and prerequisites for a full improvement

The principal boundaries of the evidence are explicit:

1. The literal active source schedule was not executed, and the finite replacement uses $a = 1$, impossible at a literal active source root. The original global r, size-dependent extension, $M/8$-type shrinkage, and sufficient packing inequalities are not reproduced.
2. The benchmark has only four edges per root, five active fixtures, and at most two active levels. One is a weighted contraction rather than SAT; the nested counting fixture is factorized and easy.
3. Real packet size is one, with only binary zero-slot modes. Higher moments, unequal slot positions, general dyadic tiles, and large-arrangement symmetries remain untested.
4. The old fixed D4/R2 decoders fail on W^2. Degree and family certificates must be regenerated at every node. Precision optimality of a selected map is not optimality of every realizable-family decoder.
5. Symmetry discovery can cost more than it saves and increases peak memory in the nested case. General inputs may have only the identity; their worst-case improvement cannot rely on nontrivial automorphisms.

6. Bounded caching and precomputed interpolation bases are common engineering choices. The measured memory does not establish the source's space bound. Removing caches and using streaming evaluators requires a new time/space comparison.

7. Exhaustive small-instance agreement does not independently verify the paper's full reductions, constants, parameter interfaces, or Lean dependency closure. Nor does it establish a new general SAT bound.

To improve the full algorithm, one would need: a certificate valid for the actual multi-packet local polynomial class; a bound on the resulting precision increments uniform over all recursive children and fixed lifts; an explicit recurrence including interpolation, packet construction, arithmetic bit complexity, and orbit-discovery costs; and a compatible space analysis. Avoiding full polynomial reconstruction by computing only the required target channels is promising but needs an exact evaluation algorithm and a cost proof. Merely selecting fewer weights does not supply such an algorithm. Adaptive symmetry screening should reject unpromising nodes cheaply before expensive group discovery, while retaining an exact fallback. These are proposed research directions, not measured improvements to the paper's full solver.

3.15 Ram Calculus interpretation and conventional comparators

The useful Ram viewpoint is to make the measurement space, degree/family certificate, target functional, precision loss, and transported symmetry explicit. In these tests it leads to a concrete node-specific decoder and a checkable orbit quotient. The advantages demonstrated are exact correctness under narrower arithmetic, useful runtime reductions on symmetric fixtures, and an explicit diagnostic boundary for invalid decoder transfer. The costs are certificate construction, stored orbit data, and strict hypotheses that exclude most of the literal source parameter regime from a tiny experiment.

These mechanisms are also classical Hermite interpolation, valuation error propagation, finite-field interpolation, tensor contraction, and finite-group orbit aggregation. A conventional implementation using the same formulas and certificates would obtain the same identities and arithmetic savings. There is no evidence here of a performance advantage unique to Ram terminology or unavailable to conventional methods. The contribution is the specified integration, its proofs, tests, and failure controls; it is not a historical-priority claim for those underlying mathematical tools.

3.16 Reproduction and preservation of existing material

All new implementation and raw evidence are in `eth-benchmark-v5/`. The principal files are `recursive_v5.py`, `precision_plans.py`, and `run_v5.py`. The frozen v4 ring implementation is imported read-only with bytecode writing disabled. Run from the v5 support directory:

```
python -B run_v5.py --selftest
python -B run_v5.py --out results-new-run
```

The output directory must not already exist. The delivered run is `results-20261027/`; it contains the self-tests, complete instance and oracle records, all 300 worker outputs, and the aggregate results with interpreter/platform data and SHA-256 hashes. The source code, data, and old reports are not silently overwritten to rerun a benchmark. The additional `verify_results_v5.py` audit checks that every active saved output contains both complete local polynomials, with the expected coefficient counts: 400 root local polynomials are present and correct across the 200 active worker

outputs. It also validates every input network and rechecks all recorded source hashes. This closes a defensive-validation gap in the engine's permissive result-check helper without altering the measured implementation or any saved benchmark result.

The original v4 technical body and bibliography are retained verbatim after this section, while the title/abstract identify the new revision. An assembly manifest checks this retention and hashes the unchanged v4 LaTeX/PDF inputs. Compilation and visual inspection address document integrity and layout, not the truth of asymptotic claims.

Conclusion of the new experiment. The exact finite recursive evaluator is implemented and tested, including nontrivial child lifts and two active levels. Node-specific precision reduces the nested working path from $5 \to 261 \to 389$ bits to $5 \to 23 \to 32$ bits. Combined precision and certified symmetry give an observed $1.93\times$ nested speed ratio with a 10.6% traced-memory increase, but almost no benefit on an asymmetric weighted case and no advantage over direct counting at this scale. Literal active-schedule and full-algorithm benchmarks remain unavailable; no improved general ETH/SAT complexity theorem is asserted.

Retained analysis and experiments from revisions 1–4

The following material preserves the earlier report, including its original limitations and experiment descriptions. The new finite recursive tests and corrected transfer conditions are those in Section 3; older statements about unexecuted full recursion concern the literal source algorithm, which remains unexecuted.

4 Scope, source status, and principal conclusions

4.1 The manuscript actually examined

The primary source is OpenAI's manuscript dated October 17, 2026 and revised October 25, 2026, in the repository folder named `The-Exponential-Time-Hypothesis-is-False-October-25-2026` [1]. The attached `paper(2).pdf` has 887,884 bytes. Its Git blob hash, computed locally, is

`0503566e95bc587a4076573a45132d97531b8ba6`.

This matches the repository PDF. The four TeX files used for the analysis are pinned to commit

`42907c9e999cc00d6abe7a50bfc1023250edd1b9`.

They are `main_full_version.tex`, `quantitative-reduction.tex`, `quantitative-contraction.tex`, and `quantitative-circuits.tex`. This fixes the object of analysis even if the live repository changes. The supplied PDF contains 66 pages.

The source announces a deterministic exact-counting algorithm for every fixed clause width k. Its headline is therefore much stronger than a heuristic SAT solver or a result on a selected instance distribution. The repository's general description also distinguishes different stages of verification and does not warrant treating every hosted manuscript as a fully independently audited publication [2]. Our analysis is correspondingly explicit about its own scope.

4.2 Three separate evidential questions

The following questions have different answers.

1. *Is Ram Calculus relevant?* Yes, particularly its corrected finite-algebra, moment, jet, state-aggregation, and certified-inversion interfaces. These can be connected to exact counting without an analogy to continuous PDEs.
2. *Can useful refinements be proved?* Yes. Sections 11–16 prove a reduction in the number of restricted-count calls, a weighted support extension, exact summary reductions under stated hypotheses, and a target-specific precision rule.
3. *Does this report improve the source's general exponent or independently establish its headline?* No. The new call-count result applies to the source's Section 3 routine, whereas its sharpest bound uses Appendix D. The local precision improvement has no proved better worst-case growth rate. The full low-degree construction, reduction, and recursive implementation have not been reproduced end to end in software or in Lean here.

The word *new* below means an addition to this comparative report, with a supplied proof or computation, not a verified claim of historical priority. Finite differences, generating functions,

Hermite interpolation, quotient algebras, and valuation error propagation have substantial classical antecedents. Their use in Ram terminology does not erase that provenance.

4.3 What was checked, and what was not

The investigation follows the main proof dependencies: the counting identity, design interface, low-degree replacement mechanism, sparse polynomial parity lift, greedy parity tests, sparsification, exact arithmetic, and the sharper contraction route. Independent corpus research and a separate mathematical audit were used. The detailed proofs in this report establish their own stated lemmas, rather than assuming that searching the large corpus constitutes verification of every report in it.

The initial experiments are exact small-instance identity checks. Revision 2 adds the matched finite implementation benchmark in Section 19. Neither experiment is a performance test of the full ETH algorithm or certifies all large-n parameter inequalities. The finite timings do not establish general practical crossover sizes. No inconsistency found in a proposed Ram shortcut is attributed to the source when the source itself preserves the necessary condition.

5 The problem and its exact computational contract

5.1 State space, arithmetic, and unresolved quantifiers

Let F be a CNF formula on n *declared* Boolean variables with m clauses, each of width at most a fixed integer $k \geq 2$. Its exact count is

$$Z(F) = \sum_{x \in \{0,1\}^n} \mathbf{1}_{\{F(x)=1\}} \in \{0, 1, \ldots, 2^n\}. \tag{14}$$

Unused variables contribute a factor of two each. An empty clause gives zero; an empty conjunction has count 2^n. Tautologies may be removed. Clause and literal normalization must preserve these conventions.

There is no spatial boundary condition or gauge constraint in this problem. The state space is a finite cube, and the observation of interest is the unnormalized sum functional. The relevant operator domains are finite polynomial spaces, Boolean quotient algebras, truncated coefficient rings, and explicitly described tensors. The complexity model counts bit operations, including construction, coefficient access, exact arithmetic, and output, rather than treating arbitrarily long rationals as unit-cost scalars.

ETH, in its usual bounded-width form, posits an exponential lower-time barrier for 3-SAT. Its precise relationships to sparse formulas depend on sparsification and size-preserving reductions [4]. The source claims the following bounds for each *fixed* k:

$$T(n, m) = (n + m)^{O_k(1)} 2^{O_k(n \log\log n / \sqrt{\log n})}, \tag{15}$$

$$T(n, m) = (n + m)^{O_{k,D}(1)} 2^{O_{k,D}(n/\sqrt{\log n})} \quad (m \leq Dn), \tag{16}$$

with polynomial bit space. Its earlier interpolation route gives

$$(n + m)^{O_k(1)} 2^{O_k(n(\log\log n)^2/\sqrt{\log n})}. \tag{17}$$

These are source claims, not improved bounds obtained in this report. For fixed parameters their exponents are $o(n)$. They do not give a polynomial-time algorithm, a useful finite-size threshold, or uniform constants when k or D grow with the input.

5.2 The available formalization and its limits here

The public Lean library has a solution entry point `OAI.Algorithms.SubexponentialSAT.Main`, importing `SqrtLogSpace.Final`. Its theorem `exact_count_and_sat` states exact counting, decision, a fixed primitive machine, and polynomial all-prefix visited-tape space. The comparator statement uses the runtime scale

$$n\frac{(\log\log(n+16))^2}{\sqrt{\log(n+16)}}.$$

Thus the inspected interface supports the weaker square-root-logarithmic scale (17), not by itself the sharper Appendix D bounds (15)–(16). The comparator specification contains an intentional `sorry`; its JSON points to the separate solution module. That placeholder is not evidence of a gap in the solution module. The inspected solution theorem invokes other proved declarations rather than a local placeholder [3].

The downloaded interface files were pinned to repository head `7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a` during this investigation. We did not compile their complete dependency closure, run the comparator, audit all permitted axioms, or check every encoding-to-paper correspondence. Accordingly, we report the existence and visible scope of the formalization, not a new independent machine-checked verification.

6 Relevant Ram corpus and the exactness boundary

The closest sources are listed in Table 1. Later corrected versions are used where they explicitly restrict earlier claims. The survey covered targeted parts of the requested directories, not every page of every book or report. This distinction matters because some older statements about universal nilpotence or automatic inversion were subsequently corrected.

Table 1: Corpus connections and non-transferable claims.

Source	Useful result or interface	Limitation for this problem
RamOperator Foundations; revised Series [9, 10]	Exact finite polynomial operators, nilpotent inverses, coefficient duality, projection leakage, coordinate invariance.	Finite inversion does not imply cheap global Boolean counting or safe arbitrary projection.
TR-item-010 [11]	Boolean quotient, moment aliasing, exact closure with a supplied tree decomposition.	Bounded clause width is not bounded treewidth; low moments need not identify the count.
M012 [12]	Exact finite-alphabet moment coordinates and explicit support-size cost.	An exact tensor basis can still have exponentially many coefficients.
M231 [13]	Hasse jets, transported multiplication, structured Toeplitz actions, field-aware bit accounting.	Noncommutative PIT is not common-zero counting; its special recurrences require their own hypotheses.
Corrected TR166 [14]	Separates coordinate conversion from model inversion and requires rank and residual closure.	A finite matrix need not be unit triangular or invertible.

Source	Useful result or interface	Limitation for this problem
Revised Volume 1 [15]	Newton/divided-difference coordinates and the correct product rule.	Confluent nodes and modular nonunits require separate arithmetic; changing basis alone is not a speedup.
RMT extension TR1 [16]	Exact signed aggregate-state recursion and explicit attainable-state cost.	Counting adds multiplicities; the optimization rule of retaining a minimum does not preserve counts.
RMT extension TR2 [17]	Multiplicity, jet information, and terminal polynomial degree budgets.	Complex-order endpoint models are not Boolean counting algorithms.
RMT extension TR3 [18]	Ordered hypercomplex channels, physical metrics, projection and finite-bank losses.	Physical tensor reconstruction is not scalar combinatorial tensor contraction.
RMT extension TR4 [19]	Range-consistent nonlinear representations, transported products, residual and root-isolation limits.	Local root certificates do not count every Boolean solution without complete coverage.

6.1 A concrete finite Ram representation

For a finite alphabet $A_i \subset \mathbb{Q}$ of distinct values, put

$$\mathcal{B}_A = \mathbb{Q}[X_1, \dots, X_n] / \left(\prod_{a \in A_i} (X_i - a) : 1 \le i \le n \right). \tag{18}$$

Evaluation identifies this algebra with functions on $\prod_i A_i$. Univariate Lagrange interpolation proves the identification in one coordinate; tensoring the maps proves the multivariate statement. Its dimension is $\prod_i |A_i|$, not the number of input factors.

In the Boolean case, a unique representative is $f(X) = \sum_{S \subseteq [n]} c_S X_S$, where $X_S = \prod_{i \in S} X_i$. The exact counting functional is

$$\Lambda(f) = \sum_{S \subseteq [n]} 2^{n-|S|} c_S. \tag{19}$$

Each X_S is one on exactly $2^{n-|S|}$ assignments, proving the formula. Normalized expectation is $2^{-n}\Lambda$, so an expectation is not a count.

Ram moment rows give one way to organize these functionals. An invertible coordinate map R produces an equivalent representation only if

$$u \star v = R\big((R^{-1}u)(R^{-1}v)\big), \qquad \Lambda_R(u) = \Lambda(R^{-1}u). \tag{20}$$

Entrywise multiplication in arbitrary transformed coordinates is generally wrong. Equation (20) preserves the same finite algebra and does not reduce its dimension. Over $\mathbb{F}_2$ the analogous scalar sum gives only parity, not the integer count.

6.2 Which information is being represented?

Three objects must not be conflated: moments of assignments on the Boolean cube; jets in a scalar deformation parameter; and point jets of a physical field. The first may require 2^n independent

coordinates. The second can be small because a *separately proved* polynomial-degree bound applies. The third requires its own regularity and domain assumptions. The source's algorithm exploits the second together with specially constructed summaries; it does not recover an arbitrary Boolean function from a few moments.

7 Proof architecture of the source and opportunities for Ram methods

Table 2: Dependency map. Source section numbers refer to the pinned manuscript.

Stage	Substantive source step	Ram connection and audit obligation
Sections 2–3	Encode the count as a product-sum and deform incidence rows while restoring each original variable at parameter zero.	Exact moment functional; degree, measure, and multiplication must be retained.
Section 3	Condition on bounded integer summaries so factors separate by groups; use Boolean finite differences to recover jets.	Exact aggregate states; new call aggregation in Section 11.
Section 4	Build low-degree rows meeting many partition/measurement constraints by auxiliary arrays, spectral structure and annihilation.	No replacement construction supplied by existing Ram frames; certify row kernels and coefficient heights exactly.
Section 5	Lift evenness modulo 2^{ℓ} and truncate a common order variable, keeping total degree linear.	Filtered coefficient algebras; truncation is global, and final coefficient summation is not multiplicative.
Section 6; Appendix A	Use parity checks and disjoint sparsification to reduce general formulas to sparse polynomial zero counts.	Preserve solution multiplicities and query size; do not assume tests already isolate the solutions before proving they do.
Appendix B	Bound denominators, intermediate bit sizes, and streamed storage.	Exact structured linear algebra; no unit-cost rational shortcut.
Appendix D.1–D.3	Encode sparse and general systems as Boolean tensor networks with controlled wires and evaluable entries.	Transported tensor coordinates preserve scalar contraction, but not automatically small size.
Appendix D.4	Cauchy deformation, scaled integral jets, tapered Hermite precision, weighted modes, and recursive contraction.	Finite inverse and valuation certificates; target-specific precision in Section 16.
Appendix C	Consequences for circuit lower bounds and other reductions.	Consequences require the upstream algorithm and the precise reductions; not new Ram theorems here.

7.1 What the hard construction accomplishes

For factors Φ_f on scopes I_f, each incidence (f, v) receives its own polynomial row $P_{f,v}(T) \in \mathbb{Q}[T]^n$. Different occurrences of the same variable may have different rows. The requirements include $P_{f,v}(0) = e_v$, degree at most $R < Q$, and, at each interpolation node, small groups and few bounded integer measurements through which every off-group dependence factors. Simply interpolating arbitrary prescribed rows at Q nodes would usually require degree of order Q, defeating the intended R/Q saving.

The source instead arranges auxiliary coordinates, uses a structured matrix/spectral construction, and annihilates selected bad-node contributions with low-degree factors that preserve the value at zero. The matrix-rigidity background includes Alman and Williams [5]; their results by themselves are not a complete construction of the source's simultaneous incidence design. Scope size, variable incidence, group balance, row range, common denominators, and the construction's own time are all essential. Replacing the construction with a generic tight frame or a low-rank approximation does not establish these properties.

The rational design in Section 4 and the recursive two-adic deformation in Appendix D are also different constructions. A refinement of one must not be silently advertised as a refinement of the other.

7.2 The swap construction in explicit operator form

Here is a closer derivation of the central Section 4 mechanism. Index an auxiliary array by pairs of labels $(a_u, d_u) \in [r]^Q \times [r]^Q$, with $D = r^{2Q}$ positions. Let Π_s exchange the two symbols in coordinate s. These are commuting rational permutation involutions. Set

$$V_0 = \prod_{s=1}^{Q} \Pi_s, \qquad W_0 = \frac{1}{2} \sum_{s=1}^{Q} (I - \Pi_s). \tag{21}$$

If a simultaneous eigenvector has j minus signs, then W_0 acts by j and V_0 acts by $(-1)^j$. At one coordinate the minus eigenspace has dimension $r(r-1)/2$ out of r^2; therefore the distribution of j among a tensor-product eigenbasis is $\mathrm{Bin}(Q, (r-1)/(2r))$.

Choose an interval $\mathcal{I} = \{A_0, \ldots, A_0 + R - 1\}$ and define

$$f_{\mathcal{I}}(z) = (-1)^{A_0} \sum_{j=0}^{R-1} (-2)^j \binom{z - A_0}{j}, \qquad M_0 = V_0 f_{\mathcal{I}}(W_0). \tag{22}$$

At $z = A_0 + h$, $0 \le h < R$, the binomial theorem gives $f_{\mathcal{I}}(z) = (-1)^z$. Thus M_0 is identity on the spectral subspace with $j \in \mathcal{I}$. Select the excluded orthogonal basis vectors as integer measurement rows $C^{\mathcal{B}}$. Their entries are $0, 1, -1$, and their number is $D \Pr[\mathrm{Bin}(Q, (r-1)/(2r)) \notin \mathcal{I}]$. It follows over $\mathbb{Q}$ that $M_0 \xi = \xi$ whenever $C^{\mathcal{B}} \xi = 0$.

The second requirement is support, not just rank. A degree-at-most-$R-1$ polynomial in the commuting Π_s is a sum of products involving at most $R - 1$ distinct swaps, because $\Pi_s^2 = I$. Multiplication by V_0 swaps every remaining coordinate. Consequently

$$M_0[u, y] \neq 0 \implies |\{s : a_y(s) \neq d_u(s)\}| \le R - 1.$$

Write that mismatch set as $E_{u,y}$, and let

$$\varphi_{u,y}(T) = \prod_{s \in E_{u,y}} (1 - T/s),$$
$$P_u(T) = e_u^T(I - M_0) + \sum_y M_0[u, y]\varphi_{u,y}(T)e_y^T. \tag{23}$$

The sum need only include nonzero entries of M_0. At zero, $P_u(0) = e_u^T$. At node t, a term using a coordinate outside the assigned group vanishes because its mismatch set contains t. The remaining nonlocal term $(I - M_0)\xi$ is determined by the shared measurements, since it vanishes on their kernel. This proves the desired separation on the auxiliary array.

This construction is legitimately an exact finite operator calculus, but its effectiveness depends on the simultaneous spectral and support properties. A Ram polynomial representation of $f_{\mathcal{I}}(W_0)$ does not establish those properties anew; it records the source's proof in compatible coordinates. The low-degree interpolation polynomial is not an approximate spectral filter on the retained eigenspaces: it has exact prescribed values there.

7.3 Pullback to actual variables, coefficient size, and parameters

The auxiliary slots contain copies of original variables, with unused slots set to zero. Copies are not independently summed Boolean variables. All copies of a variable retain the same group label, so pulling a row back through the embedding preserves its off-group kernel implication. Identifying columns cannot increase a row's absolute sum. The source's label assignment uses a conditional-expectation argument that includes dependencies from both shared variables and shared factors; treating all incidences as independent would not justify its packing bounds.

The factor $(-2)^j$ in (22) clears the halves in W_0; the remaining interpolation denominators divide $j!$. The annihilation factors in (23) have denominators dividing $Q!$. Thus $(R-1)!Q!$ clears all row coefficients. The source bounds their bit length using row-sum and polynomial coefficient norms. This is the arithmetic information missing from a mere statement that a basis change exists.

For $\lambda = \lceil \log_2 n \rceil$, $\nu = \lceil \log_2 \lambda \rceil$, and a fixed incidence exponent β, the source balances parameters of the form

$$r = \left\lfloor \frac{\sqrt{\lambda}}{\nu^2} \right\rfloor, \quad Q = \left\lfloor \frac{\lambda}{128\nu} \right\rfloor, \quad R = (\beta + 3)\lceil \sqrt{\lambda} \rceil, \quad \tau = \frac{1}{r\lambda}, \quad D = r^{2Q}.$$

A binomial-tail estimate makes the number of excluded measurements small, and $\log_2 D \le \lambda/128$ controls their magnitude budget. All required inequalities are asserted for sufficiently large n, with finite exceptions handled by direct enumeration. These thresholds can be extremely large. For example, $6R \le Q$ with $\beta \ge 1$ already requires approximately $\sqrt{\lambda} \ge 3072\nu$. This is consistent with an asymptotic theorem but supplies no practical-size advantage.

7.4 Why the rational spectral proof cannot simply move to characteristic two

On an orbit of four slots, let two swaps flip its two orientation bits. Choose $R = 1$ and $\mathcal{I} = \{1\}$. Then $M_0 = -\Pi_1\Pi_2$. The excluded eigenspace rows are $(1, 1, 1, 1)$ and $(1, -1, -1, 1)$. Modulo two those rows coincide, so $\xi = (1, 1, 0, 0)^T$ is in their modular kernel. But $M_0\xi = (0, 0, 1, 1)^T \ne \xi$ modulo two. This does not contradict the source, whose kernel proof is over $\mathbb{Q}$; it disproves an invalid finite-field shortcut. A Ram implementation must retain the declared field and multiplication law, not only the appearance of the matrix formulas.

8 Exact shared-summary counting as a Ram finite-state formulation

8.1 Polynomial deformation and degree

Let $\mathcal{A}$ be a commutative $\mathbb{Q}$-algebra and suppose the product of the incidence-separated factors has total degree at most d. Define

$$H(T) = \sum_{x\in\{0,1\}^n} \prod_f \Phi_f((P_{f,v}(T)\cdot x)_{v\in I_f}). \tag{24}$$

Then $H(0) = Z$ and $\deg H \le dR$. The degree is measured before identifying separate incidence indeterminates. A small degree for one factor is not a bound for their full product.

Fix a node t, a partition $[n] = S_1 \sqcup \cdots \sqcup S_r$, and an integer matrix C_g on each group. Write $y_g = C_g x_g$. Assign each factor to one group g. For every off-group $h \ne g$, require

$$P_{f,v}(t)|_{S_h} \in \mathrm{rowspan}_{\mathbb{Q}}(C_h). \tag{25}$$

Equivalently, the row vanishes on $\ker C_h$. This can be certified by an explicit rational row λ satisfying $P_{f,v}(t)|_{S_h} = \lambda C_h$. A factor assigned to g is then a function of its local assignment x_g and the off-group measurements.

Theorem 8.1 (Exact fiber factorization)**.** *Let $A_g(x) = a_g(x_g, y_{-g})$ take values in a commutative ring. For $E \subseteq [n]$, define*

$$B_{g,E}(y) = \sum_{\substack{x_g\in\{0,1\}^{S_g}\\ C_g x_g = y_g,\ x_i = 1\ (i\in E\cap S_g)}} a_g(x_g, y_{-g}).$$

Then

$$\sum_{\substack{x\in\{0,1\}^n\\ x_E = 1}} \prod_g A_g(x) = \sum_y \prod_g B_{g,E}(y), \tag{26}$$

where y ranges over all attainable measurement tuples, or any containing integer box with empty fibers assigned zero.

Proof. Every assignment belongs to exactly one measurement fiber. For fixed y the permitted assignments form the Cartesian product of local fibers. Each factor in that fiber depends only on its own local assignment. Distributing the finite product of sums proves the identity. No positivity, probabilistic independence, or approximate equality is used. □

This is a rederivation of the source's measurement-separation lemma. Its Ram interpretation is an exact coupling-state representation, consistent with TR1's rule that outside completions must see the same retained state [16]. For counting, equivalent states carry the *sum* of their weights; retaining only the cheapest representative would lose multiplicity.

If each measurement row has ℓ^1 norm at most D and there are c rows altogether, a safe box has $(2D+1)^c$ tuples. A streamed implementation uses at most

$$2(2D+1)^c \sum_g 2^{|S_g|} \tag{27}$$

local-assignment examinations, allowing one scan to find representatives and one to accumulate the local sum. Factor evaluation, arithmetic, and state-description costs remain additional. Polynomial space requires polynomial-length ring elements and evaluators, not just streaming indices.

8.2 Why rank alone is the wrong parameter

The rank-one row $(1, 2, 4, \dots, 2^{n-1})$ gives 2^n distinct values on the cube. Thus rank one does not give a small summary alphabet. Conversely, rows $(1, 1, 1)$ and $(2, 2, 2)$ have redundant information: retaining the first preserves their common kernel and leaves only four attainable values. The source itself emphasizes this distinction. Ram rank-aware methods are useful only together with range and coefficient-height accounting.

9 Finite jets, interpolation, and the auxiliary-PDE question

9.1 The exact jet algebra

For any commutative ring K, a node $t \in K$, and an integer $J \geq 1$,

$$\mathcal{J}_{t,J}p = p(t + z) \pmod{z^J} \in K[z]/(z^J) \tag{28}$$

is an algebra homomorphism. Its coefficients are Hasse coefficients, $[z^j]p(t + z)$. Over $\mathbb{Q}$ they equal $p^{(j)}(t)/j!$; the coefficient definition remains meaningful when $j!$ is not invertible. Multiplication is truncated convolution, not entrywise multiplication. This is precisely the characteristic-safe interface used in M231 [13].

For distinct rational nodes t_a and multiplicities J_a, put $N = \sum_a J_a$. Jets determine every polynomial of degree less than N. Indeed, a polynomial with all jets zero is divisible by $\prod_a (T - t_a)^{J_a}$, a degree-N polynomial, and must vanish if it has smaller degree. This proves injectivity; equality of finite dimensions proves unique reconstruction. The same statement over a general ring requires suitable unit differences or comaximal ideals.

Example 9.1 (Distinct modular nodes need not be sufficient)**.** The nonzero polynomial $2T$ over $\mathbb{Z}/4\mathbb{Z}$ vanishes at both 0 and 2. The nodes are distinct, but their difference is not a unit. Rational interpolation cannot simply be reduced modulo a power of two.

9.2 Finite Ram polynomial operators and their limits

On $\mathcal{P}_D = \{p : \deg p \leq D\}$, ordinary differentiation $N_D = d/dT$ is nilpotent, so

$$p(t + z) = \exp(zN_D)p(t) = \sum_{j=0}^{D} \frac{z^j}{j!} p^{(j)}(t). \tag{29}$$

This is an exact finite identity over $\mathbb{Q}$. The Ram polynomial moment calculus organizes related actions as finite rows in N_D; a unit constant plus a nilpotent operator has a terminating geometric inverse [9, 10]. The existence of this representation is useful for checking the arithmetic, but it does not make the coefficients of H cheap to obtain.

There is an important distinction between polynomials of bounded degree as a vector space and a quotient algebra with truncated multiplication. Ordinary differentiation does *not* descend to a derivation of $\mathbb{Q}[z]/(z^J)$: the ideal (z^J) is not differentiation-invariant. For $J = 2$, $z^2 = 0$ in the quotient, while a purported Leibniz rule gives $D(z^2) = 2z \neq 0$. Canonical representative differentiation is a linear matrix but not a derivation of this truncated algebra. Substitution followed by the quotient map (28) is the safe operation.

9.3 No auxiliary PDE is present in the inspected paper

The mathematical auxiliaries used here are a scalar polynomial $H(T)$, formal variables recording order, finite differences on subsets, integer linear summaries, and finite-ring tensor networks. The source does not introduce an elliptic, parabolic, hyperbolic, or quaternionic PDE whose solution supplies the counting result.

One may define $u(t, z) = H(t + z)$ and write the formal transport relation $\partial_t u = \partial_z u$ in characteristic zero. This is a rewriting of a polynomial identity, not an auxiliary PDE solver with a new complexity bound: its initial coefficients already contain the difficult count. Calling this relation a Ram partial differential operator does not provide missing information or reduce the cost of evaluating it. The auxiliary Dirichlet equation in the earlier local-smoothing manuscript is a different matter and must not be transferred into this paper.

10 Boolean-jet recovery: a self-contained rederivation

At a node t, write $\Delta P_{f,v}(z) = P_{f,v}(t + z) - P_{f,v}(t)$ and introduce independent variables $X_1, \dots, X_n$:

$$G_x(X, z) = \prod_f \Phi_f\big((P_{f,v}(t) \cdot x + \Delta P_{f,v}(z) \cdot X)_{v \in I_f}\big). \tag{30}$$

Each occurrence of X_i costs at least one power of z. Therefore

$$\deg_X\big([z^j] G_x(X, z)\big) \le j. \tag{31}$$

Moreover $\sum_x G_x(x, z) = H(t + z)$. Holding X fixed preserves the summary condition in x, whereas setting $X = x$ prematurely can destroy the factor separation. This explains the source's finite-difference step.

Lemma 10.1 (Boolean difference identity)**.** *Let $p_x(X)$, one for each Boolean x, be polynomials over a commutative ring of degree at most q. Then*

$$\sum_x p_x(x) = \sum_{\substack{E \subseteq [n] \\ |E| \le q}} \sum_{U \subseteq E} (-1)^{|E|-|U|} \sum_{x_E = 1} p_x(\mathbf{1}_U). \tag{32}$$

Proof. Reduce positive powers of each X_i to X_i, preserving Boolean evaluations and not increasing degree. Write the resulting polynomial as $\sum_E c_{x,E} X_E$. Subset inversion gives $c_{x,E} = \sum_{U \subseteq E} (-1)^{|E|-|U|} p_x(\mathbf{1}_U)$. Coefficients with $|E| > q$ vanish. At $X = x$, X_E is the indicator of $x_E = 1$. Summing proves the result. □

Take $q = \lfloor dR/Q \rfloor$ and $J = q + 1$ at Q rational nodes. Then $QJ > dR$, so the jets determine $H(0)$. Applying Lemma 10.1 coefficientwise to (30) computes those jets with

$$C_q(n) = \sum_{j=0}^{\min(q,n)} 2^j \binom{n}{j} \tag{33}$$

restricted calls per node. With the source design, $c \le 2\tau n$ and $|S_g| \le 2n/r$, giving the assignment-examination bound

$$2Qr\, C_q(n)(2D + 1)^{2\tau n} 2^{2n/r}. \tag{34}$$

All ring and factor costs still have to be included.

For $d = O(n)$ and $R/Q \to 0$, the three exponential costs have logarithm

$$O\left(n\left[\frac{R}{Q}\log\frac{2Q}{R} + \tau\log(2D+1) + \frac{1}{r}\right]\right). \tag{35}$$

The source's parameters make this the scale in (17) for appropriate polylogarithmic incidence bounds. Merely accelerating a polynomial-sized matrix inversion does not change these three costs.

11 A proved refinement: aggregating the Boolean-pair loop

This section is the main local algorithmic addition. It eliminates the outer restriction index E by retaining a small generating jet for the number of remaining selected bits. It does not assume the truth of the source's full ETH theorem.

Theorem 11.1 (Aggregated Boolean identity)**.** *Under Lemma 10.1, define, for $u \le q$ and $h \ge u$,*

$$w_q(h,u) = \sum_{k=0}^{\min(q-u,h-u)} (-1)^k \binom{h-u}{k}. \tag{36}$$

Then

$$\sum_x p_x(x) = \sum_{\substack{U\subseteq[n]\\ |U|\le q}} \sum_{x_U=1} w_q(|x|,|U|)\,p_x(\mathbf{1}_U). \tag{37}$$

Proof. In (32), interchange the finite sums over x, E, U. For fixed $U \subseteq \operatorname{supp} x$, the admissible sets E are $U \cup K$, where $K \subseteq \operatorname{supp} x \setminus U$ and $|K| \le q - |U|$. There are $\binom{|x|-|U|}{k}$ such sets of added size k, and their sign is $(-1)^k$. Their sum is exactly (36). □

For $h > u$, the weight is zero if $q - u \ge h - u$; otherwise it equals $(-1)^{q-u}\binom{h-u-1}{q-u}$. For $h = u$, it equals one. These follow from the alternating binomial sum. The defining finite sum is preferable in rings where generalized binomial notation would obscure integrality.

11.1 Why aggregation preserves local factorization

For a fixed U put $r_U = q - |U|$ and introduce one formal variable w:

$$K_U(z,w) = \sum_{x_U=1} G_x(\mathbf{1}_U, z)(1+w)^{|x|-|U|} \pmod{z^J, w^{r_U+1}}. \tag{38}$$

Define the *linear* final functional

$$\mathcal{L}_{r_U}\left(\sum_{k=0}^{r_U} b_k w^k\right) = \sum_{k=0}^{r_U} (-1)^k b_k. \tag{39}$$

The binomial theorem and Theorem 11.1 give

$$H(t+z) \equiv \sum_{|U|\le q} \mathcal{L}_{r_U} K_U(z,w) \pmod{z^J}. \tag{40}$$

For a group S_g, the extra factor is simply $(1+w)^{|x_g|-|U\cap S_g|}$ on the restricted assignments. It is local to that group. Thus Theorem 8.1 computes (38) in

$$\mathcal{A}[z,w]/(z^J, w^{r_U+1}) \tag{41}$$

with the original measurements and with $x_U = 1$ as the sole fixed-bit restriction. No new cross-group measurements are required.

The final functional must be applied *after* group multiplication. It is evaluation at -1 of the retained representative, not a ring homomorphism of the quotient. For example, modulo w^2, the product of w and w is zero, but the product of their representative evaluations at -1 is one. Applying the functional early would invalidate the count.

Corollary 11.2 (Restricted-call bound and polynomial workspace)**.** *The Section 3 jet stage can replace $C_q(n)$ calls per node by*

$$B_q(n) = \sum_{j=0}^{\min(q,n)} \binom{n}{j}, \tag{42}$$

using at most $J(q+1)$ coefficient slots per algebra element. With the source design, the number of local-assignment examinations is at most

$$2Qr\, B_q(n)(2D+1)^{2\tau n} 2^{2n/r}. \tag{43}$$

When q, J and coefficient bit sizes are polynomial in the input size, the enlarged arithmetic and streamed workspace remain polynomial per examined assignment.

Proof. There is one call (38) for each U, and each uses precisely the fiber algorithm in the larger ring. A naive multiplication costs $O(J^2(q+1)^2)$ operations in $\mathcal{A}$. The extra binomial coefficients have at most $n+1$ bits; their products and sums obey a polynomial additional bit budget under the same finite-degree assumptions. The loops over nodes, U, measurements, and local assignments can be streamed. Only polynomially many coefficients are live; no 2^n assignment table is required by the identity's implementation. □

11.2 What improves and what does not

The inequalities $B_q \le C_q \le 2^{\min(q,n)} B_q$ are immediate. For $q = o(n)$, the leading combinatorial logarithm remains $q\log(n/q) + O(q)$ in both cases. The new identity removes the extra factor attached to enumerating subsets of E, but introduces polynomial coefficient work. It therefore proves fewer restricted calls, not an unconditional wall-clock speedup. For small q, polynomial overhead can outweigh the call reduction. Section 19 now measures this tradeoff on finite matched implementations without changing that asymptotic limitation.

When $q \ge n$, $C_q = 3^n$ whereas $B_q = 2^n$; in that case the weights select only $x = \mathbf{1}_U$, so the identity reduces to direct enumeration. It is not a subexponential algorithm in that regime. For the source's sublinear q, the improved call count does not remove the other two costs in (35), and does not improve the published big-O exponent. Its relationship to Ram Calculus is the use of a small exact moment/jet state to aggregate a repeated summation. The proof is also expressible entirely in conventional generating-function algebra.

11.3 A transparent worked example

Let $n = 3$, $q = 1$, and $p_x(X) = a_x + \sum_i b_{x,i} X_i$. For $U = \varnothing$, $w_1(|x|, 0) = 1 - |x|$; for a singleton $U = \{i\}$ the weight is one. Thus the contribution of a fixed x on the right side is

$$(1 - |x|)a_x + \sum_{i:x_i=1} (a_x + b_{x,i}) = a_x + \sum_i b_{x,i} x_i = p_x(x).$$

The source pair loop uses $1 + 2n = 7$ calls. The aggregated loop uses $1 + n = 4$, with a degree-one generating jet for $U = \varnothing$. The cancellation of the repeated a_x terms explains the mechanism.

12 A weighted-filtration extension

Suppose the correction in coordinate i is divisible by z^{ν_i}, where $\nu_i \geq 1$ is a known integer. A monomial involving a support E then costs at least $\nu(E) = \sum_{i \in E} \nu_i$ powers of z. This is a strengthened structural assumption, not a property established for every source instance.

Theorem 12.1 (Weighted support aggregation)**.** *Assume the Boolean normal form of every p_x has support only on E with $\nu(E) \leq q$. Then*

$$\sum_x p_x(x) = \sum_{\nu(U) \leq q} \sum_{x_U = 1} p_x(\mathbf{1}_U) \sum_{\substack{K \subseteq \operatorname{supp} x \setminus U \\ \nu(K) \leq q - \nu(U)}} (-1)^{|K|}. \tag{44}$$

For each U, the inner weight is the sum of coefficients through degree $q - \nu(U)$ in

$$\prod_{i \in \operatorname{supp} x \setminus U} (1 - s^{\nu_i}). \tag{45}$$

Consequently it is computable by the same group factorization with one truncated generating variable and only $|\{U : \nu(U) \leq q\}|$ restricted calls.

Proof. Subset inversion reconstructs precisely the supported coefficients; the family $\{E : \nu(E) \leq q\}$ is downward closed. Interchanging sums as in Theorem 11.1 leaves $E = U \cup K$ with the stated weight budget. Expanding (45) supplies sign $(-1)^{|K|}$ and degree $\nu(K)$. Each factor in that product is local to its variable, so it can be multiplied into its own group's factor. Final coefficient summation is linear and is applied after multiplication. □

The new call count is no larger than $B_q(n)$, since $\nu_i \geq 1$. For example, with weights $(1, 2, 3)$ and $q = 2$, only $U = \varnothing, \{1\}, \{2\}$ occur: three calls rather than seven unweighted subset calls. Establishing such higher-order vanishing in an actual replacement design could be valuable. Imposing it can also raise the replacement degree R, so an overall gain requires balancing both changes. The theorem does not license declaring weights without proving the corresponding divisibilities.

13 Rank-aware summaries and exact structural special cases

13.1 Deleting redundant rows without losing the count

Proposition 13.1 (Exact row-basis reduction)**.** *Let C be an integer measurement matrix and let C' consist of a subset of its rows spanning its rational row space. Then $Cx = Cx'$ if and only if $C'x = C'x'$. Every off-group certificate through C factors through C'. The same summary fibers and exact weighted counts are retained.*

Proof. The two matrices have the same rational kernel. Each omitted row is a rational linear combination of retained rows, so equality of retained measurements implies equality of all measurements. The reverse implication is immediate. The row-space certificate (25) is unchanged as a condition. □

Selecting existing rows avoids increasing their integer magnitudes, unlike an arbitrary rational change followed by denominator clearing. If retained row j has norm D_j, the safe box bound can be sharpened to $\prod_j (2D_j + 1)$. This is a valid instance-specific preprocessing step; it may do nothing when the source design already has independent rows.

One can reduce the box further to actual attainable summaries, but producing and enumerating that set is not free. Storing all reachable values may use exponential space. A sparse-state implementation must account for its construction and memory; the source's full-box streaming remains a safe polynomial-workspace fallback. Rank one still permits 2^n attainable states, as Section 8 shows.

13.2 A bounded-width alternative independent of the source claim

Proposition 13.2 (Exact counting with a supplied tree decomposition)**.** *Given a CNF and a tree decomposition of its primal graph of width w, the count is computable in $2^{w+1}\,\mathrm{poly}(n, m, |\mathcal{T}|)$ bit operations, with the polynomial factor covering count arithmetic. Storing all messages may require $2^{w+1}\,\mathrm{poly}(n, |\mathcal{T}|)$ bits.*

Proof. Assign each clause to one containing bag. A message indexed by a separator assignment counts the satisfying extensions in the subtree interior, excluding the separator variables from multiplicity. Running intersection makes different child interiors disjoint conditional on the bag. Multiply child counts, enforce local clauses, and sum newly eliminated variables. Induction proves exactness; every global satisfying assignment is counted once. A nice decomposition has tables of at most 2^{w+1} entries. Partial counts are at most 2^n and therefore have at most $n + 1$ bits. □

This is the counting version of the finite closure in TR-item-010 [11], expressed by standard sum-product messages. It provides a legitimate structural alternative when w is small, not a new general SAT bound. Decomposition-finding cost must be added if one is not supplied. Width $o(n)$ gives a structural subexponential bound under polynomial remaining costs; arbitrary bounded-width clauses do not ensure $w = o(n)$.

13.3 A rank identity that is exact but not efficient by itself

For reduced Boolean polynomials P_i over $\mathbb{F}_2$, let $A = \mathbb{F}_2[X]/(X_i^2 - X_i)$ and let $M_i : A \to A$ be multiplication by P_i. Then the integer number of common zeros is

$$N = 2^n - \mathrm{rank}_{\mathbb{F}_2}[M_1 \ \cdots \ M_m]. \tag{46}$$

Evaluation identifies A with a product of 2^n copies of $\mathbb{F}_2$. The ideal generated by the P_i has coordinate zero at a common zero and the whole field elsewhere. The quotient dimension is therefore the number of common zeros; rank-nullity proves (46). This is a useful small-instance reference identity, but its matrices have 2^n rows. Sparse columns do not by themselves yield a subexponential rank algorithm. Neither this identity nor a finite Ram inverse proves the source's difficult algorithmic bound.

14 Sparse equations, parity lifting, and the total-degree budget

For $P_i \in \mathbb{F}_2[X]$ given as a short sum of degree-at-most-k monomials, replace powers by first powers and combine duplicate monomials modulo two. Let L_i be the same sum with integer coefficients. Then $P_i(x) = 0$ exactly when $L_i(x)$ is even. Put $\ell = n + 1$ so that a count in $[0, 2^n]$ is determined by its residue modulo 2^ℓ.

14.1 An explicit modulus-amplifying polynomial

Define

$$p_\ell(Y) = (1 - Y)^\ell \sum_{j=0}^{\ell-1} \binom{\ell + j - 1}{j} Y^j. \tag{47}$$

The sum is the inverse of $(1 - Y)^\ell$ modulo Y^ℓ, so $p_\ell \equiv 1 \pmod{Y^\ell}$ and $p_\ell \equiv 0 \pmod{(1 - Y)^\ell}$. For even integers y the first congruence gives one modulo 2^ℓ; for odd integers the second gives zero. This is the source's complement of a modulus-amplifying construction with antecedents in Beigel–Tarui and Chan–Williams [6, 7].

Write $h(Y) = Y(Y - 1)$. Repeated division by this monic quadratic gives

$$p_\ell(Y) = \sum_{j=0}^{\ell-1} e_j(Y) h(Y)^j, \qquad e_j \in \mathbb{Z}[Y], \quad \deg e_j \le 1. \tag{48}$$

This is an exact polynomial basis change with affine remainder coefficients. Its significance is arithmetic: $h(y)$ is always even.

14.2 Why the truncation is shared across all factors

Work in $\mathbb{Q}[V]/(V^\ell)$ and define

$$B_i(X, V) = \sum_{j<\ell} e_j(L_i(X)) h(L_i(X))^j V^j. \tag{49}$$

Multiply *all* factors first, discarding only total V-degree at least ℓ, and then sum the retained coefficients by $\mathcal{S}_\ell$. At integer assignments, each omitted term is divisible by 2^ℓ. Consequently

$$\mathcal{S}_\ell \left(\prod_i B_i(x, V) \bmod V^\ell \right) \equiv \prod_i \mathbf{1}_{\{P_i(x)=0\}} \pmod{2^\ell}. \tag{50}$$

The retained input degree is at most

$$k\big(m_{\text{test}} + 2(\ell - 1)\big). \tag{51}$$

Indeed a product term indexed by j_i has degree at most $\sum_i k(1 + 2j_i)$, and $\sum_i j_i < \ell$. With $m_{\text{test}} = O(n)$ this is linear in n, unlike independent full-degree lifts multiplied without the shared filtration.

This is a strong match to the Ram discipline of retaining the exact algebra, product, and projection contract. It is not a new parity lift introduced by Ram Calculus. The valuation certificate is essential: an arbitrary low-degree projection would not preserve the count.

Example 14.1 (Order two)**.** For $\ell = 2$,

$$p_2(Y) = 1 - 3Y^2 + 2Y^3 = (1 - Y) + (2Y - 1)h(Y).$$

The product for two equations retains constant and first total order in V. The discarded cross term contains $h(y_1)h(y_2)$ and is divisible by four at integer arguments. Its input degree is six, whereas the retained part has degree at most four. The coefficient-sum functional $\mathcal{S}_2$ is not multiplicative on the truncated ring; applying it before multiplication would retain the wrong cross term.

14.3 Reduction from SAT without circular selection

Let $v(x)$ be the vector of clause violations. A true solution has $v(x) = 0$. The source constructs initial parity checks that exclude small nonzero violation supports using coding-theoretic structure related to expander codes [8]. It then adds sparse parity rows greedily. For a surviving nonzero violation vector, a random candidate list hits its support with sufficient probability and rejects it with probability at least $1/4$. Thus some row removes at least a quarter of the remaining false positives.

The candidate is chosen by minimizing the *total* number accepted, computed by the sparse polynomial counter. All candidates keep every true solution, so the unknown true count is the same additive constant in every comparison. After $3n$ rows, false positives are fewer than $(3/4)^{3n}2^n = (27/32)^n < 1$, hence zero. This explains why the greedy step need not know the answer in advance. Its count queries must work on arbitrary candidate systems, not only on systems already isolating true solutions.

Finally, a disjoint sparsification partitions satisfying assignments. Overlapping disjunctions would not permit adding leaf counts. Candidate enumeration, incidence reduction, and sparsification overheads are part of the source proof and are not supplied automatically by a Ram transform.

15 The sharper tensor route and its valuation structure

15.1 What is being contracted

A Boolean tensor network assigns one bit to each edge and a ring-valued table to each vertex. Its contraction is

$$Z_G = \sum_{x \in \{0,1\}^{E(G)}} \prod_v T_v(x|_{\delta(v)}). \tag{52}$$

For fixed clause density, clause tensors and equality trees give $O_{k,D}(n)$ edges and bounded degree. Subdividing each edge with an equality tensor makes the network bipartite without changing its scalar value. Unused variables are accounted for separately. Reduction modulo 2^{n+1} recovers the integer count uniquely.

For general formulas the source encodes retained incidence monomials of the parity lift. The total-degree budget (51) permits indexing selected incidences with $O_k(n \log\log n)$ Boolean wires. High-incidence variables and degree arrays are enumerated with separately bounded overhead. Tensor entries are implicit but must remain evaluable within the allowed time and space. A short tensor description alone is not sufficient if evaluating one entry hides an equally hard problem.

The source contraction theorem assumes $O(s)$ edges, maximum degree at most $\lceil s^{9/10} \rceil$, polynomially many vertices, initial precision and order $P_0, \ell = O(s)$, and entries over

$$\mathcal{B}_{P_0,\ell} = (\mathbb{Z}/2^{P_0}\mathbb{Z})[V]/(V^\ell)$$

with explicitly bounded evaluation cost. Its claimed contraction time is $\mathrm{poly}(s+\mathcal{L})2^{O(s/\sqrt{\log s})}$ and polynomial space. Substituting $s = O(n)$ or $s = O(n \log\log n)$ gives the two headline scales.

15.2 Cauchy deformation and scaled integral coordinates

At a recursive call with M edges, ports are packed into slots of length $L = 2^a$. Set

$$S = \{-L, \dots, L-1\}, \quad C(T) = \prod_{t \in S}(T - t), \quad \alpha = 2L, \qquad W_{xy}(T) = \frac{C(T)}{C'(0)(T - (y - x))}. \tag{53}$$

The apparent pole cancels because $y - x \in S$. The entries are polynomials of degree $2L - 1$, with $W(0) = I$ on the original matched slots. If $l_0(z) = 1 - z$ and $l_1(z) = z$, the deformed contraction is

$$F(T) = \sum_{X,Y \in \{0,1\}^M} A(Y)B(X) \prod_x l_{Y_x}((W(T)X)_x). \tag{54}$$

Its value at zero is the desired contraction and its degree is at most $M(2L - 1)$. Nonzero values need not be positive or count assignments.

The grid S is a full residue system modulo 2^{a+1}. At any node t, the valuations of nonzero differences $t - q$ balance the valuation of $C'(0)$. Substituting $T = t + \alpha z$ allows the known powers of two in each factor to be canceled before division. The resulting coefficients are integral at two, and the remaining inverses are of odd units. This is the source's scaled-integrality mechanism. It is not the statement that every rational polynomial has integral local Taylor coefficients.

The source also uses an unramified extension of the two-local rationals, with residue field large enough to provide many interpolation nodes with unit differences. Ordinary integer nodes alone would not supply that property modulo 2^p. Canonical lifts of tensor entries are fixed at their creation precision; a later recursive call cannot arbitrarily relift algebraically equivalent residues and assume that polynomial deformation commutes with the change.

15.3 Tapering, weighted modes, and recursion

For a retained valuation level $i = v_2(t) < u$, the source assigns multiplicity

$$q_t = h(u - i), \qquad h = \lceil 3M/u \rceil, \quad D_t = L/2^i, \quad p_t = p + 8hD_t + 8M(a+1). \tag{55}$$

The total multiplicity exceeds $\deg F$, and the precision budget accounts for denominators in reconstruction at zero. Near a node, crossing interactions expand into modes of weight $b + 1$ whose coefficients are divisible by 2^{b+1}. Only monomials of total mode weight less than p_t can contribute modulo 2^{p_t}.

If there are at most $6M/D_t$ modes of each weight, their number is bounded using

$$\exp\left(\theta p_t + \frac{6M}{D_t} \sum_{j \geq 1} -\log(1 - e^{-\theta j})\right).$$

The sum is at most $\theta^{-1} \sum_{h \geq 1} h^{-2}$. Choosing $\theta = \sqrt{M/(D_t p_t)}$ gives $\exp(O(\sqrt{M p_t / D_t})) = \exp(O(M/\sqrt{u}))$ under the source budget. Each retained monomial separates into smaller group contractions, with at most $M/8$ edges per child. The source chooses a depth-dependent $u_d \asymp (\log s)/2^d$, so the exponents sum geometrically.

The power-of-two divisibility is an exact vanishing certificate, not a small real norm. Conventional low-rank truncation or an L^2-small Ram residual does not imply it. The source's packet construction, lift consistency, precision inheritance, and implicit-entry bounds remain independent obligations. Our finite tests below probe selected local identities, not the entire recursive theorem.

16 Target-specific Hermite recovery and precision certificates

16.1 The target functional rather than the whole polynomial

Let distinct rational nodes t have multiplicities q_t, with $N = \sum_t q_t > \deg F$, and use scaled jets $h_{t,j} = [z^j]F(t + \alpha z)$. There are unique rational coefficients $\lambda_{t,j}$ for the canonical degree-less-than-N Hermite map such that

$$F(0) = \sum_t \sum_{j<q_t} \lambda_{t,j} h_{t,j}. \tag{56}$$

If A is the confluent matrix with entries $A_{(t,j),k} = \binom{k}{j} t^{k-j}\alpha^j$ for $k \geq j$ and zero otherwise, then $A^T\lambda = e_0$. Exact rational elimination computes the weights. Equivalently, one may use cardinal Hermite polynomials. The source already uses a target-at-zero Hermite formula in Appendix D; the contribution here is to use its *actual* coefficient valuations as a certificate.

Theorem 16.1 (Coordinatewise two-adic precision certificate)**.** *Suppose $F(0)$ is integral at two and approximations to its scaled jets satisfy $v_2(\widehat{h}_{t,j} - h_{t,j}) \geq b_{t,j}$ coordinatewise in the declared coefficient basis. If*

$$b_{t,j} + v_2(\lambda_{t,j}) \geq p \quad \textit{for every nonzero } \lambda_{t,j}, \tag{57}$$

then the reconstructed value is correct modulo 2^p. A sufficient common precision at node t is

$$p_t^{\mathrm{cert}} = p + \ell_t, \qquad \ell_t = \max(0, \max_{j:\lambda_{t,j}\neq 0} -v_2(\lambda_{t,j})). \tag{58}$$

For a recursion requiring strictly increasing creation precisions, use $p + \max(1, \ell_t)$ instead.

Proof. Each weighted jet error belongs to 2^p times the integral coefficient module by (57). Their finite sum also belongs to that module, since the two-adic valuation of a sum is at least the minimum of the valuations of its summands. Subtracting (56) proves the claim. The nodewise bound enforces the coordinatewise inequalities. □

For fixed nonzero weights and independently variable coefficient errors, the coordinatewise requirement is sharp: changing just one coefficient at valuation $p - v_2(\lambda_{t,j}) - 1$ changes the target at valuation $p - 1$. This is not a global optimality theorem for interpolation. The source polynomial may have degree well below $N - 1$, allowing other reconstruction functionals, and correlated errors can obey extra relations.

The source's conservative precision bound (55) is justified by bounding the same reconstruction map through an integral reciprocal series and rational evaluation. Directly computing the weights can give smaller losses. Their calculation has polynomial cost in the number of jets and their explicit bit sizes. It does not remove the need for the source's polynomial bound on those sizes. Within a recursive implementation, one must retain the source's fixed-lift convention and use a certified budget before changing any child precision.

For completeness, the weights can be obtained without inverting the entire matrix. Put

$$E_t(T) = \prod_{v \neq t} \left(\frac{T - v}{t - v} \right)^{q_v}, \qquad b_{t,m} = [z^m] \frac{1}{E_t(t + \alpha z)}.$$

The cardinal Hermite construction gives

$$\lambda_{t,j} = E_t(0) \sum_{k=j}^{q_t - 1} b_{t,k-j} (-t/\alpha)^k. \tag{59}$$

Indeed, multiply the truncated quotient $[F(t + \alpha z)/E_t(t + \alpha z)]_{<q_t}$ by $E_t(T)$ and substitute $z = (T - t)/\alpha$. Summing over nodes reproduces each jet, so uniqueness of interpolation proves the formula; collecting each $h_{t,j}$ at $T = 0$ gives (59).

In the tapered construction $b_{t,m}$ are integral at two because $E_t(t + \alpha z)$ is integral with constant term one. Counting valuations of grid differences yields

$$v_2(E_t(0)) = D_t \sum_{j=1}^{a-i} \frac{j}{2^j} (q(i + j) - q(i)) \geq -6 D_t h,$$

where $q(i) = h \max(0, u - i)$ and $\sum_{j \geq 1} j^2/2^j = 6$. Evaluation at $-t/\alpha$ loses at most $4M(a + 1)$ additional powers of two because $q_t \leq 4M$. Therefore

$$\ell_t \leq 6 h D_t + 4M(a + 1) \leq 8 h D_t + 8M(a + 1). \tag{60}$$

Thus the proposed direct certificate is bounded by the source allowance for the same map. It can exploit cancellations in the actual weights, but (60) alone has the same asymptotic scale.

16.2 A finite demonstrated reduction in precision

We computed the exact rational weights of (56) for two small grids. The comparison in Table 3 lists extra bits *above* the desired output precision p. For $a = 1$, $L = 2$, $h = 3$, and $q_{-1} = q_1 = 3$.

a	u	M	Nodes	Source allowance	Certified loss
1	1	1	$-1, 1$	64	7
2	2	2	$-3, -1, 1, 3$	144	33
2	2	2	$-2, 2$	96	7

Table 3: Exact finite Hermite precision calculations.

For $a = 2$, $L = 4$, $h = 3$, the odd nodes have multiplicity six and the valuation-one nodes have multiplicity three. The scaling is $\alpha = 2L$. These configurations satisfy the local interpolation hypotheses but not the large-size packing conditions of the full recursive algorithm. They establish a local certificate improvement, not a measured acceleration of that algorithm.

16.3 Archimedean residuals require a different certificate

For ordinary absolute errors $|e_{t,j}| \leq \delta_{t,j}$,

$$|\widehat{F}(0) - F(0)| \leq \sum_{t,j} |\lambda_{t,j}| \delta_{t,j}. \tag{61}$$

If the target is known to be an integer and the right side is strictly less than $1/2$, nearest-integer rounding is correct. This follows from the triangle inequality and separation of distinct integers. It is a useful Ram-style residual certificate, but the weights can be large. For a normalized expectation over 2^n assignments, a raw expectation error must be less than 2^{-n-1} to ensure count rounding. Ordinary relative accuracy or a small unweighted coefficient residual does not suffice. Exact modular arithmetic is therefore not casually replaceable by floating point.

17 Complex, quaternionic, and nonlinear extensions: relevant limits

The source's factorization and reordering require an associative commutative coefficient algebra. Complex coefficients are compatible with that algebraic condition, but exact bit complexity still needs an explicit finite representation. A complex phase encoding does not reduce the number of assignments or certify cancellation for free.

Quaternion-valued Ram channels in corrected TR60 and RMT extension TR3 require specified left/right multiplication, physical metrics, and ordered operators [18]. They are useful for vector or spinor observables, but substituting quaternions for the scalar coefficient ring here invalidates unrestricted factor reordering: $ij = k$ whereas $ji = -k$. Realification represents a fixed linear quaternion operation by a real matrix and keeps composition ordered. It does not make those matrices commute, reduce the network's width, or prove an exact-counting speedup. Octonions additionally require explicit parentheses; an associative scalar-ring proof cannot be imported unchanged.

Similarly, TR4's nonlinear range-consistency theorems [19] explain how to avoid false roots in transformed systems. They do not guarantee exhaustive isolation of every root of a general Boolean polynomial system. A local Newton certificate, an invertible projected Jacobian, and a small residual each answer a different question from the exact cardinality in (14). For this source, the scalar finite-algebra branch of Ram Calculus is substantially more directly relevant than its hypercomplex or continuous nonlinear PDE branch.

18 Exact tests and reproducible evidence

The accompanying program `eth-support/exact_identity_tests.py` uses Python integers and `fractions.Fraction`, with random seed `20261026`. It writes `eth-support/test-results.json`. No external private data or numerical fitting is used. The tests are small enough to compare against direct enumeration, which is a reference method here, not the proposed asymptotic implementation.

Table 4: Executed exact checks; all passed.

Check	Cases	What the equality establishes
Boolean rational polynomials	132	Direct diagonal sum, original subset-pair formula, aggregated formula, and generating-jet evaluation agree for $1 \le n \le 6$, $0 \le q \le n+1$.

Check	Cases	What the equality establishes
Weighted support filtration	120	The weighted subset aggregation equals the direct sum for weights in $\{1, 2, 3\}$ and varied budgets.
Shared-fiber generating jets	24	Two groups of three bits with cross-group sum dependence give identical global and factored coefficient vectors.
Rational Hermite recovery	36	Exact reconstruction with two unequal multiplicities, degrees one through nine.
Parity lift and global truncation	360	Modulus-amplifying indicators, quadratic remainder expansions, and common-order products agree modulo 2^ℓ, for $1 \le \ell \le 8$.
Cauchy scaled-integrality rows	1,168	All slot pairs and nodes for $L = 2, 4, 8$ have odd-denominator scaled coefficients; origin identity and cross-group divisibility hold in the checked configurations.
Tapered Hermite configurations	2	Exact weight valuations give Table 3; ten further polynomial reconstructions agree with the true origin value.
Negative controls	4	Detect invalid early functional application, rank-only state counting, floating cancellation, and a false derivative rule in a truncated ring.

18.1 A complete small SAT deformation

Consider

$$F = (x_1 \vee x_2) \wedge (\neg x_1 \vee x_3).$$

Its satisfaction polynomial is $(x_1 + x_2 - x_1x_2)(1 - x_1 + x_1x_3)$. Replacing each variable by $(1 - T)x_i$ and summing over all eight assignments gives, exactly,

$$H(T) = 4 - 3T + T^2 - T^3 - T^4. \tag{62}$$

Thus $H(0) = 4$, matching direct counting, and

$$H(1 + z) = -8z - 8z^2 - 5z^3 - z^4.$$

Reconstruction at zero gives $8-8+5-1 = 4$. At $T = 2$, however, $H(2) = -22$: deformed evaluations are not nonnegative counts. This example checks the algebra but does not meet the asymptotically useful $R < Q$ design conditions and is not presented as a fast solver.

18.2 The exact call counts, not elapsed-time claims

The large entries in Table 5 were computed by binomial arithmetic; those many calls were *not executed*. The table quantifies the proven loop reduction, not observed runtime or memory. In particular, the $n = 80$ entry remains enormous and says nothing about practical utility.

n	q	Original C_q	Aggregated B_q
20	2	801	211
40	4	1,544,481	102,091
80	8	7,847,456,478,273	32,490,460,747
6	6	729	64

Table 5: Combinatorial counts evaluated exactly from the formulas.

18.3 Failure cases that must remain visible

Four exact controls illustrate specific invalid substitutions.

1. In $\mathbb{Q}[w]/(w^2)$, multiplying first gives $w^2 = 0$; evaluating each retained factor at one first gives one. A linear final functional must not be treated as a ring homomorphism.
2. An eight-coordinate rank-one binary-weight row has 256 distinct Boolean measurements. Low rank need not mean a small state space.
3. Integer arithmetic gives $(2^{60} + 1) - 2^{60} = 1$, while binary64 floating arithmetic gives zero. Exact cancellation needs a precision certificate or exact arithmetic.
4. Canonical differentiation of the zero product z^2 modulo z^2 is zero, while a purported Leibniz calculation yields $2z$. Truncated coefficient multiplication is not compatible with that derivative.

Another analytic obstruction, used in the corpus, is moment aliasing: uniform even- and odd-parity distributions have identical proper-coordinate marginals but disjoint supports. Low-order assignment moments cannot identify arbitrary constraints. None of these examples disproves the source, which uses more carefully specified objects.

19 Revision 2: matched implementation, runtime, and memory

19.1 What is implemented and what is not

Revision 2 goes beyond the earlier identity tests: it implements the source's Section 3 restricted subset-pair routine and the aggregated routine of Section 11 on identical exact inputs. Both compute all required jets and reconstruct the entire Hermite polynomial, whose constant term is checked against independent exhaustive CNF counting. The implementation is original Python code derived from the published formulas, not an OpenAI-supplied executable and not a timing of its Lean development. The paper directory currently contains its README, PDF, and four TeX build files; the checked PDF and main TeX Git hashes match the versions already analyzed in this report.

The fixtures satisfy the finite Section 3 interface, including $R < Q$, integer summary bounds, row-kernel certificates, and exact denominator control. They are *not* outputs of the source's general Section 4 asymptotic construction. The first 16 are deliberately small, structurally easy CNFs with bounded cross-group interfaces. Their factors already separate at $T = 0$, so a direct fiber calculation is available without jets. This stronger conventional control is measured separately below. Five additional node-varying fixtures remove that particular shortcut: their factor assignments change between interpolation nodes, and their clauses couple unmeasured variables on both sides at zero. The comparison answers whether aggregation helps the specified jet routine on matched instances,

not whether it improves the fastest SAT solver or the source's sharper Appendix D bound.

19.2 Construction and exact certificates for every fixture

In the first suite, the n declared variables are split into two contiguous groups, each with one distinguished boundary bit b_g. A clause assigned to group g may contain arbitrary variables of that group and the other group's boundary bit, but no other off-group variable. Its summary is $C_g x_g = x_{b_g}$, so $D = 1$, the total number of rows is $c = 2$, and $\tau = 1/n$ satisfies $c = 2\tau n$ and $\tau \le 1/2$. The group sizes satisfy the source's required bound.

For every clause-variable incidence, the implementation constructs

$$P_{f,v}(T) = e_v + T a_{f,v}, \qquad \operatorname{supp} a_{f,v} \subseteq S_{\sigma_f} \cup \{b_{1-\sigma_f}\}. \tag{63}$$

The coefficients on that support are seeded signed values in $\{-2, -1, 1, 2\}$; the rational fixture divides them by two. Thus $R = 1$, and $Q \ge 2$ ensures $R < Q$ and that the source's denominator factor $(R-1)!Q! = Q!$ clears the coefficients. Their integer heights meet the stated finite coefficient bound. At zero the rows restore e_v. At every node $t = 1, \dots, Q$, every off-group coordinate other than the measured boundary bit is exactly zero. This proves the row-kernel condition; the code checks it for every incidence and node.

For clause widths k_f, the code uses the valid degree bound $d = \sum_f k_f$, $q = \lfloor d/Q \rfloor$, and $J = q + 1$. Hence $QJ > dR$. The satisfaction factor for a positive literal uses the unsatisfied factor $1 - y_v$, and a negative literal uses y_v; one minus the product of unsatisfied factors is the clause indicator. Products are computed with exact integers, or `Fraction` objects for the rational fixture. Hermite reconstruction uses exact rationals for both methods.

The second suite uses $Q = 2, R = 1$ and the same boundary-bit measurements, but assigns every factor to group zero at node one and group one at node two. For a nonboundary variable v, the replacement is

$$P_v(T) = \begin{cases} (1 - T/2)e_v + TB_v, & v \in S_0, \\ (1 - T)e_v + TB_v, & v \in S_1, \end{cases}$$

where B_v is a small signed row supported only on the two boundary bits. Boundary variables use $e_v + TB_v$. At $t = 1$, the nonboundary coordinates of S_1 vanish; at $t = 2$, those of S_0 vanish. This proves the two nodewise kernel conditions, while restoration at zero is exact. Each clause contains nonboundary variables from both groups, so the constant-partition direct fiber shortcut is not certified there. These are finite input-specific designs, not a new asymptotic construction. With only two or three clauses in the second suite, elementary clause inclusion–exclusion is another small conventional counting alternative. Node-varying separation tests the mechanism more directly; it does not make these fixtures hard SAT instances.

Table 6 describes all 21 fixtures. Each suite was fixed before its own run. The node-varying suite was added after the independent audit identified the all-T separation of the first family; it is an explicitly disclosed follow-up, not a hidden replacement of unfavorable trials. A preliminary two-repetition pilot on S08 is retained separately and excluded from the reported statistics. Empty and contradictory formulas, $q = 0$, rational rows, higher jet orders, and a $q \ge n$ case are retained rather than screened out for poor performance. No result is extrapolated to larger input sizes.

Table 6: Certified fixtures and exact counts. Calls are totals over all nodes. S cases use fixed layouts; N cases use node-varying factor assignments.

ID	n	m	d	Q	q	Z	Calls O/A
S01	4	0	0	2	0	16	2/2
S02	6	2	2	3	0	0	3/3
S03	6	2	4	5	0	32	5/5
S04	4	2	4	3	1	10	27/15
S05	6	3	6	4	1	30	52/28
S06	8	3	6	4	1	96	68/36
S07	6	2	6	3	2	50	219/66
S08	8	2	6	3	2	192	387/111
S09	10	2	6	3	2	784	603/168
S10	6	2	6	2	3	48	466/84
S11	8	3	9	3	3	168	1,731/279
S12	6	2	6	3	2	50	219/66
S13	10	3	9	3	3	704	3,483/528
S14	8	3	9	2	4	168	3,394/326
S15	4	4	12	2	6	8	162/32
S16	8	6	12	5	2	24	645/185
N17	4	2	4	2	2	8	66/22
N18	6	2	4	2	2	40	146/44
N19	6	2	6	2	3	52	466/84
N20	8	2	6	2	3	192	1,154/186
N21	6	3	9	2	4	44	946/114

19.3 Matched computational work and fairness controls

Both algorithms use the same factor evaluator, exact polynomial convolution, full measurement box $\{-1, 0, 1\}^2$, node order, local assignment order, and full Hermite elimination. For each measurement tuple they first scan for representatives and then scan for local restricted sums, as in the source proof. Unattainable tuples are skipped. There is no precomputed table of all global assignment values in either timed solver. Exhaustive enumeration is an untimed independent oracle.

The original loop is ordered with U outermost so its fixed correction rows can be prepared once per U, exactly as for aggregation. It still executes every pair (E, U), with $U \subseteq E$ and $|E| \leq q$. This benign loop interchange is an optimization of the reference implementation, not an application of the aggregated identity. The aggregate implementation performs one restricted fiber calculation per U over the enlarged (z, w) coefficient array. The final alternating functional is applied only after group multiplication. Both kernels skip zero scalar products; neither uses global memoization, rank compression, weighted-support pruning, or target-only Hermite recovery.

These controls isolate the change under study. They do not establish that either Python implementation is optimal. More sophisticated shared work across restricted calls could improve the original algorithm too.

19.4 Measurement protocol and reproducibility

The main suite uses seven matched repetitions per fixture. Each method runs in a fresh sequential Python subprocess; their order within each repetition is randomized with seed `20261026`. Wall

time includes row construction, validation, Hermite-matrix construction, all node jets, and full polynomial recovery. Process launch, imports, input JSON reading, oracle construction, post-run comparisons, and result serialization are excluded. Preparation, kernel, and recovery times are also recorded separately. The runtime samples do not use `tracemalloc` or the optional detailed arithmetic counters.

The machine is Windows 11 with an Intel Core i7-1165G7, four physical cores/eight logical processors, and CPython 3.12.14. Execution is single-process per solver, without CPU-affinity pinning or control of background activity and power management. The observed timing spread is substantial. Medians and ranges are descriptive measurements, not statistical confidence intervals or universal speed guarantees.

Memory is measured in separate fresh workers. The primary measure is peak Python-traced allocation from preparation through recovery; it excludes the already loaded interpreter and input object, but includes the constructed layout, coefficient arrays, and recovery data. Windows peak working set is also recorded and includes the interpreter and tracing overhead. Additional phase profiles separate kernel allocations from the common recovery peak. Kernel incremental peak means peak traced allocation during the kernel minus the live traced allocation immediately after preparation, not an operating-system RSS difference.

Raw instances, every timing sample, exact oracle coefficients, diagnostic counters, code hash, and memory profiles are retained under `eth-benchmark-v2/matched-run01/` and `eth-benchmark-v2/matched-run02/`. The original run used the frozen `benchmark.py`; the second used `benchmark_v2.py`, which adds node-varying factor assignments and rejects noncontiguous partitions unsupported by its bit embedding. Both routines within each run use exactly the same code version. The first suite's contiguous fixtures were unaffected by that validator limitation. The new entry point also rejects Python's assertion-disabling `-O` option. The supported node-varying fixture generator always uses $Q = 2$ and complete assignments for both nodes. The frozen validator is not a general input-schema validator: a manually edited incomplete node map can pass its fallback check and fail during jet evaluation. Imported calls also require ordinary, nonoptimized Python because the assertion guard is at the CLI entry point. Neither limitation is exercised by the recorded invocations. They should be hardened before broader reuse. `phase_profile.py` and `phase_profile_v2.py` supply the separately executed phase and direct-fiber controls. The README gives commands that require new output directories and refuse to overwrite prior measurements.

19.5 Exact agreement and measured runtime

All 21 fixtures produced identical exact counts under both methods. More strongly, every jet coefficient and every reconstructed polynomial coefficient agreed with an independently enumerated deformation polynomial; unused high-degree recovery coefficients were zero. The direct Boolean truth-table count independently checked the constant term. All 294 timed solver executions, 42 separate memory runs, and 42 diagnostic runs passed these comparisons. The code's initial self-test also passed 100 convolution checks and 24 matched method checks, and rejected an intentionally invalid off-group layout. The deformation oracle has independent enumeration and convolution, but shares row construction and the row-dot-product helper with the tested implementation. Its independence is therefore not total; the explicit layout proofs and independent static review address that shared part. A separate post-timing regression run passed 100 mixed (z, w) convolution checks, 20 method/oracle comparisons including odd group sizes and an empty clause, and four alternating node-reentry checks. It also confirmed rejection of two corrupted node rows, a noncontiguous partition, and the CLI's optimized-Python invocation. Those runs are not added to the timing

statistics; their records are in `eth-benchmark-v2/regression-results.json`.

Table 7: Matched pipeline wall time in milliseconds: seven fresh-worker samples per method. O is original, A is aggregated; ratio is median O divided by median A. Ranges are observed minima and maxima.

ID	O median	A median	O/A	O range	A range
S01	0.466	0.539	0.87	0.259–0.612	0.312–0.626
S02	1.069	1.135	0.94	0.932–1.331	1.062–2.115
S03	3.804	3.779	1.01	1.913–5.723	2.121–4.487
S04	4.843	3.756	1.29	4.444–6.908	3.283–4.475
S05	18.564	12.461	1.49	17.223–21.319	11.570–15.463
S06	48.059	33.333	1.44	42.129–195.158	25.539–115.111
S07	196.037	70.523	2.78	143.456–268.470	38.056–94.696
S08	250.717	86.236	2.91	228.720–309.371	80.580–187.967
S09	1,889.863	546.129	3.46	878.564–2,927.368	283.576–852.221
S10	398.996	119.954	3.33	250.171–455.454	89.963–195.760
S11	3,042.474	607.451	5.01	1,878.445–3,416.439	238.840–828.481
S12	907.584	354.187	2.56	540.101–1,539.021	298.076–543.378
S13	9,833.618	1,643.942	5.98	7,881.177–11,138.673	805.800–3,016.550
S14	3,895.762	291.173	13.38	1,945.820–5,340.162	261.424–455.089
S15	140.890	38.303	3.68	102.853–156.422	35.260–55.006
S16	1,987.977	655.549	3.03	1,790.097–3,074.709	608.762–843.923
N17	50.535	24.907	2.03	23.570–59.050	10.283–92.737
N18	208.674	131.844	1.58	146.835–308.129	72.790–162.887
N19	880.379	225.510	3.90	709.463–1,015.560	188.610–706.896
N20	3,488.846	501.599	6.96	2,354.011–5,856.887	373.619–917.970
N21	2,667.340	473.529	5.63	2,419.055–3,238.591	337.451–816.178

Every one of the 18 cases with $q > 0$ had a shorter aggregate median in these runs, with median ratios from 1.29 to 13.38. The largest ratio occurs at S14; it is descriptive and not a forecast of scaling. The two smallest controls S01 and S02 were slower with aggregation, while S03 was nearly tied. These three have $q = 0$ and exactly the same number of restricted calls. The wide raw ranges, particularly on the busier runs, preclude a strong claim based on the precise ratio digits. Exact call and assignment-visit reductions corroborate the mechanism independently of wall-time noise.

The gains are *observed finite implementation gains*, in contrast to the earlier report's call-count-only evidence. They do not strengthen the asymptotic theorem. The $q = 0$ cases have no pair-loop reduction, so the aggregate representation can be pure overhead. The $q \geq n$ case remains an exponential enumeration regime despite its finite speed ratio. The largest tested fixture has only ten variables. The suite is not a sample of difficult unrestricted random SAT instances.

19.6 Memory, arithmetic work, and the structural control

Table 8: Memory in KiB (1 KiB = 1,024 bytes). Whole-run peaks come from separate tracing workers. Kernel increments are phase peaks minus the live post-preparation baseline, not allocation-exclusive coefficient sizes.

ID	Whole O	Whole A	Kernel increment O	Kernel increment A
S01	4.48	4.48	2.66	2.51
S02	6.73	6.66	3.61	3.27
S03	9.10	9.12	3.86	3.55
S04	9.51	9.77	3.86	3.92
S05	14.52	14.63	4.52	4.43
S06	15.17	15.25	4.62	4.52
S07	16.53	16.69	4.56	5.05
S08	17.11	17.27	4.62	5.21
S09	17.62	17.65	5.05	5.35
S10	13.59	14.05	4.63	6.29
S11	25.62	25.85	5.37	7.25
S12	18.78	18.94	6.00	7.52
S13	26.29	26.37	5.75	7.31
S14	20.41	20.96	5.54	8.65
S15	29.78	30.73	6.11	11.34
S16	38.55	38.38	6.28	6.85
N17	11.29	11.67	4.93	5.98
N18	11.37	11.64	4.95	5.88
N19	14.91	15.37	5.73	7.45
N20	15.23	15.65	6.17	8.70
N21	21.21	21.84	6.98	12.19

Whole-pipeline traced peak ratios A/O range from 0.991 to 1.033; there is no uniform memory reduction. Common layout and dense Hermite-recovery allocations can dominate the peak. For $q > 0$, observed kernel incremental-peak ratios range from 0.98 to 1.86. These are net phase measurements, not a count of just the extra generating coefficients; released buffers and stored output jets affect them. The logical coefficient capacity per ring element grows from J to at most $J(q+1)$, so a speed benefit is compatible with a larger kernel memory footprint. Untraced timing workers had lifetime process peaks between 24.56 and 25.24 MiB, including interpreter and imports. Traced-worker process peaks, retained in the raw data, also contain tracer overhead; neither process measure is an isolated solver-workspace bound.

Table 9: Diagnostic work on selected cases, measured outside timing. Products means nonzero convolution coefficient products only; bits are maximum inspected kernel coefficient numerator/denominator lengths.

ID/method	Assignment visits	Factor evaluations	Products	Bits N/D
S03 O	595	160	478	19/1
S03 A	595	160	478	19/1
S08 O	86,301	14,400	109,927	22/1
S08 A	24,753	4,320	43,732	22/1
S13 O	1,501,173	291,456	3,081,055	37/1
S13 A	227,568	47,376	644,412	37/1
S14 O	756,862	110,592	1,400,047	29/1

ID/method	Assignment visits	Factor evaluations	Products	Bits N/D
S14 A	72,698	12,492	205,851	29/1
S15 O	10,854	2,304	36,576	34/1
S15 A	2,144	576	13,276	34/1
N20 O	257,342	32,640	307,459	22/4
N20 A	41,478	5,760	76,747	22/4
N21 O	112,574	16,512	215,200	28/5
N21 A	13,566	2,472	43,290	28/5

The diagnostic scalar-product count covers nonzero products in the implemented polynomial convolutions, not every CPU instruction or the rational operations in Hermite elimination. Reported coefficient bit lengths describe inspected kernel polynomial values, not a proof of a uniform bit bound for all instances. Traced peaks are measured in separate runs and should not be conflated with the timed-run working set.

Because the first 16 fixtures already satisfy separation at zero, one direct shared-summary sum computes the count without jets or Hermite reconstruction. A separate control executes exactly that calculation, using the same local evaluator with $J = 1$. It matches every applicable exact count. The control is not applied to the five node-varying fixtures because its separation premise fails there. It is measured in three later fixed-order runs, including preparation of an unnecessary Hermite matrix, and is not part of the seven-repetition paired timing protocol. The direct-control medians range from 0.145 to 4.390 ms (all individual samples: 0.139–9.446 ms). They are shorter than the corresponding aggregate medians in every first-suite fixture, although these later, unpaired measurements are not suitable for a precise speedup estimate. The control even includes unnecessary Hermite-matrix preparation, so it is not a fully optimized direct implementation. This is an important limitation, not a failed correctness test: aggregation can improve the chosen jet routine while an easier structural method is better for this particular input family.

19.7 What has advanced and what remains open

Revision 2 establishes a reproducible implementation-level comparison, not merely agreement of abstract identities. It demonstrates the local call reduction in actual streamed fiber calculations, measures its coefficient-state cost, and exhibits both gains and overhead cases. An independent review of the new implementation and fixture assumptions is included in `eth-benchmark-v2/independent-audit.txt`, with the node-varying follow-up in `independent-node-audit.txt`. Its scope is static code and finite-layout reasoning, not independent execution or a global proof of the source paper.

The next substantial step is to generate instances from the source's actual general construction at feasible parameter values, or develop a different nontrivial design with equivalent certified guarantees. Additional baselines should share more work across the original calls and use optimized exact arithmetic. Larger controlled benchmarks would need power/load controls, more independent instances per parameter setting, and a complete account of construction cost. Incorporating the precision certificates into Appendix D remains unimplemented here. No result establishes a stronger general SAT exponent, an advantage unique to Ram terminology, or an optimality lower bound.

20 Revision 3: degree-aware recovery and certified quotient compression

20.1 What the 13-volume investigation adds

The 627-topic inventory and targeted mathematical reading are documented in Appendix C. The expanded entries often make explicit an assumption or remainder that was only implicit in a shorter corpus report. This is valuable, but explicit exposition is not by itself a new theorem. We distinguish (i) retained results already proved in the preceding report, (ii) classical consequences made explicit by the books, and (iii) additional application-level deductions proved below. No worldwide priority claim is inferred from a result's absence in the preceding report. The useful Ram contribution is the connected treatment of exact representation, information retained, physical norms, defects, and the conditions under which a finite computation answers the original question. The same proved bounds are available by conventional methods under the same hypotheses.

The additional deductions are exact target-identifiability tests, sharp finite degree-aware precision budgets, and orbit compression under proved permutation symmetry. The first two attach to Hermite recovery; the third attaches to the Section 3 Boolean-jet routine. They do not replace the source's general circuit/measurement construction or Appendix D recursive tensor contraction. The auxiliary-PDE question is unchanged: no such PDE occurs in the inspected ETH manuscript.

Corpus connection	Additional deduction	Essential limit
M225, R106, R112, R69	Row-space target recovery and sharp 1/4/7-bit losses for three certified degree classes	Fixed finite grid; the degree-three case is already licensed by the local source lemma
M310, R117, R198	One Boolean-jet call per certified permutation orbit	Equivariance of the entire deformed family is required
R103, R111, R135, R152, R189	Exact closure and complete coefficient contracts; explicit failure witnesses	Reencoding a state does not remove distinguishable information

20.2 Recovering the target is weaker than recovering every coefficient

M225 treats confluent Hermite interpolation, while R106 and R112 describe compatible displaced polynomial jets and full-state rank. R69 and R165 motivate goal-sensitive measurement choices, but their real-norm error and variance formulas are not modular precision theorems. This makes the degree-freedom remark in Section 16 into a precise usable criterion. Let V be a specified finite-dimensional rational polynomial space, with coefficient vector $c \in \mathbb{Q}^d$. Let the available exact measurements be $h = Ac \in \mathbb{Q}^m$ and the desired scalar be ℓc. The matrix A may be rectangular or rank deficient. In the ETH application the rows are scaled Hermite jets, and ℓ evaluates at zero.

Theorem 20.1 (Target identifiability and a certificate for deleting jets)**.** *The target is determined by the measurements for every $c \in \mathbb{Q}^d$ if and only if*

$$\ker A \subseteq \ker \ell, \qquad \textit{equivalently} \qquad A^T\lambda = \ell^T \textit{ for some } \lambda \in \mathbb{Q}^m.$$

For a selected subset S of the measurement rows, deletion of the other rows preserves exact target recovery if and only if $A_S^T\lambda_S = \ell^T$ is solvable. If it is not solvable, a vector $c_* \in \ker A_S$ with $\ell c_* \neq 0$ is an explicit impossibility witness. These tests concern exact data and are separate from precision or runtime optimization.

Proof. Equal measurements of c and c' mean $c - c' \in \ker A$. Their targets agree exactly when ℓ annihilates this kernel. Finite-dimensional row-space/kernel duality gives the displayed linear equation. A solution gives $\ell c = \lambda^T h$ for every c. Applying the same argument to A_S proves the deletion criterion and its witness statement. In particular 0 and c_* are indistinguishable from the retained data when deletion fails. □

There is an important limit for ordinary prefix jets. Suppose the retained nodes are all nonzero, with consecutive derivatives $j = 0, \dots, r_t - 1$, and put $N_S = \sum_t r_t$. On the full space $\deg F \le D$, these data determine $F(0)$ if and only if $N_S > D$. If $N_S \le D$, the polynomial $\prod_t (T - t)^{r_t}$ has all retained jets zero but a nonzero constant term. Conversely, if $N_S > D$, any polynomial in the measurement kernel is divisible by that product and has smaller degree, hence is zero. Target-only recovery therefore does not permit arbitrary deletion of ordinary Hermite data; its advantage comes from the actual degree class, a target-specific functional, or other certified structure.

For integer nodes and scaling, the Hermite rows on the monomial basis $1, T, \dots, T^D$ are

$$A_{(t,j),k} = \begin{cases} \binom{k}{j} t^{k-j} \alpha^j, & k \ge j, \\ 0, & k < j. \end{cases}$$

Every proposed degree reduction must be proved for the *actual* polynomial being recovered. Observing small high-degree coefficients numerically is not a certificate. When more rows than needed are available, the family of target formulas is the affine solution set of $A^T \lambda = e_0$. This is the appropriate place to choose cheaper channels or better precision budgets, without requiring an inverse for the entire ambient Hermite space.

For any chosen formula, the already proved condition $b_i + v_2(\lambda_i) \ge p$ remains necessary and sufficient for a uniform guarantee against independently variable errors of valuations at least b_i in each retained coordinate. A vanishing weight needs no measurement. Row-space membership does not by itself minimize the valuation loss or the cost of obtaining the retained rows.

Theorem 20.2 (A precision-plan criterion over the two-local rationals)**.** *Let $\mathcal{O} = \{a/b \in \mathbb{Q} : b \text{ odd}\}$, and suppose the exact jet coordinates and target are integral in this ring. For proposed integer channel precisions b_i and desired output precision p, an exact rational linear decoder on the specified coefficient space can guarantee the target modulo 2^p for all independent coordinate errors in $2^{b_i}\mathcal{O}$ if and only if*

$$\ell^T \in A^T \operatorname{diag}(2^{p-b_i}) \mathcal{O}^m. \tag{64}$$

When A and ℓ are integral at two, the least nonnegative common extra loss among exact linear decoders is

$$\delta_* = \min\{\delta \ge 0 : 2^\delta \ell^T \in A^T \mathcal{O}^m\},$$

provided the target-row equation is solvable.

Proof. A fixed decoder is exact when $A^T \lambda = \ell^T$. Its output error always belongs to $2^p\mathcal{O}$ exactly when $\lambda_i \in 2^{p-b_i}\mathcal{O}$ for each coordinate. Sufficiency follows by summing integral errors; necessity follows by taking a single nonzero coordinate error 2^{b_i}. Substituting $\lambda = \operatorname{diag}(2^{p-b_i})\mu$, $\mu \in \mathcal{O}^m$, gives (64). For $b_i = p + \delta$, multiply the target equation by 2^δ. A rational decoder has finite denominator valuations, so the minimum exists whenever one decoder exists. □

This extends the fixed-decoder certificate to all exact decoders on a known space. It is a classical module-membership criterion, not a new general interpolation theorem or a proved cheapest mixed-precision schedule. Smith normal form over $\mathcal{O}$ can decide membership after clearing denominators.

A nilpotent coefficient algebra is not a discrete valuation ring: the scalar certificate is used coefficientwise, as in the source, not by asserting a Smith form over that algebra. No general Smith-form optimizer was implemented in the new tests. R78's real-norm mixed-precision iteration and R165's Gram-variance formula are useful conceptual comparisons but are not substitutes for this divisibility argument. R43 describes Gaussian orthogonal Hermite panels, not confluent Hermite interpolation; the shared word does not make its formulas applicable here.

20.3 A sharp finite degree-aware precision improvement

Use the same local six-channel grid as the first precision example in Section 16: nodes $t = -1, 1$, scaling $\alpha = 4$, and available jets $h_{t,j} = [z^j]F(t + 4z)$ for $j = 0, 1, 2$. The following formulas are exact on the indicated degree classes:

$$\deg F \le 1: \quad F(0) = \tfrac{1}{2}(h_{-1,0} + h_{1,0}), \tag{65}$$

$$\deg F \le 3: \quad F(0) = \tfrac{1}{2}(h_{-1,0} + h_{1,0}) + \tfrac{1}{16}(h_{-1,1} - h_{1,1}), \tag{66}$$

$$\deg F \le 5: \quad F(0) = \tfrac{1}{2}(h_{-1,0} + h_{1,0}) + \tfrac{5}{64}(h_{-1,1} - h_{1,1}) + \tfrac{1}{128}(h_{-1,2} + h_{1,2}). \tag{67}$$

One verifies them by substitution of $F(T) = T^k$ for every k in the specified class; linearity then proves the all-polynomial claim. Odd terms cancel in the symmetric combinations. The remaining even terms determine the coefficients in the displayed triangular system.

Proposition 20.3 (Optimal common losses for this grid and these classes)**.** *For exact rational linear target formulas on the six available channels, the smallest nonnegative common extra-precision losses for degrees at most* $1, 3, 5$ *are respectively* $1, 4, 7$ *bits. Thus a certified degree-three class reduces the degree-five certificate by three bits, and a certified degree-one class by six bits, in addition to deleting two or four unnecessary channels.*

Proof. Equations (65)–(67) give the upper bounds: all weighted errors are in $2^p\mathbb{Z}_{(2)}$ when every retained jet is known modulo $2^{p+1}, 2^{p+4}, 2^{p+7}$, respectively. For the lower bounds use

$$F_m(T) = (T + 1)^m(T - 1)^{m+1}, \qquad m = 0, 1, 2.$$

These have degrees $1, 3, 5$ and odd constant term. Their six scaled jets are all divisible by 2^{1+3m}. Direct expansion at -1 and 1 gives at least one jet of valuation exactly $1 + 3m$. If every coefficient of an exact target formula had valuation strictly greater than $-(1 + 3m)$, its value on F_m would be even, contradicting the odd target. Hence a coefficient with at least the claimed denominator loss is unavoidable, even if all six channels are retained. This is optimality within the stated exact linear recovery and independent-error contract, not a lower bound for all counting algorithms or for correlated-error encodings. □

An indistinguishability argument strengthens the lower bound for this data contract. The admissible polynomials 0 and $2^{p-1}F_m$ have identical six-channel residues modulo $2^{p+1+3m-1}$, but their targets differ modulo 2^p. Therefore one fewer common bit cannot suffice even for a nonlinear decoder using only these residues. This does not assert that the witnesses are realized by every source tensor network; additional tensor-realizability information changes the data contract.

For completeness, the optimal common losses for all six degree classes $D = 0, 1, 2, 3, 4, 5$ are $0, 1, 3, 4, 6, 7$. Besides the odd-degree formulas, upper weights in the six-channel order

$(-1,0),(-1,1),(-1,2),(1,0),(1,1),(1,2)$ are

$$\lambda^{(0)} = (1,0,0;0,0,0),$$
$$\lambda^{(2)} = (1/4,0,0;3/4,-1/8,0),$$
$$\lambda^{(4)} = (11/16,1/8,1/64;5/16,-1/32,0).$$

They are checked on the monomial bases. The witnesses 1, T^2-1, and $(T^2-1)^2$ have odd constants and minimum six-jet valuations $0,3,6$, proving the matching lower bounds. In particular, forcing weights to be symmetric can miss the degree-four optimum.

The distinction from revision 2 is substantive but local. Revision 2 computed the exact weights for the full degree-five space and obtained seven extra bits rather than the source's conservative 64-bit allowance. For the particular source-lemma fixture $a=1, L=2, u=1, M=1$, the degree bound $M(2L-1)$ is already three. Thus four extra bits are certified for that same local configuration, without assuming an additional degree reduction. Degree one requires an additional certificate. This reads the surplus channels against the actual source degree bound, rather than against the larger canonical interpolation space. For a genuinely degree-five polynomial the seven-bit result remains unchanged.

These small parameters are not a complete instance of all large-size packing conditions in the source's recursion. Creation-time canonical lifts must remain fixed, and a child precision must still obey any inherited strictly increasing requirement. Moreover, Appendix D obtains the entire polynomial at each active node using $M(2L-1)+1$ evaluations. Deleting two jet channels does not by itself delete those evaluation calls. Smaller valid node precision can reduce arithmetic cost, but no full recursive runtime improvement has been measured here.

Misidentifying the degree gives immediate failures. Applying (65) to T^2 returns 1 instead of 0; applying (66) to T^4 returns -1; applying (67) to T^6 returns 1. All three failures are exact, not floating-point effects.

20.4 Symmetry can remove more Boolean-jet calls, but only with a certificate

M310 (Krawtchouk polynomials), R117 (quotient closure), and R198 (lumpability) suggest looking for exact permutation orbits in the aggregated U loop. The relevant fact is symmetry of the complete deformed family, not just equal Hamming weights of the selected sets.

Theorem 20.4 (Orbit aggregation of the existing Boolean identity)**.** *Let a permutation group $G \le S_n$ act on assignments and subsets. Assume the family in Theorem 11.1 obeys*

$$p_{gx}(gX) = p_x(X) \qquad (g \in G)$$

as an exact polynomial identity in the declared coefficient algebra. Put

$$S_U = \sum_{x_U=1} w_q(|x|,|U|)p_x(\mathbf{1}_U).$$

Then $S_{gU} = S_U$, and the count or weighted target is

$$\sum_x p_x(x) = \sum_{\mathcal{O}\subset\{U:|U|\le q\}} |\mathcal{O}| S_{U_\mathcal{O}},$$

where the sum is over G-orbits and one representative is chosen in each orbit. For full permutation symmetry this needs only $\min(q,n)+1$ representative restricted calls. The same conclusion holds coefficientwise for the truncated z,w jets when the full deformed $G_x(X,z)$ is equivariant.

Proof. Replace x by gy in S_{gU}. This is a bijection between the restricted assignments, preserves cardinalities, and transforms $p_{gy}(\mathbf{1}_{gU})$ into $p_y(\mathbf{1}_U)$. Thus the summands agree. Group the finite U sum in Theorem 11.1 by its orbits. For S_n, subsets of each fixed size are one orbit, with cardinality $\binom{n}{|U|}$. All manipulations preserve integer multiplicities; no averaging by a possibly noninvertible group order is required. □

The permutation condition must hold at the actual interpolation node, for its factor assignment, summary measurements, correction jets, and output functional. Generic labelled 3-CNF formulas need not have such symmetry. Discovering a useful group and certifying equivariance has a cost. Even after call reduction, each representative call can remain expensive; this is a restricted-instance refinement, not a new general SAT exponent. The source's general construction has not been shown to satisfy the condition.

20.5 Why a small summary or Hamming weight alone is insufficient

In a finite state algebra let J be a retained linear measurement map and T_a the operator for multiplication by a factor a. There is an exact coarse update $\widehat{T}_a$ for *every* state, with $JT_a = \widehat{T}_a J$, exactly when $\ker J$ is invariant under T_a. Indeed, kernel invariance makes $\widehat{T}_a(Jv) = JT_a v$ well-defined; the converse follows by applying the intertwining to $v \in \ker J$. For a sequence of factors, this condition is needed for each update, and the final target must factor through J as well. This is R117 applied to the paper's exact finite algebra. A complete exact quotient may offer no compression: if multiplication and the target distinguish every state, all of those distinctions must remain represented.

On two labelled bits, states $(1,0)$ and $(0,1)$ have the same Hamming weight but multiplication by x_1 distinguishes them. Thus a Hamming-weight summary is not closed for an arbitrary labelled factor. M310's exact Krawtchouk closure relies on the symmetry of its specific operators. It does not license replacing the source's carefully designed shared measurements by weight alone. R103 and R152 supply a related warning: one coordinate-line or total-sum check cannot certify all mixed coefficients of a multivariate jet. The full coefficient agreement retained from the v2 tests is therefore more informative than agreement of a single recovered count.

20.6 New exact tests and their limited computational meaning

The standard-library script `mathworld-v3-support/test_eth_refinements.py` checks all basis monomials for the three Hermite formulas, their precision certificates, and the three out-of-class failures. The separate lower-bound checks verify all six jets of each F_m. The script `mathworld-v3-support/test_degree_table.py` also checks all 21 basis monomials across degrees zero through five, the even-degree weights, and the residue-indistinguishability controls. Its record is `mathworld-v3-support/degree-table-results.json`. For the symmetry test it uses

$$p_x(X) = (1 + |x|)\left(1 + \sum_{i=1}^{n} X_i\right)^q.$$

This is an exactly symmetric, integer-valued, degree-q family. Direct assignment summation, the ordinary aggregated identity, and the orbit-reduced identity give identical *weighted sums*:

n	q	Exact weighted sum	Source pair calls	v2 U calls	Orbit calls
6	2	5,248	73	22	3
8	3	239,616	577	93	4
10	4	14,029,824	4,521	386	5

These are not SAT solution counts, and the reference implementation enumerates assignments inside the calls. The table tests identities and counts representative calls; it reports no measured wall-clock or memory advantage. Such a benchmark would require the same factorized implementation, symmetry-detection cost, and bit arithmetic on each method.

For a negative control, let $n = 4, q = 1$ and $p_x(X) = 1 + 3X_1 + X_2$. The true weighted sum is 48. Invalidly treating all singleton subsets as symmetric and using $\{1\}$ as their representative returns 112. A valid implementation must reject that compression, rather than accept an apparent reduction in work. The results are recorded in `mathworld-v3-support/eth-refinement-results.json`.

20.7 Incremental conclusions, costs, and remaining work

The preceding v2 runtime and memory results are retained without change. The new tests establish exact local identities, smaller certified precision on specified degree classes, and fewer representative calls on symmetric fixtures. They do not add a measured speedup on generic instances. Rank tests and exact target solves cost rational linear algebra; symmetry must be certified; a target-optimal formula may have expensive coefficients; and the source's recursive lifting and valuation contracts must still be obeyed.

A next implementation should derive certified degree bounds inside each actual recursive node, measure the complete cost of a selected target formula, and detect only symmetries that survive all correction and measurement maps. It should compare all coefficients and exact counts, not just floating-point target values, and retain negative controls like the ones above. These are concrete routes to additional local improvements, not completed improvements to the source's asymptotic exponent or an independent proof of its full ETH claim.

21 Revision 4: matched symmetry and degree-aware precision experiments

21.1 Scope and the two different original routines

Revision 4 implements the v3 reductions rather than inferring elapsed-time gains from operation counts. It conducts two separate experiments. The first compares the source's Section-3 subset-pair routine, the earlier aggregated Boolean-jet routine, and a certified orbit reduction, on exact CNF counting instances with the same complete polynomial-recovery output. The second compares the source's conservative tapered-Hermite precision allowance with smaller certified budgets in a local Appendix-D contraction calculation. They are not stages of one newly implemented general SAT solver. Combining their measured ratios would be invalid.

"Original" in the tables means an independent implementation of the specified source routine, not a benchmark of an OpenAI-supplied executable. The Section-3 kernel is imported unchanged from the frozen v2 benchmark. The modular contraction implements the displayed Cauchy deformation, evaluation/interpolation, and recovery steps from the pinned source in Section 4. The source's large-instance construction, recursive packet reduction, full finite-field parameter schedule, Lean

dependency closure, and claimed worst-case counting bound are not executed here. The v2 and v3 results earlier in this report are retained as historical results; the new measurements below have their own dated records.

21.2 Protocol, exact validation, and comparison contract

All timings are fresh-worker measurements using Python 3.12 on Windows 11, with an Intel Family 6 Model 140 processor and eight logical CPUs. Each case/method has seven sequential timing samples in seeded randomized, interleaved order. Two separate fresh workers measure peak traced Python allocations; another records diagnostic counters. Import time, JSON output, independent exhaustive or rational-oracle validation, and process startup are excluded from elapsed time. Preparation, certificates, orbit catalogue construction, arithmetic, and recovery are included. No CPU affinity is set; medians and observed ranges are descriptive, not statistical confidence intervals or forecasts of scaling. No separate benchmark runs in parallel with these measurements.

All worker outputs, not only a selected sample, are checked against an oracle. CNF tests compare the truth count, every jet coefficient, and every coefficient of the recovered deformation polynomial, including required zero padding. The oracle enumerates Boolean assignments and uses separate coefficient convolution, but shares the specified layout constructor. Accordingly, the layout certificate and mathematical derivation remain necessary; this is not two wholly independent implementations of the entire problem description. Modular tests compare all retained integer output coefficients and both complete local cubics against an independent `Fraction` polynomial computation, reducing the oracle coefficients at the actual working precision. Non-base extension-field coordinates must vanish. Checks are stronger than agreement of one final count.

The peak-memory figures are `tracemalloc` measurements of Python allocations during the pipeline. They include output arrays and preparation, not just coefficient storage, and do not count every native allocation. Inputs are deserialized before measurement, so their already-resident allocation is not part of the traced peak. Windows process working-set peaks are also retained in the raw records, but include interpreter/import history and, for memory runs, tracer cost. They are not isolated solver-workspace bounds. Source/case hashes and all individual samples accompany the report.

21.3 Certified symmetry of the complete finite layout

For each accepted permutation, the implementation checks the entire symbolic multiset of signed factors, including literal multiplicities, exact deformation-row coefficients, fixed boundary measurements, group membership, and the complete assignment of factors to groups at every interpolation node. It searches the subgroup generated by accepted adjacent transpositions inside each group's nonboundary coordinates. It does not claim to find every possible automorphism.

These checks are a sufficient certificate for the covariance used in Section 20.4. Permuting a restricted assignment gives a bijection preserving each factor, the correction rows, the number of free ones, and the mixed (z, w) coefficients. Therefore an orbit contributes its cardinality times a representative's entire coefficient block. Integer orbit multiplicities require no group-order division. The original and aggregated methods stream their subsets; the orbit method materializes a subset set and a catalogue. Its extra memory is measured. A single traversal constructs the catalogue, avoiding repeated minimum-set scans that would otherwise be quadratic when all orbits are singletons.

Let $\mathcal{O}_q$ be the certified orbits on subsets of size at most q, and let Q be the number of interpolation

nodes. The respective restricted call counts are

$$Q \sum_{j \le \min(q,n)} 2^j \binom{n}{j}, \qquad Q \sum_{j \le \min(q,n)} \binom{n}{j}, \qquad Q|\mathcal{O}_q|. \tag{68}$$

The identical exact arithmetic, restricted-count kernel, node set, truncation order, and full Hermite solve are used for the three methods. Thus additional gains of orbit over aggregation are not obtained by asking the orbit method for fewer output coefficients.

Fixtures and exact counts. The suite includes an empty formula, an unsatisfiable formula, fixed-group symmetric families, a partially broken symmetry, fully asymmetric deformation rows, and both asymmetric and symmetric node-varying layouts. In the main fixed-group family, $n = 2(k+1)$, each group has one boundary variable, and the clauses are $X_i \vee X_{b_{1-g}}$ for every nonboundary i in group g. Its truth count is $(2^k+1)^2$. This closed form is a useful additional check and also makes clear that this is not a hard SAT family. Asymmetric rows can destroy deformation symmetry even when the original CNF still has a simple symmetric form.

The final case has $n = 6$, boundaries $0, 3$, and all four clauses $X_i \vee X_j$ with $i \in \{1,2\}$, $j \in \{4,5\}$. Its count is $4(2^2+2^2-1) = 28$. The deformation rows have own-coordinate coefficients $-1/2$ in the first group and -1 in the second, plus the measured combination $X_0 - X_3$. At node 1 all factors are assigned to the first group; at node 2 all are assigned to the second. The relevant off-group unmeasured coefficients vanish at those nodes. The two within-group swaps remain certified. Here $d = 8, q = 4, J = 5$; 31 orbits cover 57 subsets. The three methods make 946, 114, and 62 restricted calls, respectively. This tests a real orbit reduction with rational rows and changing factor assignments, rather than only a layout already separable at zero.

Table 10: Exact CNF counts, accepted generators, and total restricted calls across all nodes. Counts and full polynomial outputs agree for O, A, and G.

ID	n	q	Count	Gen.	O calls	A calls	G calls
S01	6	0	64	2	2	2	2
S02	6	2	0	2	292	88	52
S03	6	2	25	2	292	88	52
S04	8	2	81	4	645	185	65
S05	8	3	81	4	2308	372	100
S06	10	2	289	6	1206	336	78
S07	8	4	81	4	5091	489	117
S08	8	3	81	3	2308	372	148
S09	6	2	25	0	292	88	88
S10	6	2	40	0	146	44	44
S11	6	4	28	2	946	114	62

Runtime and memory. The following results include symmetry discovery and catalogue construction in total time. O denotes the original pair routine, A the aggregated routine, and G the certified orbit routine. "Direct" is a separate truth-table count-only control, not a full-polynomial workload.

Table 11: End-to-end median milliseconds. Ratios greater than one favor G. Direct produces only the truth count and is not a matched full-polynomial workload.

ID	O ms	A ms	G ms	O/G	A/G	Direct ms
S01	0.486	0.592	0.625	0.78	0.95	0.047
S02	164.518	55.744	50.041	3.29	1.11	0.260
S03	231.618	95.140	70.163	3.30	1.36	0.401
S04	1,168.400	251.432	104.235	11.21	2.41	1.564
S05	3,609.031	594.488	120.167	30.03	4.95	1.716
S06	5,744.987	1,527.460	524.953	10.94	2.91	7.170
S07	6,960.293	453.487	185.596	37.50	2.44	1.884
S08	2,578.626	655.570	211.188	12.21	3.10	1.691
S09	217.112	75.735	65.543	3.31	1.16	0.322
S10	136.587	72.493	73.123	1.87	0.99	0.303
S11	1,398.999	378.203	256.360	5.46	1.48	0.388

Table 12: Observed timing ranges in milliseconds; seven samples per method, not confidence intervals.

ID	O range	A range	G range
S01	0.310–0.541	0.410–1.264	0.455–0.888
S02	140.222–309.382	50.953–70.630	30.692–67.670
S03	115.836–302.201	55.827–486.549	51.450–121.483
S04	617.266–1210.942	206.929–473.691	82.679–201.239
S05	3005.578–4354.493	367.473–1180.226	106.305–354.555
S06	4833.502–6260.226	912.579–1921.027	252.676–586.238
S07	6056.210–7373.114	411.937–1078.258	114.646–314.599
S08	2053.590–4069.640	369.284–1503.054	163.816–572.775
S09	142.032–306.069	49.389–138.241	49.239–92.769
S10	76.948–282.307	32.927–107.207	34.880–92.471
S11	936.397–1919.113	206.568–454.870	150.814–295.475

Table 13: Median peak traced Python allocation in KiB from two separate workers. Ratios below one favor G; these are whole-pipeline allocation peaks.

ID	O KiB	A KiB	G KiB	G/O	G/A	Direct KiB
S01	4.359	4.492	8.664	1.99	1.93	1.039
S02	27.937	27.897	41.788	1.50	1.50	2.312
S03	26.241	26.202	38.108	1.45	1.45	2.312
S04	39.065	38.815	55.925	1.43	1.44	2.312
S05	42.737	42.659	59.862	1.40	1.40	2.312
S06	57.026	56.651	79.956	1.40	1.41	2.344
S07	37.640	37.835	54.960	1.46	1.45	2.312
S08	42.733	42.655	59.921	1.40	1.40	2.312
S09	26.241	26.202	38.624	1.47	1.47	2.312
S10	12.006	12.271	20.541	1.71	1.67	2.312
S11	21.990	22.623	34.205	1.56	1.51	2.312

G has a lower median than A in 9 of the 11 cases, and than O in 10. The A/G median ratios over all cases range from 0.95–4.95; O/G ranges from 0.78–37.50. On the 8 cases with a strict certified orbit-call reduction, A/G ranges from 1.11–4.95. The observed G/A traced-peak ratios span 1.40–1.93, so memory savings are not inferred from call savings. The direct count-only control is faster than G in 11 of the 11 cases. These are comparisons of the recorded medians, not claims that every individual timing sample or every implementation has the same ordering.

Case S09 has no certified call reduction. Its slightly lower G median therefore must not be attributed to symmetry compression. S01 and S10 are controls in which G is slower than A. The strongest timing separation is on the deliberately structured cases, not on arbitrary formulas.

Reducing calls does not guarantee reducing peak memory. The catalogue, mixed jet state, recovery matrix, and returned coefficient vectors can dominate different cases. A trivial group returns no reduction in (68), while still incurring certificate overhead. Direct counting is included to prevent a gain over the original jet routine from being mistaken for a gain over the best way to solve these small formulas.

21.4 A matched modular contraction with degree-aware precision

The precision experiment uses $a = u = M = 1$, $L = 2$, $\alpha = 4$, and $S = \{-2, -1, 0, 1\}$. There is one real port in its packet. The source's Cauchy entry is the polynomial

$$W(T) = \frac{(T+2)(T+1)(T-1)}{-2} = 1 + \frac{T}{2} - T^2 - \frac{T^3}{2}. \tag{69}$$

The four tensor entries A_0, A_1, B_0, B_1 have fixed, nonnegative, creation-time integer coefficients in $\mathbb{Z}[V]/(V^\ell)$. The actual deformed contraction evaluated by the code is

$$F(T) = \sum_{x,y\in\{0,1\}} A_y B_x \, l_y(W(T)x), \qquad l_0(z) = 1 - z, \quad l_1(z) = z. \tag{70}$$

Consequently $F(0) = A_0B_0 + A_1B_1$, with products truncated modulo V^ℓ. Its coordinates are *weighted integer contraction values*, not numbers of satisfying assignments of a CNF. The CNF counts belong to the separate symmetry experiment.

For each $t = -1, 1$, all methods evaluate $F(t + 4\zeta)$ at $\zeta = 0, 1, U, 1 + U$ in

$$(\mathbb{Z}/2^b\mathbb{Z})[U, V]/(U^4 + U + 1, V^\ell). \tag{71}$$

The reduction of U^4+U+1 is irreducible over $\mathbb{F}_2$, and these four points have pairwise unit differences. Ordinary Lagrange interpolation therefore recovers the *entire cubic* at each node. Contents are canceled from the integer Cauchy factors before modular evaluation; powers of two are never inverted as units. Odd parts and extension-ring units are inverted exactly, with Newton lifting and verification. Every non-direct method performs eight genuine contraction evaluations, 32 Boolean summands, and two full cubic interpolations. No evaluation-call savings are attributed to deleting unused derivative outputs.

The local jets $h_{t,j} = [z^j]F(t + 4z)$ are integral even though F can have half-integral coefficients in T. Creation-time representatives are fixed across precision changes. An output precision p is chosen so that

$$2^p > \ell(\max A_0 \max B_0 + \max A_1 \max B_1). \tag{72}$$

Therefore every retained nonnegative coefficient is uniquely recovered as an integer, not merely compared as a residue. Coefficients of degree at least ℓ in V lie outside the specified algebra and are not claimed to be recovered.

Four principal ablations. With the six jets ordered as in Section 20.3, compare:

1. **S64:** the source allowance $b = p + 64$ and the six-channel canonical Hermite target formula with denominator 128.
2. **C7:** the identical formula and full evaluation workload at $b = p + 7$, isolating the earlier improvement in the precision certificate.
3. **D7:** the degree-three formula with denominator 16, still at $b = p + 7$, isolating the change of decoder from the change of working bits.
4. **D4:** the same degree-three formula at $b = p + 4$, implementing the v3 degree-aware precision reduction.

The two formulas are

$$F(0) = \frac{64(h_{-1,0} + h_{1,0}) + 10(h_{-1,1} - h_{1,1}) + h_{-1,2} + h_{1,2}}{128}, \tag{73}$$

$$F(0) = \frac{8(h_{-1,0} + h_{1,0}) + h_{-1,1} - h_{1,1}}{16}. \tag{74}$$

Division is performed on the exactly divisible integer numerator before reduction modulo 2^p. The source's literal truncated-reciprocal Hermite formula is separately checked against the compiled six-channel weights on the monomial basis through degree five and on sufficient-precision diagnostic runs. Compiling its fixed linear weights avoids unfairly charging the reference for repeated symbolic derivation. A six-channel diagnostic below its certified budget is not treated as a correctness reference.

21.5 A stronger but narrower two-bit control

This one-edge realization occupies a two-dimensional polynomial family, not the whole space of cubics. Direct expansion of (70) gives

$$F(T) = H + CW(T), \qquad H = A_0(B_0 + B_1), \quad C = (A_1 - A_0)B_1. \tag{75}$$

Since $W(-1) = 0$ and $[z]W(-1 + 4z) = 4$, one has the sharper identity

$$F(0) = h_{-1,0} + \frac{1}{4}h_{-1,1}. \tag{76}$$

The additional method **R2** uses $b = p+2$ and this formula while retaining the same eight evaluations and both full cubic interpolations. It is a restricted-family control, not a replacement for D4 on a general cubic. Its correctness follows coefficientwise from the displayed two-dimensional family and the valuation of $1/4$.

Two extra bits are also necessary for recovery from the six-channel residue data on this family. The zero function and $2^{p-1}W(T)$ have different targets modulo 2^p but all six jets agree modulo 2^{p+1}. The latter function is realized by nonnegative scalar tensors with $A_0 = B_0 = 0, A_1 = 1, B_1 = 2^{p-1}$. This is a lower bound for the declared observation contract, not for algorithms given the tensor entries: an algorithm may directly compute their contraction. Applying (76) to the generic cubic-class member T^2 returns -1 instead of zero. Thus R2's extra structural condition is essential.

This strengthens the local implementation without contradicting the v3 four-bit lower bound on *all* cubics. It also cautions against claiming optimality for a tested realizable family merely from a lower bound on a larger polynomial space.

21.6 Precision results: exact values, runtime, and memory

Eight cases cover tensor-entry sizes of 8, 64, 512 and 2048 bits and nilpotent dimensions $\ell = 1, 4, 16$. Every retained integer coefficient and every full local polynomial is checked for every method. Complete decimal output vectors, including the large values not practical to typeset here, are retained in the per-case oracle and result JSON files. Reading these files must preserve arbitrary-size integers (for example, Python's standard JSON decoder); conversion to floating-point numbers in some spreadsheet or JavaScript tools would lose exactness. The method **Direct** computes $A_0B_0 + A_1B_1$ by ordinary integer convolution; it returns the same target coefficients but not the local deformation polynomials.

Table 14: Weighted-contraction cases. Input bits are creation precision; output bits describe the exact retained integer coefficients, not an error tolerance. All five modular methods and Direct agree on every retained coefficient.

ID	Input bits	ℓ	p	Output bits	Exact comparison
P01	8	1	17	17–17	all equal
P02	64	1	129	129–129	all equal
P03	512	1	1025	1025–1025	all equal
P04	2048	1	4096	4096–4096	all equal
P05	8	4	19	17–19	all equal
P06	64	4	131	129–131	all equal
P07	8	16	21	17–21	all equal
P08	64	16	133	129–133	all equal

For example, P01 returns the exact integer 75637. All larger scalar values and nilpotent coefficient vectors are in `comparison-v4.json` and the raw oracle records.

Table 15: Median end-to-end milliseconds for the precision ablations. S64/C7 use six jets; D7/D4 use the cubic decoder; R2 uses only the certified one-edge family. All non-direct methods still compute eight evaluations and two full cubics.

ID	S64	C7	D7	D4	R2	Direct
P01	3.480	3.078	2.599	2.228	2.302	0.081
P02	3.739	2.883	3.427	3.724	3.543	0.088
P03	5.146	5.755	5.546	5.193	5.109	0.091
P04	19.963	26.865	25.878	23.571	24.848	0.129
P05	7.972	5.800	8.655	8.917	9.020	0.107
P06	4.725	5.071	4.601	5.524	4.297	0.075
P07	51.089	66.823	49.844	40.387	57.809	0.206
P08	68.413	69.285	74.091	61.415	59.586	0.274

Table 16: Observed ranges in milliseconds for the principal precision comparisons.

ID	S64 range	D7 range	D4 range	R2 range
P01	2.954–7.684	1.675–3.545	1.771–6.823	1.851–4.888

ID	S64 range	D7 range	D4 range	R2 range
P02	2.565–7.272	1.945–9.045	1.872–4.212	1.858–3.812
P03	3.641–8.212	3.851–7.483	3.333–6.670	3.474–7.894
P04	16.332–30.681	17.075–51.304	16.872–53.162	14.710–65.923
P05	6.224–13.871	4.744–27.771	8.576–21.861	6.376–12.257
P06	4.220–6.156	4.121–7.429	4.329–9.384	4.137–7.437
P07	28.762–94.863	28.788–119.971	29.650–74.533	29.743–80.638
P08	35.527–80.860	38.580–153.141	37.400–108.107	41.269–72.164

Table 17: Median peak traced Python allocation, KiB, for the same precision cases. Preparation, complete polynomial storage, and returned result objects are included.

ID	S64	C7	D7	D4	R2	Direct
P01	56.610	54.677	54.677	54.677	54.677	2.125
P02	58.216	57.880	57.880	57.880	57.860	2.188
P03	67.102	66.633	66.633	66.637	66.641	2.656
P04	100.004	99.457	99.457	99.434	99.426	4.250
P05	130.188	128.028	128.028	128.028	128.028	2.609
P06	132.544	131.927	131.927	131.942	131.985	2.812
P07	211.383	208.539	208.539	208.539	208.539	5.703
P08	220.609	216.562	216.562	216.566	216.578	6.469

D4 has a lower median than S64 in 4 of eight cases; S64/D4 ranges from 0.85–1.56. The isolated three-bit comparison D7/D4 ranges from 0.83–1.23, with D4 lower in 5 cases. The restricted-family D4/R2 ratio ranges from 0.70–1.29. These finite timing outcomes do not alter the proved precision budgets; a ratio near one or below one can reflect implementation overhead or timing variability and does not imply a failure of exact recovery. D4/S64 traced-peak ratios range from 0.97–1.00; D4/D7 ratios range from 1.00–1.00. Direct integer contraction is faster than D4 in 8 of eight cases.

In these measurements the three-bit D7–D4 change yields neither a consistent runtime advantage nor an appreciable traced-memory reduction. The nearly equal allocation ratios must not be interpreted as a substantial memory saving. The certified reduction in required bits is a mathematical result; a robust performance benefit from that reduction remains unestablished by this benchmark.

The source algorithm already handles sufficiently small networks by direct summation. In particular, this one-edge fixture is not an active example satisfying the source's large-size packet inequalities. Its deformation workload is deliberately exercised to measure a local arithmetic component. A gain for D4 or R2 over S64 therefore does not establish a gain over the full source algorithm's best path on that input. The direct control makes this limitation operational rather than merely verbal.

All methods except Direct evaluate the same eight points. Any change in elapsed time comes from modular arithmetic width, the unit-lifting schedule, decoder arithmetic, and implementation effects, not fewer recursive calls. The D7–D4 comparison isolates the extra three bits from the degree-aware certificate; it need not give a detectable or consistent speedup in Python. Allocation granularity and common polynomial storage can similarly mask a three-bit memory change. The scalar-ring operation counters are present in all precision variants, so their overhead is shared, but these remain research demonstrators rather than optimized production kernels.

21.7 Negative tests, review, and what they rule out

The symmetry selftest checks six layouts under all three methods and 176 mixed-coefficient comparisons under accepted generators. It rejects a nonpermutation and rejects the candidate swaps for fully asymmetric rows. Bypassing certification on the recorded asymmetric fixture changes the true count 25 to 2,333,861,705. Both coefficient corruption and that wrong constant term are enforced negative controls. A separate read-only audit confirmed the full-family certificate and the node-varying count-28 case; its catalogue-scaling finding was corrected before the main timings.

The precision selftests check all case/method combinations with complete Fraction-oracle validation, cardinal interpolation identities, the literal Hermite formula, invalid degree use, insufficient precision, nonunit inversion, insufficient integer-output range, and corruption of a cubic coefficient that a count-only test would miss. They also demonstrate why recomputing a creation-time expression at a larger modulus is not the same as lifting its stored canonical representative. For example, an entry stored as $(1 + 1) \bmod 2 = 0$ must remain the integer lift zero, not become two when later working at higher precision.

The main runs contain 644 timed executions, 184 memory executions, and 92 diagnostic executions, all validated. This totals 920 fresh worker outputs, including the explicitly labelled direct controls. The two final selftest files record additional correctness checks, which are excluded from these timing statistics.

These checks do not certify all source proof dependencies or arbitrary user-created layouts. The orbit certificate is deliberately sufficient and conservative, and the precision degree certificate follows from the implemented one-edge construction. There is no heuristic degree guessing or unchecked approximate symmetry in a timed solver.

21.8 Interpretation and requirements for a general algorithm

The experiments advance v3 in two concrete ways. Orbit reductions now operate inside an exact CNF pipeline and are measured with discovery, recovery, and memory included. Degree-aware precision now controls actual modular contractions and interpolation, rather than merely the size of formal weights applied to already available jets. The additional R2 identity exploits the precisely stated realizable family. None of these steps proves a better unrestricted SAT exponent.

The arithmetic and symmetry mechanisms are expressible through conventional Hermite interpolation, two-adic divisibility, generating functions, and group actions. Ram Calculus supplies the organizing target/jet/closure framework used to expose and audit them. The comparisons do not establish that an equivalent conventional implementation would be less accurate or slower. All reported answers are exact; there is no approximate-accuracy advantage to claim.

A full-algorithm improvement would require certified degree and precision contracts at every active recursive node, creation-time lift consistency, the complete source field and packet schedule, and a bit-complexity analysis including preprocessing and interpolation. A symmetry reduction would have to survive all node-dependent constructions and continue to save representative inner work after certification costs. Larger difficult instances, appropriate established counting baselines, and full-recursion memory measurements are still needed. The present narrow controls must not be extrapolated to instances lacking their structure.

21.9 Files and reproduction

All new implementation and raw evidence are in `eth-benchmark-v4` beside this report. The code uses the Python standard library; existing source modules and reports are read only and bytecode writing is disabled. The two suites run sequentially, and output directories must not already exist. Example commands, run from that new directory, are:

```
python -B run_benchmarks.py selftest --module symmetry --out new-sym-test.json
python -B run_benchmarks.py selftest --module precision --out new-prec-test.json
python -B run_benchmarks.py run --module symmetry --out new-sym-run
python -B run_benchmarks.py run --module precision --out new-prec-run
```

Use normal Python, not `-O` or `-OO`. The frozen Section-3 dependency uses assertions and the new launcher rejects optimized Python. The final raw directories are `matched-symmetry` and `matched-precision`; the separately labelled pilot is excluded from reported statistics. Summary CSV/JSON files contain exact outputs and all method comparisons, and the report-generation script derives its numeric tables from the raw summaries. Existing LaTeX, PDF, benchmark and public source files are not overwritten.

22 Equation comparison and attribution of improvements

Table 18: Equivalent representations and their actual benefit.

Conventional/source form	Ram-compatible form or refinement	Assessment
Sum over Boolean assignments	Boolean quotient plus moment functional $\Lambda(X_S) = 2^{n-\lvert S\rvert}$	Exact rederivation; no automatic compression.
Condition on shared linear measurements	Exact aggregate-state/fiber operator with multiplicity sums	Same source identity; clearer certificate and state-cost accounting.
Subset-pair Boolean jets	One restricted call per U plus a local Hamming generating jet	Proved fewer calls; polynomial arithmetic overhead; no better headline exponent.
Uniform first-order correction budget	Weighted downward-closed support family	Proved conditional extension when higher correction orders exist.
Conservative Hermite precision envelope	Actual valuation of the target-functional weights	Proved certificate; strict finite examples; no general asymptotic gain shown.
Arbitrary summary row list	Exact row-basis selection and row-specific integer boxes	Proved instance-dependent reduction without changing fibers.
Tensor basis replacement	Dual inverse transforms on contracted legs; transported product	Exact equivalence; dimensions and hidden contraction costs remain.

Conventional/source form	Ram-compatible form or refinement	Assessment
Ordinary low-rank or L^2 truncation	Quotient-ideal or valuation-certified truncation; otherwise explicit target error	Prevents unsupported exact-count claims; may require more computation.
Physical quaternionic channels	Ordered real block representation	Methodological relevance only; no demonstrated SAT improvement.

22.1 Advantages of the Ram-based organization

The strongest advantage is a disciplined modular description: state space, coefficient algebra, multiplication, observation functional, and error certificate are all explicit. The new generating-jet aggregation exemplifies how an extra small coefficient state can eliminate a repeated combinatorial index. Weighted order budgets make stronger local vanishing useful without discarding necessary contributions. The precision certificate targets the count itself rather than an unnecessarily uniform coefficient accuracy.

The corpus also contributes useful safeguards: projection leakage is distinguished from quotient reduction; rank is distinguished from attainable state count; finite inverses are distinguished from unrestricted inverses; and nonlinear transformed products retain their range constraints. These are valuable for an implementer deciding where a local optimization remains mathematically exact.

22.2 Disadvantages and comparisons with existing methods

The scalar proof can already be expressed concisely using conventional finite differences, quotient rings, generating functions, and Hermite interpolation. Calling these objects Ram operators does not confer a unique speed advantage. More coefficient dimensions can increase constants and bit arithmetic. Exact row compression may be redundant, sparse-state storage can destroy polynomial-space guarantees, and complex or hypercomplex coordinates can make a purely rational problem harder to implement.

The source's main innovations are precisely where a generic representation is insufficient: simultaneous low-degree incidence designs, arithmetic degree control, and a valuation-sensitive contraction recursion. Those ingredients are retained rather than replaced. A fair implementation comparison must use the source's streamed and structured baseline, not only a naive dense interpolation algorithm or exhaustive assignment sum.

23 Research directions with explicit success criteria

23.1 Implement the aggregated routine inside the actual construction

Revision 2 implements both routines on checked finite rows, including node-varying layouts, and records the matched work, runtime, and memory in Section 19. A next implementation should exercise the source's actual general Section 4 construction and more optimized baselines, rather than only these small input-specific designs. A useful result would identify the range of q, J and group sizes in which the saved calls outweigh the extra generating dimension. Correct output must be compared to exact counts on every tractable instance.

23.2 Search for higher-order correction designs

Theorem 12.1 becomes useful if real designs give $\nu_i > 1$ for many coordinates. The optimization problem is to reduce the weighted subset family without increasing R, the number or range of measurements, or construction cost enough to cancel the saving. An asserted improvement to (35) must display all three terms after the redesign. A formal higher-order zero is not free.

23.3 Deploy exact target weights in recursive two-adic recovery

The nodewise certificate can be inserted into a prototype of Appendix D while preserving immutable creation-time lifts. A first goal is a measured reduction in working precision and mode enumeration on nontrivial recursive instances. A stronger goal would be an improved *uniform* upper bound on the valuations of the Hermite weights. Only the latter, combined with the recursion and packing bounds, could justify a better asymptotic exponent. The finite numbers in Table 3 do not establish it.

23.4 Finite-domain and weighted-count extensions

The finite-alphabet algebra (18) and weighted factors permit formal extensions to fixed-domain constraint satisfaction and rational partition functions. The source Section 3 already allows commutative $\mathbb{Q}$-algebras, so that observation alone is not new. A new complexity theorem would require a compatible low-degree design, bounds on domain encoding, factor scopes, coefficient sizes, and a final modulus sufficient to recover the weighted answer. Positivity and an $n + 1$-bit count bound cannot be assumed for arbitrary signed weights.

23.5 Formal verification and negative results

The aggregation and weighted-support theorems are finite-sum identities well suited to Lean formalization. Their implementation should be linked to explicit coefficient arrays and the streamed state machine, rather than proving only an abstract equality. The target-precision theorem similarly needs a verified valuation calculation and fixed-lift interface.

No argument here proves that the source's exponent is optimal. A lower bound for a selected moment representation, an example with many summary values, or an inability to improve an interpolation estimate is not a lower bound for all algorithms. Proving a genuine general lower bound would require a separate computational-model theorem. Conversely, an equivalent Ram formulation does not by itself overcome such a barrier.

24 Conclusion

Ram Calculus is substantively relevant to the source's finite algebra and verification interfaces, not because the paper secretly solves an auxiliary PDE. The most concrete addition is the aggregated Boolean-jet identity, which replaces a subset-pair enumeration by a subset enumeration with a small exact generating state. The weighted version and exact row-basis reduction give conditional instance-sensitive extensions. Target-specific two-adic Hermite weights yield provably valid precision budgets, with strict improvements in the executed finite examples.

These results improve selected calculations and clarify their limitations. The added finite benchmarks demonstrate exact agreement and local implementation speed gains with a memory

tradeoff, but they do not supply a stronger unrestricted SAT time bound, certify performance on general hard instances, or replace the source's difficult construction and recursive proof. The mathematical contribution is therefore a collection of exact, testable refinements and a precise separation of representation, information, arithmetic precision, and computational complexity. The positive research opportunity is to integrate these refinements into the actual algorithm and determine, with full bit and memory accounting, which local advantages survive at scale.

A Bounded-audit findings and arithmetic safeguards

The separate source-level audit did not find a demonstrated error in the examined Section 3–5 identities, Section 4 construction, Appendix B arithmetic accounting, and Appendix D precision and recursion arguments. This statement is deliberately narrower than global certification. Appendix A's complete combinatorial proof, all external prerequisites, and the circuit-lower-bound consequences were not independently re-proved; the Lean dependency graph was not rebuilt.

Two arithmetic details deserve emphasis. First, parity divisibility in Section 5 is used at the restored integer inputs at $T = 0$, not at the rational deformed inputs at other nodes. Second, the source bounds intermediate numerators, not merely the final answer. A common denominator for a full contribution can be fixed in advance. Adding exponentially many signed contributions increases the numerator bit length by the logarithm of their number, provided each contribution already has a uniform height bound. It is not necessary to multiply every accumulated term into the next one. Polynomial-sized interpolation then admits exact fraction-free elimination with bounded minors. "Fraction-free" includes exact integer division and is not synonymous with division-free inversion.

The contraction audit likewise preserves the distinction between the scalar two-local field and the truncated coefficient algebra containing nilpotents. Only scalar units are inverted. Canonical lifting is not a ring homomorphism: the canonical lift of $1 + 1$ modulo two is zero, while the sum of the two lifts is two. The creation-time convention avoids needing that false homomorphism property. Reducing precision without preserving this convention would not be justified by Theorem 16.1 alone.

These observations explain why the report offers refinements rather than claiming repairs to proven source errors. The negative controls reject tempting but invalid alternative implementations. They are not findings against the original manuscript.

B Source register and reproducibility

Revision 2 benchmark code, frozen instances, per-trial data, and independent static audits are retained in `eth-benchmark-v2/`. The first timed code version and its data remain unchanged; the second version adds node-varying fixtures and a stricter partition contract. Public source TeX files are stored in `eth-support/source-42907c9/`. Extracted PDF text is `eth-support/source-paper-layout.txt`. The small test program and its JSON output are in `eth-support/`. Independent research notes record corpus paths and bounded-audit findings there. Existing research reports and supplied files were not changed.

The mathematical claims proved in this report do not require the entire private corpus as an axiom system. The specific imported ideas and their limitations are identified in Section 6; the main new identities have self-contained proofs. In particular, the aggregation theorem uses only finite subset inversion and distributivity, and the precision theorem uses only a fixed exact recovery functional and the valuation inequality for sums.

C Revision 3 corpus register, provenance, and reproducibility

The 13 source volumes were inventoried by their stable topic markers, not by inferred page headings. The inventory contains exactly 627 unique entries: 428 MathWorld-linked topics and 199 Ram-specific topics. The corpus reference is [20]. A complete topic register, with volume, source line anchors, and separate dispositions for the two OpenAI papers, is supplied as `mathworld-v3-support/topic-relevance-register.csv`. It includes topics not selected for a further application. That disposition is not a theorem that a topic can never be useful.

All 627 entries were included in the structural and broad relevance screen. Mathematical inspection was targeted to the potentially useful formulations, hypotheses, examples, and limits, rather than an attempted fresh proof audit of every result in all 13 books. Broad keyword matches are not treated as evidence of substantive applicability or novelty. For this paper the register distinguishes 32 explicit crosswalk entries, 81 related background entries, and 514 entries from which no additional application was selected. The two papers' categories overlap and are not additive.

The following precise locators support the mathematical crosswalk. A line number is the start of the topic entry in `Ram-Calculus-MathWorld-Volume-NN.tex`; it is not a PDF page number. Topic IDs remain the preferred stable locators when a volume is edited.

Topic	Title	Volume / line
M024	Finite Difference	01 / 3772
M025	Binomial Transform	01 / 3956
M026	Generating Function	01 / 4131
M027	Moment	01 / 4303
M028	Moment-Generating Function	01 / 4481
M097	Divided Difference	02 / 8775
M101	Umbral Calculus	03 / 175
R43	Hermite Data Panels	05 / 7811
R69	Goal-Oriented Ram Error Budgets	06 / 2866
R78	Mixed-Precision Ram Error Budgets	06 / 4222
M225	Hermite's Interpolating Polynomial	07 / 3502
M248	Walsh Transform	07 / 7028
M249	Hadamard Transform	07 / 7180
M310	Krawtchouk Polynomial	09 / 2100
R103	Mixed-Derivative Generation of the Full Jet Matrix	11 / 5040
R106	Polynomial Compatibility Map Between Displaced Jets	11 / 5490
R111	Degree-Closure Conditions for Variable-Coefficient Jet Rows	11 / 6241
R112	Tall Forward Jet Map of Polynomial-Coefficient Differential Operators	11 / 6393
R113	Graded Monomial Bandedness of Multi-Index Ram Operators	11 / 6549
R117	Quotient Closure of Retained Information and the Invariant Lifting Equation	11 / 7160
R123	Composing Coefficients Force Binomial-Type Bases	12 / 175
R135	Finite Closure Characterizes the Taylor Jet; Scale-b Jets Do Not Close	12 / 2905
R152	Coefficient-Sum Spectral Constraint	12 / 6259
R165	One-Channel Pruning Identity for Ram Measurement Banks	12 / 8325
R173	Noncharacteristic Jet Recurrence and the Boundary Closure Determinant	13 / 175
R185	Difference-Jet Representation of Discrete Shift-Variant Kernels	13 / 1948
R187	Shifted Leibniz Rule for Forward Differences	13 / 2236
R189	Graph-Local Moment Matrices and Multilinear Exactness	13 / 2517
R190	Nilpotent Policy-Graph Inverse and the Soft-Value Path Partition	13 / 2662
R195	Hop-Shell Aggregation with Covariance Remainder	13 / 3364

Topic	Title	Volume / line
R198	Lumpability Intertwining Defect	13 / 3780
R199	Dynamic Kron Elimination as Graph Volterra Memory	13 / 3939

Public-source verification was repeated on October 26, 2026. The current GitHub PDF blob identifiers for both manuscripts were unchanged from the previously retained versions. The wave manuscript's two root source files and all 19 section files matched the local pinned blobs; the ETH manuscript's four build TeX files also matched. This checks version identity, not the correctness of the source theorems. The records are `mathworld-v3-support/public-source-provenance.json` and `mathworld-v3-support/public-wave-sections.json`. Only public repository identifiers were requested online; private corpus text was not included in public search queries.

The original 13 volume sources and the two baseline reports and their PDFs are protected by the SHA-256 register `mathworld-v3-support/input-sha256.json`. Revision 3 is a new, standalone TeX source retaining the preceding report's full technical body, examples, comparison tables, and recorded checks. The additional material is separately identified; the earlier experiments were not silently relabelled as new tests. No original source or PDF was replaced. The code and JSON outputs in the new support directory distinguish exact arithmetic tests from floating-point manufactured diagnostics. Neither kind of test substitutes for the stated analytic proofs or for execution of the source's full algorithm/proof construction.

References

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[2] OpenAI. *Math repository: description and verification status*. Accessed October 26, 2026. https://github.com/openai/math/blob/main/README.md.

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are given in Appendix C. Private corpus source, not a claim of external endorsement or an independent novelty certification.